Lagrangian and Hamiltonian Mechanics

Hannah Price, based on notes by NK Wilkin, MR Dennis,
DM Gangardt and IV Lerner,

University of Birmingham UK

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1 Lagrangian approach to classical mechanics

The first part of the course will focus on the reformulation of Newton's Laws of Motion in terms of Lagrangian mechanics. As you will remember, Newton's theory is phrased in terms of three "Axioms or Laws of Motion":

- First Law (Inertia): An object remains at rest or moves with constant velocity unless acted upon by a net external force.
- Second Law (Dynamics): The net force, $\mathbf{F}$, on an object equals the rate of change of its momentum, i.e.

$$\mathbf{F} = \frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} \equiv \dot{\mathbf{p}}$$

- Third Law (Action–Reaction): For every force exerted by one object on another, there is an equal-magnitude, opposite-direction force exerted back on the first object, i.e $\mathbf{F}_{12} = -\mathbf{F}_{21}$.

Aside: Isaac Newton (English, 1643-1723), is acknowledged as one of the greatest scientists of all time, especially for his mathematical work on mechanics and gravitation (for which he introduced a method of calculus, known as "fluxions", including the dot notation for t derivatives), and his experimental work on optics. He spent a long period as a civil servant, as Warden and Master of the Royal Mint. He also wrote books about alchemy and theology, and is remembered for his unpleasant temper (and long dispute with Leibniz over the discovery of calculus)

All that is left is then to apply these laws to any system of material bodies by writing and (somehow) solving the appropriate system of differential equations:

$$\sum_j \mathbf{F}_{ji} + \mathbf{F}_i^{\mathrm{ext}} = \dot{\mathbf{p}}_i\,, \qquad i = 1, \ldots, N \tag{1.1}$$

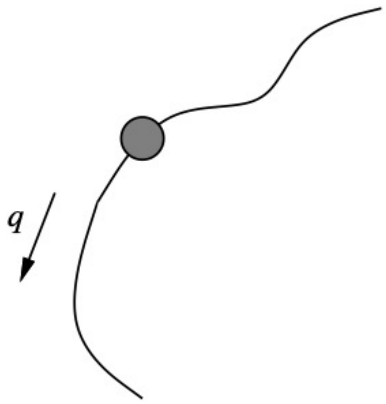
where N is the number of bodies in the system, and where we include both forces between two bodies and external forces. Impressively, this is all that was needed for Newton to prove Kepler's laws concerning the motion of astronomical bodies, such as planets, moons and comets, which otherwise had long mystified humanity.

1.1 Difficulties of the Newtonian approach

However, there are three main problems with the Newtonian approach:

1. ***The Newtonian equations of motion (EoM) are excessive***: the system 1.1 may contain a lot of constraining forces that are (i) difficult to determine and (ii) not interesting by themselves. Even the simple (albeit contrived) example of a bead moving on a bent piece of wire is not that simple in the Newtonian approach: there, the equation of motion includes forces which keep the bead on the wire and it is not easy to derive them. On the other hand, as the only role of these forces is to ***constrain*** the motion of the bead to the wire, they may be of

no interest in solving the problem. Thus, ***the EoM contain* redundant *information***.



A bead on a bent wire: q is the "generalised coordinate" – the meaning of this will be explained later – corresponding to the distance along the wire.

2. ***The Newtonian equations of motion are not invariant*** with respect to coordinate transformation. These equations are vector equations, and vectors are nontrivially transformed when changing coordinates. It is often not obvious what would be the best coordinate frame for a particular problem as the EoM may look very different in different frames.

3. ***It's difficult to write down the EoM*** except in the simplest cases, as expressions for $\dot{p}$ in any coordinate system but Cartesian are rather involved; even the equation for the Earth's motion around the Sun – one of the simplest – is already rather complicated, in spite of the (approximate) circular symmetry of the problem. Constraints, when exist, lead to further pronounced difficulties.

In this section, I will first remind you what conservative systems and potential forces are. Then I will cast Netwon's 2$^{\text{nd}}$ law for this case into the form of ***Lagrange's equations***. Finally, I will explain the concept of generalised coordinates, which forms the cornerstone of the Lagrangian approach, and illustrate how they allow one to avoid an explicit consideration of constraints.

1.2 Conservative systems and potential forces in Newtonian mechanics

Loosely, conservative systems are those with no dissipative forces such that the full mechanical energy is conserved. Simple non-dissipative forces are potential, i.e. they can be derived from a potential energy function. Let me now remind you what all this means.

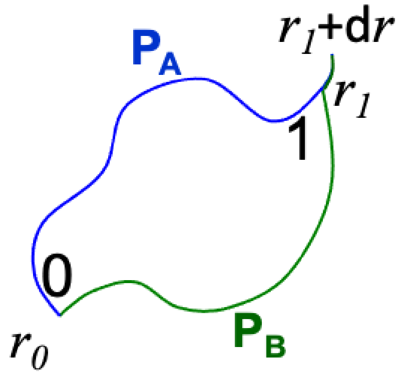
The conservation of energy follows from Newton's laws. Consider the work done by an external force $\mathbf{F}$ on a particle:

$$\begin{aligned} W = \int \mathbf{F}\cdot \mathrm{d}\mathbf{r} &= m\int_{\mathbf{r}_0}^{\mathbf{r}_1} \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}\cdot \mathbf{v}\,\mathrm{d}t \\ &= \frac{m}{2}\int_{\mathbf{r}_0}^{\mathbf{r}_1} \frac{\mathrm{d}}{\mathrm{d}t}\left(v^2\right)\,\mathrm{d}t = \frac{m}{2}(v_1^2 - v_0^2) \equiv T_1 - T_0\,, \end{aligned} \tag{1.2}$$

where v_0 and v_1 are velocities at point $\mathbf{r}_0$ and $\mathbf{r}_1$. The kinetic energy $T = mv^2/2$ is conserved when no work is done, *i.e.* the force is perpendicular to the displacement.

Note that this is typical for constraint forces - they never produce any work and do not contribute to energy.

Now we can consider the more general case of a ***conservative*** force. A force is called conservative (or potential) if the work done by it is independent of path. This is illustrated in the figure below. Here, the paths P_A and P_B are associated with the same amount of work being performed. By definition, the work of such a force depends only on endpoints and so can be written as



$$W = \int_{\mathbf{r}_0}^{\mathbf{r}_1} \mathbf{F} \cdot \mathrm{d}\mathbf{r} \equiv V(\mathbf{r}_0) - V(\mathbf{r}_1). \tag{1.3}$$

The reason for the choice of sign is convention – it turns out that $V(\mathbf{r})$ is the ***potential energy*** of the particle at $\mathbf{r}$. Indeed, since Eq. (1.2) is always valid, one arrives at the mechanical energy conservation law by combining Eqs. (1.2) and (1.3):

$$T_1 + V_1 = T_2 + V_2\,.$$

Let us now change infinitesimally the destination point: $\mathbf{r}_2 = \mathbf{r}_2 + \mathrm{d}\mathbf{r}$. The resulting change in work is entirely due to the infinitesimal change of the potential at the destination (omitting the index 2):

$$\begin{aligned} dW = V(\mathbf{r}) - V(\mathbf{r}+\mathrm{d}\mathbf{r}) &= V(x,y,z) - V(x+\mathrm{d}x, y+\mathrm{d}y, z+\mathrm{d}z) \\ &= -\frac{\partial V}{\partial x}\mathrm{d}x - \frac{\partial V}{\partial y}\mathrm{d}y - \frac{\partial V}{\partial z}\mathrm{d}z \equiv -\boldsymbol{\nabla} V \cdot \mathrm{d}\mathbf{r} \end{aligned}$$

and on the other hand $\mathrm{d}W = \mathbf{F}\cdot\mathrm{d}\mathbf{r}$, so that

An example of the most known non-conservative force is friction: in its presence, work obviously does depend on path.

$$\boxed{\mathbf{F} = -\boldsymbol{\nabla} V} \tag{1.4}$$

The conservative force is just the gradient of the ***potential energy*** V. For most of this course, we will be dealing only with conservative forces.

We can then put this into Newton's second law of motion as:

This is a first slightly "magic" way to get to the notion of a Lagrangian. We will do this in better ways later on!

$$-\boldsymbol{\nabla} V = \mathbf{F} = \frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} = m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t}.$$

But let us also note that the velocity of such a particle can be related back to the kinetic energy as $T = \frac{1}{2}m\mathbf{v}^2$, which we can also write as $m\mathbf{v} = dT/d\mathbf{v}$. We can then re-express Newton's equation in terms of energies as

$$\boldsymbol{\nabla} V + \frac{d}{dt}\frac{\partial T}{\partial \mathbf{v}} = 0.$$

However, for conservative forces, the potential energy must also be independent of the velocity i.e. $\partial V/\partial \mathbf{v} = 0$. Similarly, the kinetic energy is independent of position (as $\mathbf{r}$ and $\mathbf{v}$ are independent), i.e. $\partial T/\partial \mathbf{r} = 0$.

Note: the gradient operator acts only on spatial coordinates, so $\mathbf{F} = -\boldsymbol{\nabla} V$ implies that V is a function of position alone and does not depend on velocity.

Joseph-Louis Lagrange (Italian-French, 1736-1813) is best remembered for his work on Mechanics, Calculus and Probability. He had an uncompromising mathematical style; he famously wrote, in the preface to his Méchanique Analytique, "the reader will find no figures in this work. The methods which I set forth do not require either constructions or geometrical or mechanical reasonings: but only algebraic operations, subject to a regular and uniform rule of procedure". His lectures were famous for being difficult to follow!

This implies that we can rewrite the above equation with the derivatives acting on only a single function, either the sum $T+V$ (which we recognise as the total energy) or the difference $T-V$. Although it may seem a bit strange at first, it turns out we can do a lot more with the second choice; this we define as the ***Lagrange function*** most often called simply ***Lagrangian*** (writing now $\dot{\mathbf{r}}$ instead of $\mathbf{v}$):

$$\mathcal{L} \equiv T - V = \frac{1}{2}m\dot{\mathbf{r}}^2 - V(\mathbf{r}). \tag{1.5}$$

We then cast Newton's 2[nd] law into the form of ***Lagrange's equation*** which is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{\mathbf{r}}}\right) - \frac{\partial \mathcal{L}}{\partial \mathbf{r}} = 0 \tag{1.6}$$

Indeed you can work backwards and check,

$$\begin{cases} \dfrac{\mathrm{d}}{\mathrm{d}t}\left(\dfrac{\partial \mathcal{L}}{\partial \dot{\mathbf{r}}}\right) = \dfrac{\mathrm{d}}{\mathrm{d}t}(m\dot{\mathbf{r}}) = m\ddot{\mathbf{r}} \\ \dfrac{\partial \mathcal{L}}{\partial \mathbf{r}} = -\dfrac{\partial V}{\partial \mathbf{r}} \equiv -\boldsymbol{\nabla} V \end{cases} \Rightarrow \quad \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{\mathbf{r}}}\right) - \frac{\partial \mathcal{L}}{\partial \mathbf{r}} = 0 \quad \Leftrightarrow \quad m\ddot{\mathbf{r}} + \underbrace{\boldsymbol{\nabla} V}_{\equiv -\mathbf{F}} = 0\,.$$

We have therefore taken a simple, known problem and made it look maybe more complicated without any obvious benefit. However, the power of the Lagrangian approach will hopefully become apparent after a few more examples.

1.3 Constraints and generalised coordinates

Note: Constraints that cannot be written in the form $\varphi_n(\mathbf{r}_1, \ldots \mathbf{r}_N, t) = 0$ are "non-holonomic", and typically involve velocities (e.g. rolling without slipping). Such systems may still be easy to solve, but there's no general theory so you need a different method for each problem.

The main practical advantage in introducing the Lagrangian is that it remains in the same form in the presence of any constraints, choosing convenient coordinates to write T and V, and thus $\mathcal{L}$. Moreover, we can choose the number of coordinates to not be larger than the number of ***degrees of freedom***.

The number of degrees of freedom for a system are the number of independent coordinates needed to specify its configuration at an instant. For example, a free particle can move in three dimensions and has 3 degrees of freedom. However, when its motion is constrained – *e.g.*, to a plane or spherical surface – the number of degrees of freedom reduces to 2, or even to 1 if the motion constrained even further (*e.g.*, for a train moving along railway, or a bead on a wire, or a mass rotating at the end of a rope, *etc*). The motion of two bodies may also be constrained by the presence of strong interactions between them. Generically, many constraints are present in the mechanical motion of a many-body system. The essence of Lagrangian mechanics is to take account of ***all the constraints***, reducing the number of equations of motion for the system.

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Constraints which can be written down explicitly, are called *holonomic*

For example, if there exist k holonomic conditions that relate the coordinates of N bodies in different moments of time,

$$\varphi_n(\mathbf{r}_1, \ldots \mathbf{r}_N, t) = 0\,, \qquad n = 1, \ldots, k$$

they can be (in principle) resolved to eliminate k out of $3N$ coordinates, so that the system has $3N - k$ ***degrees of freedom***.

Such an elimination can be captured by introducing new independent variables, $q_1, \ldots, q_{3N-k}$, so that the old variables are expressed via the new as

$$\mathbf{r}_n = \mathbf{f}_n(q_1, \ldots, q_{3N-k}, t)\,, \qquad n = 1, \ldots, N$$

The parameters q_i, together with their time derivatives $\dot{q}_i$ are then the ***generalised coordinates and velocities***. , which fully describe the mechanical motion in the system with constraints.

Aside: Properly, we should include holonomic constraints by introducing "Lagrange multipliers" (a concept you previously met in Maths) to write a new "constrained Lagrangian". However, provide we are only interested in the dynamics of generalised coordinates, we can forget about this subtlety and just work with our original unconstrained Lagrangian, making calculations much easier.

Important: generalised coordinates are not unique! Any convenient choice describing the same configuration is equally valid.

Examples:

- A particle moving on a spherical surface: The constraint, $\phi_1 = 0$, is $\varphi_1(x, y, z) = x^2 + y^2 + z^2 - R^2$. Two angles, longitude θ and latitude ϕ, are 'natural' generalised coordinates, so that $x = R\cos\theta\cos\phi$, $y = R\cos\theta\sin\phi$ and $z = R\sin\theta$. A second constraint, $\varphi_2 = 0$, would confine motion to a one-dimensional curve (*e.g.*. $\varphi_2 \equiv z$ would constrained motion to that along the equator). For a *moving* surface – *e.g.*the Earth moving around the Sun – such constraints would be time-dependent.

Note: here R is a constant, not a coordinate!

- A bead on a straight spoke is obviously a one-dimensional system - so it should have two constraints. One of them could be $z = 0$ (the spoke is in the xy plane), and the other, choosing the coordinate origin to be on the spoke, would be $y/x = \tan\theta$. But clearly the most convenient "generalised coordinate" would be just a position of the bead on the spoke.

A bead on a wire of arbitrary shape is again a one-dimensional system. To write constraints explicitly is more difficult, but again the bead's position on the wire is a natural generalised coordinate.

1.4 Lagrangian and Lagrange's equations

In general, a Lagrangian for a conservative (non-dissipative) system with s degrees of freedom is defined in the same way as for the simple example above:

$$\boxed{\mathcal{L}(q_i, \dot{q}_i) \equiv T(q_i, \dot{q}_i) - V(q_i)} \tag{1.7}$$

$$\dot{q}_i \equiv \frac{\mathrm{d}q_i}{\mathrm{d}t}\,, \quad i = 1 \ldots, s. \tag{1.8}$$

Here T and V are the kinetic and potential energies of the system, $\{q_i\}$ and $\{\dot{q}_i\}$ are sets of ***generalised*** coordinates and velocities characterising all the s degrees of freedom in the system. Generalised

Example: for the bead on a wire in Fig.1, the distance along the wire from some fixed point to the bead could be chosen as a generalised coordinate.

coordinates may be the standard cartesian (or polar, *etc*) coordinates but in general this is $\{q_i\}, \{\dot{q}_i\}$ could be any set of numbers fully characterising positions and velocities of all bodies in the system.

Then ***Lagrange's*** equations of motion for any conservative mechanical system are given by

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) - \frac{\partial \mathcal{L}}{\partial q_i} = 0\,, \qquad i = 1, \dots, s \tag{1.9}$$

Lagrange's equations are, like Newton's, still 2$^{\text{nd}}$ order differential equations.

I will derive these equations from Newtonian mechanics in week 2, and from a much more general approach in the middle of the course.

Note that in the most trivial case, $V = 0$, Lagrange's equations become

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{q}_i}\right) - \frac{\partial T}{\partial q_i} = 0\,, \qquad i = 1, \dots, s \tag{1.10}$$

which if T, as is usual, is a function only of $\dot{q}_i$, then the second term vanishes and we can deduce that $\ddot{q} = 0$ which is the first Newton's law.

1.5 Summary: advantages of the Lagrangian approach

- It is based on the *energy* of the system, which is (i) more readily calculable than the forces exerted on the system and (ii) invariant with respect to the choice of coordinates.
- By picking convenient generalised coordinates, it both allows one to reduce the number of degrees of freedom and removes the need to consider forces of constraints.
- If there exist conservation laws, they may be accounted for by a proper choice of generalised coordinates.
- The generality of the Lagrangian approach makes it very convenient in electrodynamics, general relativity and (modern) quantum mechanics.

In the second part of the course we will consider the ***Hamiltonian approach*** which allows one, amongst other things, to include all conservation laws automatically, independently of the choice of generalised coordinates.

1.6 Kinetic energy in generalised coordinates

For a system of n parcicles, we want to express the kinetic energy $T = \sum \frac{1}{2} m \dot{\mathbf{r}}_n^2$ via q and $\dot{q}$. To do this, let us consider a general coordinate change:

$$\mathbf{r}_n = \mathbf{r}_n(q_1, \dots, q_s; t), \quad n = 1 \dots N, \quad s = 3N - k.$$

Here k is the number of constraints such that s is the number of degrees of freedom. Using the chain rule, we write down the Cartesian velocities $\dot{\mathbf{r}}_i$ in terms of generalised velocities $\dot{q}_i$:

$$\mathrm{d}\mathbf{r} = \frac{\partial \mathbf{r}_1}{\partial \dot{q}_1} \mathrm{d}q_1 + \dots + \frac{\partial \mathbf{r}_s}{\partial \dot{q}_1} \mathrm{d}q_s \quad \Rightarrow \quad \dot{\mathbf{r}}_n \equiv \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} = \sum_{i=1}^{s} \frac{\partial \mathbf{r}_n}{\partial \dot{q}_n} \dot{q} + \frac{\partial \mathbf{r}_n}{\partial t}.$$

Therefore, the kinetic energy of the n^{th} particle is

Note: We use the "dummy" index in the summation $\sum_i A_i \equiv \sum_j A_j = A_1 + A_2 + \dots + A_k$ to write $\dot{\mathbf{r}}^2 \equiv \mathbf{r} \cdot \mathbf{r}$ as a scalar product of two identical sums.

$$\begin{aligned}
T &= \sum_{n=1}^{N} T_n = \sum_{n=1}^{N} \frac{1}{2} m_n \dot{\mathbf{r}}_n^2 \\
&= \sum_{n=1}^{N} \frac{1}{2} m_n \left(\sum_i^s \frac{\partial \mathbf{r}_n}{\partial q_i} \dot{q}_i + \frac{\partial \mathbf{r}_n}{\partial t} \right) \cdot \left(\sum_j^s \frac{\partial \mathbf{r}_n}{\partial q_j} \dot{q}_j + \frac{\partial \mathbf{r}_n}{\partial t} \right) \\
&= \sum_{ij} \underbrace{\sum_{n=1}^{N} \frac{1}{2} m_n \left(\frac{\partial \mathbf{r}_n}{\partial q_i} \cdot \frac{\partial \mathbf{r}_n}{\partial q_j} \right)}_{\equiv h_{ij}} \dot{q}_i \dot{q}_j + \sum_i \underbrace{\sum_{n=1}^{N} m_n \left(\frac{\partial \mathbf{r}_n}{\partial q_i} \cdot \frac{\partial \mathbf{r}_n}{\partial t} \right)}_{\equiv b_i} \dot{q}_i \\
&+ \underbrace{\sum_{n=1}^{N} \frac{1}{2} m_n \left(\frac{\partial \mathbf{r}_n}{\partial t} \right)^2}_{\equiv c} \qquad (1.11) \\
&\equiv \sum_{ij}^{s} h_{ij} \dot{q}_i \dot{q}_j + \sum_{i}^{s} b_i \dot{q}_i + c. \qquad (1.12)
\end{aligned}$$

So, the kinetic energy is always a quadratic form in the generalised velocities. The coefficients can be calculated directly by resolving constraints, and in general they depend also on generalised coordinates.

Note that the summation in Eq. (1.12) goes over the number of degrees of freedom, while in $T = \sum_n T_n$ it ran over the number of particles. As the latter summation is trivial, let's write down explicitly the coefficients in Eq. (1.12) in the case of one particle:

Aside: The h_{ij} terms are directly related to the Jacobian of the coordinate transformation from the original coordinates to the generalized coordinates, and are (up to factors) the metric components.

$$h_{ij} \equiv \frac{m}{2} \frac{\partial \mathbf{r}}{\partial q_i} \cdot \frac{\partial \mathbf{r}}{\partial q_j} \qquad b_i \equiv m \frac{\partial \mathbf{r}}{\partial q_i} \cdot \frac{\partial \mathbf{r}}{\partial t} \qquad c \equiv \frac{m}{2} \left(\frac{\partial \mathbf{r}}{\partial t} \right)^2 \qquad (1.13)$$

If all the constraints are time-independent (such constraints are called ***scleronomic***), then $\mathbf{r}$ does not explicitly depends on time, so that both b and c in Eq. (1.13) vanish. In this case $T = \sum_{ij}^{s} h_{ij}\dot{q}_i\dot{q}_j$ is purely quadratic.

This can be further simplified for orthogonal coordinates (e.g. Cartesian, polar, cylindrical, spherical) where

Note: we are following a common convention to keep the factor of $m/2$ outside of h_i. These h_i are often called the scale factors or Lame' coefficients (e.g. in Eigenphysics or mathematics in general).

$$\frac{\partial \mathbf{r}}{\partial q_i} \cdot \frac{\partial \mathbf{r}}{\partial q_j} = h_i^2 \delta_{ij} \quad \Rightarrow \quad T = \frac{1}{2} m \sum_i h_i^2 \dot{q}_n^2 \,. \tag{1.14}$$

In other words, in orthogonal coordinates, the kinetic energy becomes a sum of independent quadratic terms in each generalised velocity, greatly simplifying calculations. This is why such orthogonal coordinate systems are often so helpful in solving mechanics problems.

1.6.1 Example: Kinetic Energy in Plane Polars

As an example, let us consider plane polars, showing explicitly that these are an orthogonal coordinate system and then calculating the form of the kinetic energy. We have

$$\mathbf{r} = \begin{pmatrix} x \\ y \end{pmatrix} = r \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix} \quad \Rightarrow \quad \frac{\partial \mathbf{r}}{\partial r} = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix} \,, \quad \frac{\partial \mathbf{r}}{\partial \theta} = r \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}$$

We then have

$$\begin{aligned} &\frac{\partial \mathbf{r}}{\partial r} \cdot \frac{\partial \mathbf{r}}{\partial \theta} = 0 \,, \qquad \frac{\partial \mathbf{r}}{\partial r} \cdot \frac{\partial \mathbf{r}}{\partial r} = \cos^2\theta + \sin^2\theta = 1 \\ &\frac{\partial \mathbf{r}}{\partial \theta} \cdot \frac{\partial \mathbf{r}}{\partial \theta} = r^2(\sin^2\theta + \cos^2\theta) = r^2, \end{aligned}$$

showing this is an orthogonal coordinate system as expected. By comparing with Eq. 1.14, we can then read off

It is easy to derive but you can often save time by remembering this expression for T!

$$h_r = 1 \,, \qquad h_\theta = r, \quad \Rightarrow \quad T = \frac{1}{2} m \left(\dot{r}^2 + r^2 \dot{\theta}^2 \right) \,.$$

Of course, this is trivial to check by a direct substitution of polars into $\dot{x}^2 + \dot{y}^2$. The meaning of the two terms in T is also clear; the first term corresponds to the kinetic energy due to radial motion, while the second describes the kinetic energy from angular motion. The latter scales with the radius, as a given angular speed corresponds to a larger linear speed at larger radius (i.e. $v_\theta = r\dot{\theta}$).

1.6.2 Example: Kinetic Energy in Spherical Polars

We now do the same thing for spherical polars, which is another common orthogonal coordinate system. The corresponding results are

$$\mathbf{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = r \begin{pmatrix} \sin\theta\cos\phi \\ \sin\theta\sin\phi \\ \cos\theta \end{pmatrix} \quad \Rightarrow$$

$$\frac{\partial \mathbf{r}}{\partial r} = \begin{pmatrix} \sin\theta\cos\phi \\ \sin\theta\sin\phi \\ \cos\theta \end{pmatrix}, \quad \frac{\partial \mathbf{r}}{\partial \theta} = r \begin{pmatrix} \cos\theta\cos\phi \\ \cos\theta\sin\phi \\ -\sin\theta \end{pmatrix}, \quad \frac{\partial \mathbf{r}}{\partial \phi} = r \begin{pmatrix} -\sin\theta\sin\phi \\ \sin\theta\cos\phi \\ 0 \end{pmatrix}.$$

It is then straightforward to show that

$$\frac{\partial \mathbf{r}}{\partial r}\cdot\frac{\partial \mathbf{r}}{\partial \theta} = \frac{\partial \mathbf{r}}{\partial r}\cdot\frac{\partial \mathbf{r}}{\partial \phi} = \frac{\partial \mathbf{r}}{\partial \theta}\cdot\frac{\partial \mathbf{r}}{\partial \phi} = 0\,, \quad h_r = 1\,, \quad h_\theta = r\,, \quad h_\phi = r\sin\theta,$$

and hence that

Again quite a useful expression to remember

$$\Rightarrow \quad T = \tfrac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\dot{\phi}^2\right).$$

This now corresponds to the sum of kinetic energies from radial, polar and azimuthal motion respectively.

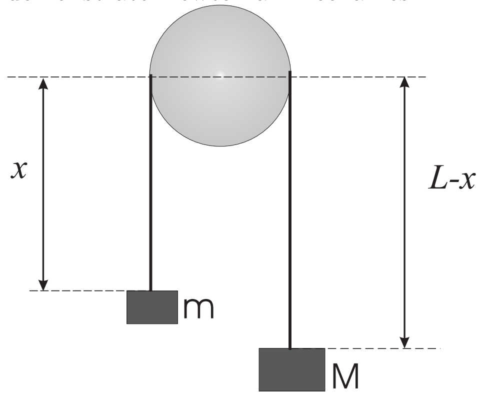
1.7 Lagrange's equations for simple systems with constraints: examples

In this section, we consider a few particular examples, illustrating how to choose generalised coordinates, $\{q_i\}$, in a convenient way. The idea is always to use as few parameters as possible, such that the number of coordinates is equal to the number of degrees of freedom in the system.

1.7.1 Example: Atwood's Machine

Two masses, m and M, are connected by a string hung over a smooth peg. There is only one independent coordinate, x, whose choice is determined up to a constant. We set x to be the distance between the centre of the peg and the mass m, while $L-x$ is then the distance between the centre of the peg and the mass M as shown in the diagram. The corresponding kinetic and potential energies and Lagrangian are:

The Atwood machine was invented by George Atwood in 1784 to experimentally verify and demonstrate Newtonian mechanics.



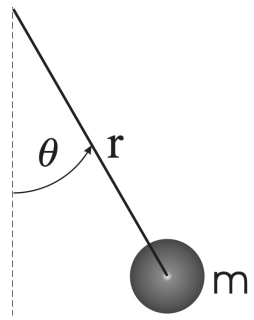
$$\begin{aligned} T &= \frac{1}{2}(m+M)\dot{x}^2\,, \\ V &= -Mgx - mg(L-x) = (m-M)gx + \text{const} \\ \mathcal{L} &= \frac{1}{2}(m+M)\dot{x}^2 + (M-m)gx + \text{const} \end{aligned}$$

From this we can use Lagrange's equation to show

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathcal{L}}{\partial \dot{x}} = (m+M)\ddot{x}\,, \quad \frac{\partial \mathcal{L}}{\partial x} = (M-m)g$$
$$\Rightarrow \quad (m+M)\ddot{x} = (M-m)g \quad \Rightarrow \quad \ddot{x} = \frac{M-m}{M+m}g\,.$$

This cleanly shows how acceleration depends on the mass difference, not total mass alone. Note that it does not matter whether we measured a fixed length L from the centre of the peg, as in the figure, or from the top (which would mean $L \to L + R/2$). Such a change will change V, and thus $\mathcal{L}$ by a constant. But what matters is Lagrange's equation, which remains the same when $\mathcal{L} \to \mathcal{L} + \text{const}$. Even considerably more involved changes in $\mathcal{L}$ will keep the equations of motion the same, as we shall learn soon.

1.7.2 Example: Simple Pendulum in a Plane



Let us now consider a simple pendulum moving in a 2D plane under the influence of gravity. We assume as usual that the radius is constant; this means that we only have one degree of freedom, i.e. the polar angle θ. Using this we can see that

$$T = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2\right) = \frac{1}{2}mr^2\dot{\theta}^2\,, \quad V = mgz = -mgr\cos\theta$$
$$\Rightarrow \quad \mathcal{L} = \frac{1}{2}mr^2\dot{\theta}^2 + mgr\cos\theta$$

which leads to

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = mr^2\dot{\theta}\,, \qquad \frac{\partial \mathcal{L}}{\partial \theta} = -mgr\sin\theta$$
$$\Rightarrow \quad mr^2\ddot{\theta} = -mgr\sin\theta \quad \Rightarrow \quad \ddot{\theta} = -\frac{g}{r}\sin\theta$$

For small angles, $\sin\theta \approx \theta$, this leads to the expected equation for a simple harmonic oscillator (SHO):

$$\ddot{\theta} + \omega_0^2\theta = 0, \quad \omega_0 \equiv \sqrt{g/r}.$$

2 A justification for Lagrange's equations and more examples of their use

2.1 D'Alembert's principle and Lagrange's Equations

This subsection on D'Alembert is *not* critical for your understanding of the course at this point, and we will derive the Lagrangian in a more powerful way later. **So why have I included it?** A number of you, as you start realising the power that the Lagrange approach has - will have started to become uncomfortable as to where it arrived from.

Much later in the course, we will go through a very general approach to the derivation, where a starting point will not be Newtonian equations as we use here.

Today we derive Lagrange's equations for the simplest case of one particle which is so constrained that it only has one generalised coordinate—such as a bead sliding on a wire in three dimensions (see Figure above), with the generalised coordinate q corresponding to the distance along the wire. In this system, we can express all x, y, z as functions of q, or equivalently $\mathbf{r} = \mathbf{r}(q,t)$. The general case with many degrees of freedom is more tedious to derive, and the result will be quoted after the simple case.

Note that if the wire is moving, then we have an example of a *time dependent* constraint, where $\mathbf{r}(t)$ depends on t both through its coordinate along the wire $q(t)$ being time dependent, *but also* through the position of the wire being time-dependent (*i.e.*even if q is constant, $\mathbf{r}$ changes). We want to calculate the work, $\mathrm{d}W = \mathbf{F}\cdot\delta\mathbf{r}$, done by an external ***conservative*** force $\mathbf{F}$ for an infinitesimal displacement, $\delta\mathbf{r}$, of the bead ***along*** the wire. Note that $\delta\mathbf{r} \neq \mathrm{d}\mathbf{r}$, as the components $\mathrm{d}x$, $\mathrm{d}y$, $\mathrm{d}z$ are not mutually independent. One can obviously use the chain rule and the fact that $\mathbf{r} = \mathbf{r}(q(t), t)$ to write

$$\delta\mathbf{r} = \frac{\partial\mathbf{r}}{\partial q}\,\mathrm{d}q\,.$$

Then we use Newton's 2$^{\text{nd}}$ law, $\mathbf{F} = \dot{p}$, to write the expression for the work $\mathrm{d}W = \mathbf{F}\cdot\delta\mathbf{r}$ as

$$\left(\mathbf{F}\cdot\frac{\partial\mathbf{r}}{\partial q}\right)\mathrm{d}q = \left(\dot{p}\cdot\frac{\partial\mathbf{r}}{\partial q}\right)\mathrm{d}q \tag{2.1}$$

Important: the convention here and elsewhere is that *dot* indicates $\frac{\mathrm{d}}{\mathrm{d}t}$ and *not* $\partial/\partial t$. The full derivative, *e.g.* of $\mathbf{r}(q,t)$, is

$$\dot{\mathbf{r}} \equiv \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} = \left(\frac{\partial\mathbf{r}}{\partial t}\right)_q + \left(\frac{\partial\mathbf{r}}{\partial q}\right)_t\frac{\mathrm{d}q}{\mathrm{d}t} \equiv \frac{\partial\mathbf{r}}{\partial t} + \frac{\partial\mathbf{r}}{\partial q}\dot{q}\,. \tag{2.2}$$

Note that in the present context, the first term here comes from the motion of the bead relative to the wire while the second term comes from the motion of the wire.

Now I intend to change the variables in Eq. (2.1) from the original coordinates $\mathbf{r}$ to the generalised coordinate q. Thus I will demonstrate that Lagrange's equations – which will arise at the end of this exercise – naturally follows from Newton's laws when one changes to generalised coordinates.

After cancelling dq, let us transform both the l.h.s. and r.h.s. of Eq. (2.1). Using $F = -\boldsymbol{\nabla} V$ and applying the chain rule, we find for the l.h.s.:

$$\mathbf{F}\cdot\frac{\partial \mathbf{r}}{\partial q} = -\boldsymbol{\nabla} V\cdot\frac{\partial \mathbf{r}}{\partial q} = -\left(\frac{\partial V}{\partial x}\frac{\partial x}{\partial q} + \frac{\partial V}{\partial y}\frac{\partial y}{\partial q} + \frac{\partial V}{\partial z}\frac{\partial z}{\partial q}\right) = -\frac{\mathrm{d}V}{\mathrm{d}q} \equiv \mathcal{F}_q\,. \quad (2.3)$$

$\boxed{\mathcal{F}_q \equiv -\frac{\mathrm{d}V}{\mathrm{d}q} \text{ is often called the generalised force.}}$

The r.h.s. of Eq. (2.1) is a bit more tricky. First note that

$$\dot{p}\cdot\frac{\partial \mathbf{r}}{\partial q} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\mathbf{p}\cdot\frac{\partial \mathbf{r}}{\partial q}\right) - \mathbf{p}\cdot\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathbf{r}}{\partial q}\,. \quad (2.4)$$

Assuming that $\mathbf{r}$ is a smooth enough function of q and t, we can change the order of differentiation in the second term, thus obtaining

$$-\mathbf{p}\cdot\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathbf{r}}{\partial q} = -m\dot{\mathbf{r}}\cdot\frac{\partial \dot{\mathbf{r}}}{\partial q} = -\frac{1}{2}m\frac{\partial\left(\dot{\mathbf{r}}^2\right)}{\partial q} = -\frac{\partial T}{\partial q}\,. \quad (2.5)$$

Important: we will be using this rule a lot in later parts of the course!

To deal with the first term in Eq. (2.4), let us note the ***"dots cancellation"*** rule:

$$\frac{\partial \mathbf{r}}{\partial q} = \frac{\partial \dot{\mathbf{r}}}{\partial \dot{q}}\,. \quad (2.6)$$

(This immediately follows from the differentiation of the chain rule, Eq. (2.2), with respect to $\dot{q}$, since the first term there does not depend on $\dot{q}$.) We substitute $\dot{\mathbf{p}} = m\dot{\mathbf{r}}$ into the first term in Eq. (2.4) and apply the identity 2.6 (introducing dots instead of cancelling them):

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\mathbf{p}\cdot\frac{\partial \mathbf{r}}{\partial q}\right) = m\frac{\mathrm{d}}{\mathrm{d}t}\left(\dot{\mathbf{r}}\cdot\frac{\partial \dot{\mathbf{r}}}{\partial \dot{q}}\right) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{2}m\frac{\partial\left(\dot{\mathbf{r}}^2\right)}{\partial \dot{q}}\right) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{q}}\right). \quad (2.7)$$

Combining Eqs. (2.5) and (2.7), we find that the r.h.s. of Eq. (2.1) equals

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{q}}\right) - \frac{\partial T}{\partial q}\,.$$

The final step of the derivation is to equate this to the l.h.s. of Eq. (2.1) given by Eq. (2.3), which gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{q}}\right) - \frac{\partial (T-V)}{\partial q} = 0 \qquad \Rightarrow \qquad \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right) - \frac{\partial \mathcal{L}}{\partial q} = 0\,, \quad (2.8)$$

where we used that V does not depends on $\dot{q}$, so that $\frac{\partial T}{\partial \dot{q}} = \frac{\partial \mathcal{L}}{\partial \dot{q}}$.

For problems with many degrees of freedom, the generalised coordinates q_i and generalised velocities $\dot{q}_i$ are, by definition, mutually independent. So the above derivation would have remained practically the same: we just choose the *virtual displacement* $\delta\mathbf{r}$ caused by an infinitesimal change, $\mathrm{d}q_i$, of one of the generalised coordinates only.

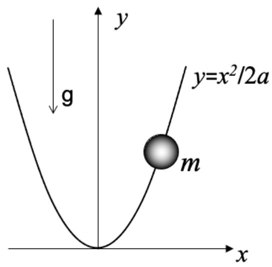
The above derivation was based on d'Alembert's ***virtual work*** principle: $(\mathbf{F} - \dot{p}) \cdot \delta\mathbf{r} = 0$.

Jean Le Rond d'Alembert (French, 1717-1783) was a French mathematician with a quarrelsome personality. His work centred around trying to make Mechanics a rigorous branch of mathematics rather than physical experiment. His work on vibration led him to write down the wave equation, or "d'Alembert equation", which is fundamental in modern physics.

2.2 Including an external force

To illustrate how to include an external force, we will again consider a simple example system with only one degree of freedom (here corresponding to a bead on a parabolic wire) so that we can set it up similarly to the examples in the previous week. Once we have written down our equations, we will add an external force; this will correspond to making the wire rotate.

2.2.1 Example: Bead on a parabolic wire



As shown in the diagram, we consider a bead of mass m moving along a parabolic wire under the influence of gravity. We choose x as the generalised coordinate. Due to the parabolic constraint, at any moment, we have $y = x^2/2a$ (a has the dimensionality of length). This means that $\dot{x} = \sqrt{a/2y}\,\dot{y}$ and the kinetic energy $T = \frac{1}{2}m(\dot{x}^2+\dot{y}^2) = \frac{1}{2}m(1+a/2y)\dot{y}^2$. The potential energy $V = mgy$. Thus, the Lagrangian and its derivatives are given by

$$\mathcal{L} = \frac{1}{2}m\left(1+\frac{a}{2y}\right)\dot{y}^2 - mgy$$

$$\frac{1}{m}\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial\mathcal{L}}{\partial\dot{y}} = \frac{\mathrm{d}}{\mathrm{d}t}\dot{y}\left(1+\frac{a}{2y}\right) = \ddot{y}\left(1+\frac{a}{2y}\right) - \frac{a\dot{y}^2}{2y^2}$$

$$\frac{1}{m}\frac{\partial\mathcal{L}}{\partial y} = -\frac{a\dot{y}^2}{4y^2} - g.$$

From this we arrive at Lagrange's equation, $2(2y^2+ay)\,\ddot{y} - a\dot{y}^2 + 4gy^2 = 0$, which cannot be analytically solved (except for small y) but is easy to analyze numerically.

2.2.2 Example: Bead on a rotating parabolic wire

We now let the wire be rotated (by an external force) at angular velocity ω. What should ω be in order for the bead to remain at a given height: $y = H$?

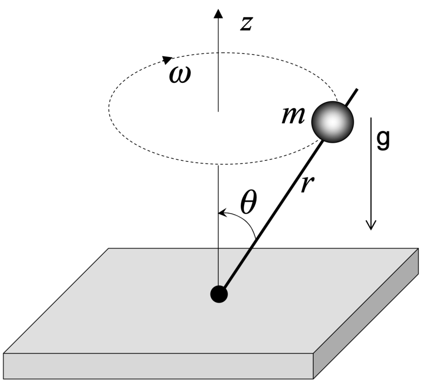
Now we should add to $\mathcal{L}$ the kinetic energy of bead's rotation, $\frac{1}{2}mr^2\dot{\theta}^2$. This, however, does not add a degree of freedom since the angular velocity is fixed: $\dot\theta = \omega$, while $r \equiv x = \sqrt{2ay}$. Thus

$$\mathcal{L} = \frac{1}{2}m\Big(1+\frac{a}{2y}\Big)\dot y^2 + may\omega^2 - mgy = \frac{1}{2}m\Big(1+\frac{x^2}{a^2}\Big)\dot x^2 - \underbrace{\big(mgy - may\omega^2\big)}_{\equiv V^{\rm eff}}$$

Note: the concept of an effective potential is one we shall be using a lot in the future.

Here we have used that since the angular contribution to T does not depend on any generalised velocity, it is convenient to include it (with the opposite sign, naturally) together with V thus arriving at the effective potential $V^{\rm eff}$. From this we can see that, in Lagrange's equation, all we now need to do is to substitute $g \to g - a\omega^2$. If we want the bean to remain at a given height, it must be in equilibrium with $\dot y = \ddot y = 0$; from this we can therefore deduce that we must have $\omega = \sqrt{g/a}$. Therefore, the Lagrangian approach has given us an easy way to solve this question.

2.2.3 Example: Bead on a rotating spoke



We now consider another example coresponding to a bead of mass m, which slides without friction along a spoke that rotates at rate $\omega \equiv \dot\phi$ around the vertical axis. The angle θ is fixed, which means that a convenient generalised coordinate is the distance r along the spoke. Importantly, the number of degrees of freedom is again only one! Although ϕ is changing, it is nothing more than a time-dependent constraint.

Using spherical polars with $\theta = $ const and $\dot\phi = \omega$, one has

$$\begin{aligned} T &= \frac{1}{2}m(\dot r^2 + r^2\omega^2\sin^2\theta)\,, \qquad V = mgr\cos\theta \\ \mathcal{L} &= \frac{1}{2}m(\dot r^2 + r^2\omega^2\sin^2\theta) - mgr\cos\theta \\ \Rightarrow \quad & \frac{\partial\mathcal{L}}{\partial\dot r} = m\dot r\,, \quad \frac{\partial\mathcal{L}}{\partial r} = mr\omega^2\sin^2\theta - mg\cos\theta \end{aligned}$$

Then Lagrange's equation becomes

$$\Rightarrow \quad \ddot r = \Omega^2 r - g_0\,, \qquad \text{with} \qquad \Omega \equiv \omega\sin\theta\,,\ g_0 \equiv g\cos\theta$$

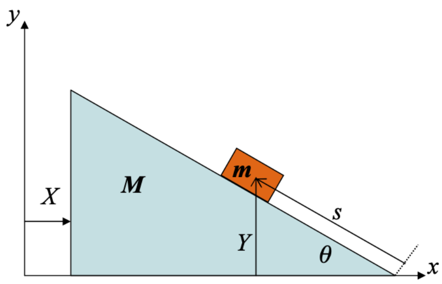
It's almost like SHO under a const force, but the sign is different. The solution for $g_0 = 0$ (rotation in horizontal plane) is $r = r_0\sinh(\Omega t + \alpha)$ (r_0 and α are const of integration). A full solution is

$$r = r_0\sinh(\Omega t + \alpha) + \frac{g_0}{\Omega^2}\,.$$

2.3 Systems with 2 degrees of freedom

We shall now go through a set of examples of writing and (sometimes) solving Lagrange's equations.

2.3.1 Example: Block on a moving inclined plane



A block of mass m slides without friction on the inclined plane of mass M which is free to move on the horizontal plane. Clearly, there are two degrees of freedom in this problem. For our generalised coordinates, a possible choice is X – the distance from the inclined plane to the origin, and s – the distance from the block to the end of the inclined plane. We will only consider $s > 0$, meaning that the block is still on the inclined plane.

The constraint that the block sits on the inclined plane is taken into account automatically if one notices that the altitude of the block is always $Y = s\sin\theta$. The kinetic energy of the plane is then $T = \frac{1}{2}M\dot{X}^2$ while that of the block is $\frac{1}{2}m\left[(\dot{X} - \dot{s}\cos\theta)^2 + (\dot{s}\sin\theta)^2\right]$. Therefore,

$$\mathcal{L} = \frac{1}{2}M\dot{X}^2 + \frac{1}{2}m\left[(\dot{X} - \dot{s}\cos\theta)^2 + (\dot{s}\sin\theta)^2\right] - mgs\sin\theta\,.$$

Lagrange's equations are then:

Note that there is nothing particularly special in this example. It is used more to show that it is not only balls on springs/strings that can be solved via Lagrangian techniques.

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial\mathcal{L}}{\partial\dot{X}}\right) - \frac{\partial\mathcal{L}}{\partial X} = 0 \qquad &\Rightarrow \quad M\ddot{X} + m\left(\ddot{X} - \ddot{s}\cos\theta\right) = 0\\
\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial\mathcal{L}}{\partial\dot{s}}\right) - \frac{\partial\mathcal{L}}{\partial s} = 0 \qquad &\Rightarrow \quad -\cos\theta\left(\ddot{X} - \ddot{s}\cos\theta\right) + \ddot{s}\sin^2\theta + g\sin\theta = 0
\end{aligned}$$

From the 1$^{\text{st}}$ equation (which is obviously the momentum conservation law for the total momentum of the inclined plane and the block), one finds $\ddot{X} = m\ddot{s}\cos\theta/(m+M)$. Then, on substituting this into the 2$^{\text{nd}}$ equation, one has

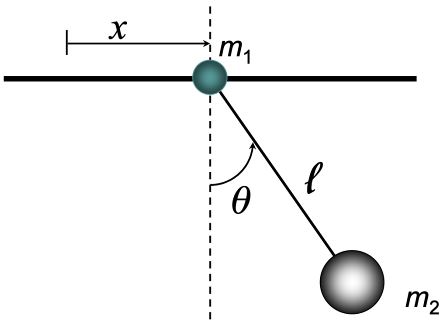
$$\begin{aligned}
&\ddot{s} - \frac{m\cos^2\theta}{M+m}\ddot{s} = -g\sin\theta\\
&\quad\Rightarrow \quad \ddot{s} = -g\sin\theta\frac{M+m}{M+m\sin^2\theta}, \ddot{X} = -\frac{1}{2}\frac{mg\sin 2\theta}{M+m\sin^2\theta}\,.
\end{aligned}$$

Note that the accelerations $\ddot{s}$ and $\ddot{X}$ are negative due to our choice of the direction of the generalised coordinates.

2.3.2 Example: Sliding pendulum

Consider a bob of mass m_1 moving without friction along a rod. Another bob of mass m_2 is attached to the first one by a weightless wire. We choose x and θ as generalised coordinates. The task is to write down Lagrange's equations without solving them. The motion of the

This is an extra example, that we haven't explicitly gone through, please work through it and let me know if you have queries.



first bob is simple and we can see that $T_1 = \frac{1}{2}m\dot{x}^2$. For the second bob,

$$\begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} x \\ 0 \end{pmatrix} + \ell \begin{pmatrix} \sin\theta \\ -\cos\theta \end{pmatrix} \Rightarrow \begin{pmatrix} \dot{x}_2 \\ \dot{y}_2 \end{pmatrix} = \begin{pmatrix} \dot{x} \\ 0 \end{pmatrix} + \ell\dot{\theta} \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix}$$

Therefore,

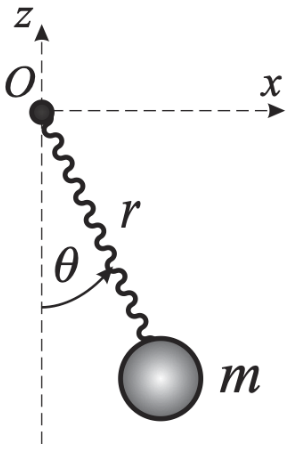
$$T_2 = \frac{1}{2}m_2\left(\dot{x}_2^2 + \dot{y}_2^2\right) = \frac{1}{2}m_2\left(\dot{x}^2 + \ell^2\dot{\theta}^2 + 2\ell\dot{x}\dot{\theta}\cos\theta\right) .$$

The potential energy $V_1 = 0$, $V_2 = m_2 g y_2 = -m_2 g\ell cos\theta$, so that the Lagrangian

$$\mathcal{L} = \frac{1}{2}m_1\dot{x}^2 + \frac{1}{2}m_2\left(\dot{x}^2 + \ell^2\dot{\theta}^2 + 2\ell\dot{x}\dot{\theta}\cos\theta\right) + m_2 g\ell cos\theta .$$

Note that this Lagrangian does not depend on x. Hence, Lagrange's equation for x becomes very simple: $\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial\mathcal{L}}{\partial\dot{\theta}} = 0$. This means that the quantity $\frac{\partial\mathcal{L}}{\partial\dot{\theta}} = \text{const}$, *i.e.*is conserved. There is a special name for this: the generalised momentum (next lecture).

2.3.3 Example: Spring pendulum



A mass m is connected to a fixed point by a spring of spring constant κ and natural length ℓ. At moment $t = 0$ the length of the spring $r = \ell$ (it is not extended), and it makes an angle θ_0 with the vertical axis. Then it is released and started to oscillate in the presence of gravity. With such initial conditions it will move in a plane - we therefore choose polar coordinates as our generalised coordinates. Clearly, there are two degrees of freedom: the angle θ and the spring length r. Here, there are two contributions to the potential energy (elastic and gravitational energy), while the kinetic energy is the standard one (previously derived) in polar coordinates. Thus one finds

$$T = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2\right) , \qquad V = -mgr\cos\theta + \frac{1}{2}\kappa\left(r - \ell\right)^2 ,$$

Lagrange's equations:

$$\text{for } r: \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial\mathcal{L}}{\partial\dot{r}}\right) = m\ddot{r} , \qquad \frac{\partial\mathcal{L}}{\partial r} = mg\cos\theta - \kappa(r - \ell),$$

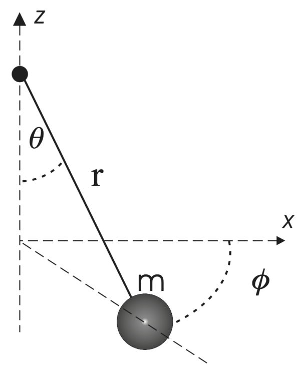
$$\text{for } \theta: \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial\mathcal{L}}{\partial\dot{\theta}}\right) = mr^2\ddot{\theta} + mr\dot{r}\dot{\theta} , \qquad \frac{\partial\mathcal{L}}{\partial\theta} = -mgr\sin\theta,$$

and hence:

$$\ddot{r} = g\cos\theta - \kappa(r - \ell), \qquad \ddot{\theta} = -\frac{\dot{r}\dot{\theta} + g\sin\theta}{r}.$$

Solving these equations analytically is not possible (but this is true even for a normal physical pendulum). We return to this example again later to see how the conservation laws may help.

2.3.4 Example: Spherical pendulum



A mass m is connected to a fixed point by a string of a fixed length ℓ. Otherwise, it is free to move in any direction. Clearly, there are two degrees of freedom: the angles θ and ϕ. T, V and $\mathcal{L}$ in polar coordinates (with angle θ chosen so that $z = r\cos\theta$ and $r = \ell$ fixed) are

$$T = \frac{1}{2}m\ell^2\left(\dot\theta^2 + \sin^2\theta\dot\phi^2\right), \quad V = -mg\ell\cos\theta\,,$$
$$\mathcal{L} = T - V = \frac{1}{2}m\ell^2\left(\dot\theta^2 + \sin^2\theta\dot\phi^2\right) + mg\ell\cos\theta$$

Note that $\mathcal{L}$ does not depend on ϕ (although it still depends on $\dot\phi$). Such a variable is called the cyclic variable, and will be considered in general in the next lecture. The Lagrange equation for such a variable is very simple:

$$p_\phi = \frac{\partial\mathcal{L}}{\partial\dot\phi} = m\ell^2\sin^2\theta\dot\phi \equiv h = \text{const.}$$

The Lagrange's equation for θ is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial\mathcal{L}}{\partial\dot\theta}\right) = m\ell^2\ddot\theta\,, \quad \frac{\partial\mathcal{L}}{\partial\theta} = m\ell^2\sin\theta\cos\theta\dot\phi^2 - mg\ell\sin\theta$$
$$\Rightarrow \ddot\theta = \sin\theta\cos\theta\dot\phi^2 - \frac{g}{\ell}\sin\theta\,,$$
$$\Rightarrow \quad \ddot\theta = \frac{h^2\cos\theta}{m\ell^2\sin^3\theta} - \frac{g}{\ell}\sin\theta\,.$$

Again, there is no exact solution to these equations. Moreover, unless $h = 0$, *i.e.*there is no circular motion, the spherical pendulum never behaves as the standard ”mathematical” pendulum: even for small θ its motion does not reduce to simple oscillations. The reason is the conservation of angular momentum, as the smaller θ, the higher angular velocity $\dot\phi$.

3 Symmetries and Conservation Laws.

Last week, in the lecture support session, we covered the example of a mass on a spring, that was free to rotate in the x-y plane. It was introduced as a simpler example of the Spring pendulum 2.3.3 - where we had arrived at differential equations that we could not solve analytically. All it required was us to apply the Lagrangian formulation, but we pointed out a few 'tricks' along the way.

We will revisit that example in a moment and emphasise what will turn out to be the key parts of the solution that will prove to be physically important, rather than 'luck' or just a 'piece of neat mathematics' for this set up.

I recommend that you set yourself a reminder to revisit this section over the next few weeks as its consequences become clearer.

First, we pointed out as we worked through the example, that ϕ was cyclic, let's define this mathematically. These are *really important* concepts for the module, and for theoretical physics in general.

3.1 Cyclic variables and generalised momenta

We start by defining the generalised momentum p_i which is *canonically conjugate* to the generalised coordinate q:

$$\boxed{p_i \equiv \frac{\partial \mathcal{L}}{\partial \dot{q}_i}}. \tag{3.1}$$

We can rewrite Lagrange's equations in terms of generalised momenta

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) - \frac{\partial \mathcal{L}}{\partial q_i} = 0, \qquad \Rightarrow \quad \dot{p}_i \equiv \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) = \frac{\partial \mathcal{L}}{\partial q_i}.$$

Examples: if $\mathcal{L}$ in Cartesian coordinates does not depend on x, it means that it is translationally invariant – the usual momentum conservation follows from this invariance. If $\mathcal{L}$ is θ independent in polar coordinates, it means rotational invariance which results in the conservation of a proper component of the angular momentum.

Now we can see that if $\mathcal{L}$ does not explicitly depend on the coordinate q_i, then the r.h.s. in the above equation vanishes, which means that p_i is conserved! Such a coordinate is called ***cyclic*** (or ***ignorable***).

Importantly, a conservation law is always a consequence of some – explicit or implicit – symmetry. In general, the conservation of a p_i conjugate to a cyclic generalised coordinate q_i follows from $\mathcal{L}$ being invariant with respect to shifts in this coordinate.

3.2 The Hamiltonian and energy conservation

There was another important factor that we didn't explore in solving the example of the spring pendulum in Section 2.3.3 - and that was that the Lagrangian itself had no time dependence. Let us now explore the consequences of this in more detail.

This section may appear quite abstract, but is fundamental for our understanding for this course, and the first mention of 'Hamiltonian'.

We assume the Lagrangian is time independent, $\frac{\partial \mathcal{L}}{\partial t} = 0$, *i.e.*$\mathcal{L} = \mathcal{L}(q_i, \dot{q}_i)$. Such a translational invariance in time clearly constitutes a symmetry which should result in a conservation law. As time and space are essentially different in nonrelativistic systems, such a conservation manifests in a way rather different to the conservation of generalised momentum. To see this, let us consider the full time derivative of $\mathcal{L}$. Then we can show that, in contrast to $\dfrac{\partial \mathcal{L}}{\partial t}$, the full time derivative does not necessarily vanish:

$$\begin{aligned}
\frac{\mathrm{d}\mathcal{L}}{\mathrm{d}t} &= \sum_i \left[\frac{\partial \mathcal{L}}{\partial q_i}\frac{\mathrm{d}q_i}{\mathrm{d}t} + \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\frac{\mathrm{d}\dot{q}_i}{\mathrm{d}t}\right] + \underbrace{\frac{\partial \mathcal{L}}{\partial t}}_{=0} \qquad \text{chain rule} \\
&= \sum_i \left[\dot{q}_i \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) + \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\frac{\mathrm{d}\dot{q}_i}{\mathrm{d}t}\right] \qquad \text{Lagrange's eqs for the 1}^{\text{st}} \text{ term} \\
&= \sum_i \left[\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\dot{q}_i\right)\right] = \frac{\mathrm{d}}{\mathrm{d}t}\sum_i \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\dot{q}_i\right) \qquad \text{product rule} \\
&\Rightarrow \frac{\mathrm{d}}{\mathrm{d}t}\left[\sum_i \dot{q}_i \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \mathcal{L}\right] = 0
\end{aligned}$$

We have therefore arrived at a new conserved quantity which is called the ***Hamiltonian*** or Hamilton's function:

Sir William Rowan Hamilton (1805 – 1865) was an Irish mathematician, physicist, and astronomer who made important contributions to classical mechanics, algebra and optics.

$$\boxed{\mathcal{H} = \sum_i \dot{q}_i \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \mathcal{L} \equiv \sum_i \dot{q}_i p_i - \mathcal{L}} \tag{3.2}$$

Strictly speaking, the Hamiltonian is a function of q_i and p_i, not q_i and $\dot{q}_i$ as the Lagrangian. The transformation from $\mathcal{L}$ to $\mathcal{H}$, Eq. (3.2), is our first example of ***Legendre transformation*** used to change variables. In more detail, it should be written as

Adrien-Marie Legendre (1752 – 1833) was a French mathematician who made numerous contributions to mathematics, including most famously the Legendre polynomials and Legendre transformations.

$$\mathcal{H}(q_i, p_i) = \sum_i \dot{q}_i p_i - \mathcal{L}\left(q_i, \dot{q}_i(q_i, p_i)\right)$$

where one uses the definition of the generalised momentum, Eq. (3.1) to express generalised velocities $\dot{q}_i$ in $\mathcal{L} = \mathcal{L}(q_i, \dot{q}_i)$ as functions of p_i and q_i.

However, throughout this course, we will also call the conserved quantity $\mathcal{H}(q_i, p_i; t)$ the Hamiltonian, even when we do not perform an explicit change of coordinates from $\dot{q}_i$ to p_i.

*All conserved quantities are also called **first integrals***

Identifying the first integrals is often the key step in solving mechanical problems, as Lagrange's equations are considerably simplified when these exist, whether due to the presence of cyclic variables or because $\mathcal{L}$ is t-independent.

For example, we have done so in the previous lecture / example classes when considering the spherical pendulum.

A recipe for approaching problems: by noticing that all the conserved generalised momenta are constants, one can exclude the appropriate generalised velocities from the remaining Lagrange's equations (which - by definition of the cyclic variables - do not depend on the appropriate coordinate), thus reducing the number of variables. Also, writing $\mathcal{H} = const$ gives the 1st rather then the 2nd order differential equation, which is especially convenient for a system with one degree of freedom.

Remember: scleronomic constraints are ones that are time-independent.

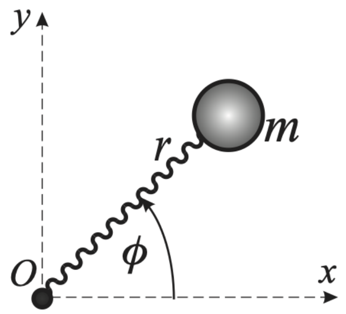
We have seen that for stationary generalised coordinates (*scleronomic constraints*) the kinetic energy is a homogenous quadratic function of $\dot{q}_i$, Eq. (1.14). In this (and only in this) case one finds

$$\begin{aligned} \sum_i \dot{q}_i \frac{\partial \mathcal{L}}{\partial \dot{q}_i} &= \sum_i \dot{q}_i \frac{\partial}{\partial \dot{q}_i} \left(\frac{1}{2} m \sum_n h_n^2 \dot{q}_n^2 \right) = \frac{1}{2} m \sum_{i,n} \dot{q}_i h_n^2 \frac{\partial \dot{q}_n^2}{\partial \dot{q}_i} \\ &= m \sum_{i,n} h_n^2 \dot{q}_i \dot{q}_n \underbrace{\frac{\partial \dot{q}_n}{\partial \dot{q}_i}}_{=\delta_{ni}} = 2T \\ \Rightarrow \quad & \mathcal{H} = 2T - \mathcal{L} = T + V \equiv E \,, \end{aligned}$$

i.e.$\mathcal{H}$ turns out to be equal to the full energy of the system.

In the case when constraints are time-dependent but $\frac{\partial \mathcal{L}}{\partial t} = 0$, $\mathcal{H}$ is conserved – as the above proof remains valid – but it is not equal to the total energy. The energy is ***not*** conserved in this case as the system is not closed: some work is being done on the system.

3.2.1 Example: Mass on a spring in the plane



A mass m is attached via a spring of spring constant κ and natural length ℓ to a fixed point O, while being free to move in the horizontal plane. The initial conditions are such that the spring starts at its natural length and the mass moves with velocity v perpendicular to the spring, *i.e.*in Cartesian coordinates $(x_0, y_0) = (\ell, 0)$ and $(\dot{x}_0, \dot{y}_0) = (0, v)$.

A set of solution steps:

To check your understanding you might find it helpful to work through the solution steps, and then compare with the worked answer that follows.

a) How many degrees of freedom (d.o.f)?

b) Can you choose convenient generalised coordinates and write down $\mathcal{L}$?

c) Identify the cyclic variable and find the conserved conjugate momentum. Then write down Lagrange's equation(s) for the non-cyclic variable(s) or –

d) Alternatively, write down $\mathcal{H}$ and use it as the first integral.

Worked Solution: Clearly, there are 2 degrees of freedom which are easy to describe in polar coordinates. The kinetic energy is as usual for polar coordinates, while the potential energy is that of an elastic spring with spring constant κ and extension $r-\ell$. Consequently, the Lagrangian is

$$\mathcal{L} = T - V = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\phi}^2\right) - \frac{1}{2}\kappa\,(r-\ell)^2 \ .$$

We can see that $\mathcal{L}$ does not depend on ϕ, which is thus a cyclic variable. The corresponding conjugate momentum is conserved:

$$p_\phi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = mr^2\dot{\phi} \equiv h$$

where h is a constant. The physical meaning of h is a (projection of) angular momentum.

For the non-cyclic variable r, one has

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{r}}\right) - \frac{\partial \mathcal{L}}{\partial r} = m\ddot{r} - mr\dot{\phi}^2 + \kappa(r-\ell) = 0 \quad \Rightarrow \quad m\ddot{r} - \frac{h^2}{mr^3} + \kappa(r-\ell) = 0\,.$$

As the Lagrangian does not explicitly depend on time *and* the kinetic energy is quadratic in the generalised velocities, the Hamiltonian is both conserved and equal to $T+V$, *i.e.*to the full energy.

Note that it won't be sufficient simply to write down $\mathcal{H} = T + V$, without qualifying this statement. Instead of the above qualification ($\mathcal{L}$ is quadratic...) one can deduce $\mathcal{H}$ from the definition, using

$$\mathcal{H} = \dot{r}\frac{\partial \mathcal{L}}{\partial \dot{r}} + \dot{\phi}\frac{\partial \mathcal{L}}{\partial \dot{\phi}} - \mathcal{L} = \frac{1}{2}m\dot{r}^2 + \frac{h^2}{2mr^2} + \frac{\kappa}{2}\,(r-\ell)^2 \ .$$

The initial conditions in polar coordinates correspond to $r_0 = \ell$, $\dot{r}_0 = 0$, $\phi_0 = 0$, and $\dot{\phi}_0 = v/\ell$. The conserved quantities can also be calculated at the initial moment, so that $h = m\ell v$ and $\mathcal{H} = T_0 = \frac{1}{2}mv^2$.

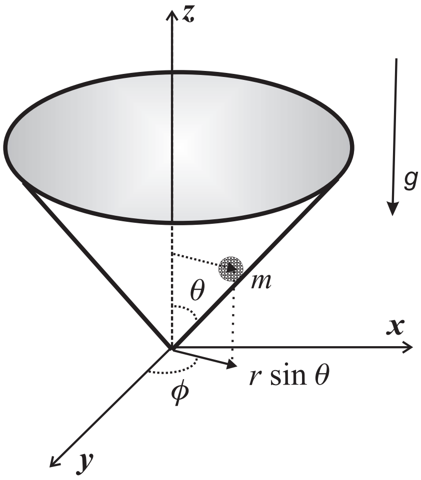
We note that solving the equation $\mathcal{H} = \frac{1}{2}mv^2$ is considerably easy than solving the 2nd order Lagrange's equation for r! However, that 2nd order equation can also be converted to the 1st order one for the case $\ddot{r} = f(r)$ (which is true for this particular problem). Indeed, noting that $2\dot{r}\ddot{r} = \mathrm{d}\dot{r}^2/\,\mathrm{d}t$ and multiplying both sides of the equation by $\dot{r}$, we obtain $\mathrm{d}\dot{r}^2 = 2f(r)\,\mathrm{d}r$. Integrating this with the proper initial conditions give the same 1st order equation as follows from $\mathcal{H} = E$.

As a contrast, let's return to the example of a bead on a rotating spike discussed previously. Then we can also construct the Hamiltonian starting from the Lagrangian as follows

$$\mathcal{L} = \frac{1}{2}m(\dot{r}^2 + r^2\omega^2 \sin^2\theta - gr\cos\theta) \quad \Rightarrow \quad \frac{\partial \mathcal{L}}{\partial \dot{r}} = m\dot{r}$$
$$\Rightarrow \quad \mathcal{H} = \underbrace{\dot{r}\frac{\partial \mathcal{L}}{\partial \dot{r}}}_{\substack{\shortparallel \\ m\dot{r}^2}} - \mathcal{L} = \frac{1}{2}m(\dot{r}^2 - r^2\omega^2\sin^2\theta + gr\cos\theta) \neq E\,.$$

Here $\mathcal{H}$ is still conserved, but E is not: rotating the spike requires some work done on the system.

3.2.2 Example: Mass on the surface of a cone



A block of mass m is constrained to move without friction on the surface of a vertical cone of half-angle α under the influence of gravity. A natural choice for the generalised coordinates are spherical polars ***subject to the constraint*** $\theta = \text{const}$, meaning that $\dot{\theta} = 0$ in T.

Therefore, we obtain a Lagrangian for the system as follows:

$$\mathcal{L} = \frac{m}{2}\left(\dot{r}^2 + r^2\sin^2\theta\dot{\phi}^2\right) - mgr\cos\theta\,.$$

Note that θ here is simply a constant, not a variable! On the other hand, ϕ is a cyclic variable, meaning that the appropriate conjugate momentum (which is just the angular momentum) is conserved:

$$p_\phi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = mr^2\sin^2\theta\dot{\phi} = \text{const}\,.$$

As $\mathcal{L}$ is time independent, $\mathcal{H}$ is conserved. Since there are no time-dependent constraints (alternatively - T is the quadratic function of generalised velocities), we also know that $\mathcal{H} = E$. Therefore, we obtain

$$\frac{m}{2}\left(\dot{r}^2 + r^2\sin^2\theta\dot{\phi}^2\right) + mgr\cos\theta = E\,.$$

Substituting here $\dot{\phi} = p_\phi/mr^2\sin^2\theta$, we obtain the 1st order equation for one degree of freedom only, r:

$$\dot{r}^2 = \frac{2E}{m} - \frac{p_\phi^2}{m^2r^2\sin^2\theta} - 2gr\cos\theta \equiv E - V^{\text{eff}}(r)\,.$$

Summary of Conservation Laws

- If there exists a cyclic (ignorable) variable q_i (satisfying the condition $\dfrac{\partial \mathcal{L}}{\partial q_i} = 0$), then the canonically conjugate momentum $p_i \equiv \dfrac{\partial \mathcal{L}}{\partial \dot{q}_i}$ is conserved
- If $\dfrac{\partial \mathcal{L}}{\partial t} = 0$, then the Hamiltonian $\mathcal{H} = \sum \dot{q}_i p_i - \mathcal{L}$ is conserved.
- If ***also*** all the constraints are t independent, then $\mathcal{H} = E$ and energy $E \equiv T + V$ is conserved

4 Central forces, and more realistic physical systems

4.1 Motion in a 1D Effective Potential

As typically it is not straightforward to solve Lagrange's equations exactly, two things become equally important: one needs to understand the qualitative characteristics of the motion even before solving the equations, and one needs to learn ways of solving equations approximately.

Remember: we briefly met the concept of an effective potential for the example of a bead on a rotating parabolic wire in Section 3.

Luckily, in many cases, the symmetries of the problem (reflected by the one degree of freedom in an "effective" one-dimensional (1D) potential where an exact solution is possible. This is also applicable in the cases where $\mathcal{H} \neq E$ in the presence of time-dependent constraints, e.g. for the bead on the rotating spoke that we previously discussed.

Another case in which effective potentials are useful is when there are a (few) cyclic variable(s) such that by using the appropriate generalised momenta conservation laws, we can make T effectively inhomogeneous in the generalised velocities. The most interesting case of this sort is that of central motion (or the two-body problem) which we consider in the next lecture.

4.1.1 Using conservation laws

Now we turn to the general case of a particle's motion in a $1d$ (effective) potential and show how this can be solved using conservation laws. We shall only consider the class of problems for which $\mathcal{L}$ is time-independent.

In the simplest case, when there are no constraints, or constraints exist but the kinetic energy is still a quadratic form in a generalised velocity, the Lagrangian and Hamiltonian can be written down as

$$\mathcal{L} = \frac{1}{2}m\dot{q}^2 - V^{\text{eff}}(q) \quad \Rightarrow \quad \mathcal{H}(q,\dot{q}) = \frac{1}{2}m\dot{q}^2 + V^{\text{eff}}(q) = \mathcal{E} = \text{const}$$

where the Hamiltonian is equal to a constant because the Lagrangian is time-independent. As we discussed above, provided all the constraints are time independent (i.e. scleronomic), then (and only then) we have that $\mathcal{E} \equiv E$ – the full energy. In any case, $\mathcal{E}$ represents a conserved quantity, whether it equals the energy or not.

An example when $\mathcal{E} \neq E$ is the bead on the rotating spoke considered before.

This conservation law is enough to fully solve the problem – one does not need to write down Lagrange's equations. To see this, let us rearrange the expression from the Hamiltonian as

$$\dot{q} = \pm\sqrt{\frac{2}{m}\left(\mathcal{E} - V(q)\right)} \quad \Rightarrow \quad \pm\frac{\sqrt{m}\,\mathrm{d}q}{\sqrt{2\left(\mathcal{E} - V(q)\right)}} = \mathrm{d}t\,, \tag{4.1}$$

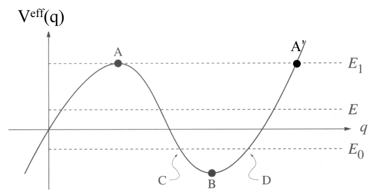
where we have dropped the superscript on the potential for simplicity. We can then integrate both sides of this equation in order to find an expression for the (positive) time that it takes to move from q_0 to some q_1 (the end coordinate) as given by the ***quadrature***:

$$t = \sqrt{\frac{m}{2}} \int_{q_0}^{q_1} \frac{\mathrm{d}q}{\sqrt{\mathcal{E} - V(q)}}. \tag{4.2}$$

In classical mechanics, "quadrature" primarily refers to solving problems by expressing solutions as integrals, a process called reduction to quadrature, crucial for systems where analytic solutions are tough but integrals can be approximated numerically.

This integral may, or may not, be calculable, depending on a concrete choice of $V(q)$.

But even without evaluating the integral, one can deduce some qualitative features of the motion by graphing the potential. An example of a generic potential is sketched in the figure. Crucially, the expression inside the square root of the integral above must be positive. This means that no classical motion is possible with energies $\mathcal{E} < V(q)$ so that on the $\mathcal{E} - q$ plane there are forbidden areas. In terms of the diagram, we can see for an energy E_0, this means that motion is only allowed within the central region for $C < q < D$. While for an energy E_1, motion is allowed for $q < A$.



A generic 1D effective potential

Aside: For values of q where $\mathcal{E} = V$, we have a classical "turning point", where the particle must turn around to avoid entering a forbidden region. This crops up again in semiclassical approximations of quantum mechanics which are taught in Quantum Mechanics 4.

4.1.2 Using Lagrange's equation

While the "energy" conversation law gave us a short cut to get the equation of motion, we could also have obtained this directly from Lagrange's equation. In our case

$$\ddot{q} = -\frac{1}{m}\frac{\mathrm{d}V^{\text{eff}}}{\mathrm{d}q} \equiv f(q)\,.$$

We have already briefly noted that equations of this type, $\ddot{q} = f(q)$ can be reduced to a 1$^{\text{st}}$ order differential equation by multiplying both parts by $2\dot{q}$:

$$\left.\begin{aligned} 2\dot{q}\ddot{q} &= \frac{\mathrm{d}\dot{q}^2}{\mathrm{d}t} \\ 2\dot{q}f(q) &= 2\frac{f(q)\,\mathrm{d}q}{\mathrm{d}t} \end{aligned}\right\} \quad \Rightarrow \quad \mathrm{d}(\dot{q}^2) = 2f(q)\,\mathrm{d}q$$

$$\Rightarrow \quad \dot{q}^2 = 2\int f(q)\,\mathrm{d}q \equiv -\frac{2}{m}\left(V^{\text{eff}}(q) + \text{const}\right)$$

If we call the constant $-\mathcal{E}$, we return exactly to what resulted from the conservation of the Hamiltonian.

4.2 Central Forces

In the examples last week, we have been creating systems where the motion of the mass has been constrained to move around a central point. In physics be it atomic or astrophysics, central forces are at the heart of our developing an understanding of the particle motions. In this section we are going to take a step back from the formal framework we have been developing (cyclic, conjugate momenta, Hamiltonian etc.) and set up the two body problem, within the Lagrangian framework.

Now may be a good moment to check that you remember what those words mean.

4.2.1 Separation of Motion for a Two-body System

In 3D, two interacting mechanical bodies with negligible internal structure have 6 degrees of freedom between them (i.e. $\mathbf{r}r_1$ and $\mathbf{r}r_2$). When two bodies interact only via a potential that depends on their separation, the nontrivial part of their motion is effectively reducible (as we will show shortly) to that of a particle with one degree of freedom in a certain ***effective potential***, where we know how to solve the problem in general. More precisely, their motion can be separated into two independent components: the motion of their centre of mass, and their relative motion.

Some common examples of these types of systems include bodies interacting via gravity, via electrostatic (Coulomb interactions) or masses joined by springs.

The Lagrangian for two bodies interacting only via a potential that depends on their separation is given by

$$\mathcal{L} = \frac{1}{2}m_1\dot{\mathbf{r}}_1^2 + \frac{1}{2}m_2\dot{\mathbf{r}}_2^2 - V(|\mathbf{r}_1 - \mathbf{r}_2|)\,. \tag{4.3}$$

We can then change to relative, $\mathbf{r}$, and centre-of-mass, $\mathbf{R}$, coordinates:

$$\begin{aligned} \mathbf{R} &= \frac{m_1\mathbf{r}_1 + m_2\mathbf{r}_2}{m_1+m_2} \\ \mathbf{r} &= \mathbf{r}_1 - \mathbf{r}_2 \end{aligned} \qquad \Rightarrow \qquad \begin{aligned} \mathbf{r}_1 &= \mathbf{R} + \frac{m_2}{m_1+m_2}\mathbf{r} \\ \mathbf{r}_2 &= \mathbf{R} - \frac{m_1}{m_1+m_2}\mathbf{r} \end{aligned} \tag{4.4}$$

This is an example of "separation of variables".

by substituting these new coordinates into the kinetic energy:

$$\begin{aligned} T &= \frac{1}{2}m_1\left(\dot{\mathbf{R}} + \frac{m_2}{m_1+m_2}\dot{\mathbf{r}}\right)^2 + \frac{1}{2}m_2\left(\dot{\mathbf{R}} - \frac{m_1}{m_1+m_2}\dot{\mathbf{r}}\right)^2 \\ &= \frac{1}{2}(m_1+m_2)\dot{\mathbf{R}}^2 + \frac{1}{2}\frac{m_1m_2}{m_1+m_2}\dot{\mathbf{r}}^2 \\ \Rightarrow \quad \mathcal{L} &= \underbrace{\frac{1}{2}M\dot{\mathbf{R}}^2}_{\text{CoM}} + \underbrace{\frac{1}{2}\mu\dot{\mathbf{r}}^2 - V(|\mathbf{r}|)}_{\text{relative}}\,. \end{aligned}$$

The centre-of-mass motion here corresponds to that of a free particle with a total mass $M = m_1 + m_2$. Indeed, $\mathbf{R}$ is a cyclic variable, so that

$$\frac{\partial\mathcal{L}}{\partial\dot{\mathbf{R}}} = M\dot{\mathbf{R}} = \text{const} \qquad \Rightarrow \qquad \text{free motion}$$

As the momentum of the center of mass is conserved, we can see that $\mathcal{L} = \text{const} + \mathcal{L}(\mathbf{r}, \dot{\mathbf{r}})$. However, we have also learned that a constant in the Lagrangian is irrelevant as it does not enter into Lagrange's equations. In other words, we can ignore this constant, meaning that the entire problem is effectively reduced to that of the relative motion. This relative motion is that of a particle of "reduced mass", μ, in the central-force potential $V(|\mathbf{r}|)$. This is effectively "one-dimensional" motion which we now need to describe in more detail.

4.2.2 Potential and Angular Momentum for a central force

In general, if a particle moves in a potential $V(r)$, that depends only on the distance to the centre of coordinate system, then the conservative force acting on it, $\mathbf{F} = -\boldsymbol{\nabla} V(r)$, is always along $\mathbf{r}$, *i.e.*is central. Indeed,

$$\mathbf{F} = -\boldsymbol{\nabla} V(r) = -\frac{\partial V}{\partial r}\left(\frac{\partial r}{\partial x}\mathbf{i} + \frac{\partial r}{\partial y}\mathbf{j} + \frac{\partial r}{\partial z}\mathbf{k}\right) = -\frac{\partial V}{\partial r}\frac{x\mathbf{i} + y\mathbf{j} + z\mathbf{k}}{r} \equiv -\frac{\partial V}{\partial r}\frac{\mathbf{r}}{r}.$$

As mentioned above, central forces are an idealisation of gravitational forces acting in star systems or electric forces in the atom, *etc.* For a two-body system interacting via fundamental forces, these forces are always central.

Let us now consider the associated angular momentum for a central force. Recall that the angular momentum is defined by

$$\mathbf{L} = \mathbf{r} \times \mathbf{p}.$$

Its rate of change is then

$$\frac{\mathrm{d}\mathbf{L}}{\mathrm{d}t} = \underbrace{\dot{\mathbf{r}} \times \mathbf{p}}_{\substack{\shortparallel \\ (\mathbf{p}\times\mathbf{p})/m = 0}} + \mathbf{r} \times \dot{\mathbf{p}} = \mathbf{r} \times \mathbf{F} \equiv \boldsymbol{\tau} \quad \text{(torque)} \tag{4.5}$$

Equation $\boldsymbol{\tau} = \dot{\mathbf{L}}$ is the angular analog of $\mathbf{F} = \dot{\mathbf{p}}$. In spherical polar coordinates, it can be reduced to a form similar to $\mathbf{F} = m\dot{\mathbf{v}}$, by noticing that only the part of acceleration perpendicular to the radial direction, $\dot{\mathbf{v}}_\perp = r\ddot{\phi}$ contributes to $\dot{\mathbf{L}}$. Therefore, $|\dot{\mathbf{L}}| = |mr\dot{\mathbf{v}}_\perp| = mr^2\ddot{\phi}$, so that the equation of motion is $\tau = \mathcal{I}\ddot{\phi}$, where the moment of inertia $\mathcal{I} = mr^2$ plays the same part in angular motion as mass in linear motion. Note also that $|L| = m|\mathbf{r} \times \dot{\mathbf{v}}| = mr|\mathbf{v}_\perp| = mr^2\dot{\phi} \equiv \mathcal{I}\dot{\phi}$. However, we will do this properly in the framework of Lagrangian mechanics.

Now, for the central force $\dot{\mathbf{L}} = \mathbf{r} \times \mathbf{F} \propto \mathbf{r} \times \mathbf{r} = 0$, *i.e.*$\mathbf{L} = \text{const}$.

Angular momentum in the presence of central forces is conserved

Note that (by definition) $\mathbf{L}$ is always perpendicular both to $\mathbf{r}$ and $\dot{\mathbf{r}}$, *i.e.*to the direction of motion (unless $\mathbf{L} = 0$ which is a special case). Therefore, for $\mathbf{L} = \text{const}$ the motion is always confined to a plane perpendicular to $\mathbf{L}$. As a consequence,

motion in the field of conservative central forces is planar

Thus we have reduced our problem from 3 to 2 degrees of freedom.

4.2.3 Equations of motion and conservation laws

As the motion is planar, it is convenient to choose to work in polar coordinates in the plane $z = 0$ (alternatively, one can think of the equatorial plane, $\theta = \pi/2$ in spherical polar coordinates). Having in mind the context of the two-body problem, we will now use μ, tge reduced mass, instead of m . The Lagrangian is then

Of course, this is also fully applicable to the motion of a single particle of mass μ in a central-force potential.

$$\mathcal{L} = T - V = \tfrac{1}{2}\mu\left(\dot{r}^2 + r^2\dot{\phi}^2\right) - V(r). \tag{4.6}$$

This does not depend on ϕ, which is therefore a cyclic variable. So it gives the second conservation law:

Remember the first conservation law is $\mathbf{L} = 0$.

$$p_\phi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \text{const} \quad \Rightarrow \quad \mu r^2\dot{\phi} \equiv \mathcal{I}\dot{\phi} = \ell \quad (\text{const}), \tag{4.7}$$

where $\mathcal{I} = \mu r^2$ is the the corresponding moment of inertia. The meaning of ℓ is the z component of the angular momentum. Note that there are only two – rather than three – independent conservation laws here (since the components are not independent because of $\mathbf{r}\cdot\mathbf{L} = \mathbf{p}\cdot\mathbf{L} = 0$): $|\mathbf{L}|$ and the direction of $\mathbf{L}$ are conserved.

Now there is only one non-cyclic variable left, r, and one can write Lagrange's equation for r as

Note that Eq. (4.7) is Lagrange's equation for ϕ: for any cyclic variable, the appropriate Lagrange equation is just a conservation law.

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{r}}\right) - \frac{\partial \mathcal{L}}{\partial r} = \mu\ddot{r} - \mu r\dot{\phi}^2 + \frac{\partial V}{\partial r} = 0 \quad \Leftrightarrow \quad = \mu\ddot{r} - \frac{\ell^2}{\mu r^3} + \frac{\partial V}{\partial r} = 0\,. \tag{4.8}$$

where the conservation law in Eq. 4.7 was used for the last equality.

However, there is one more conservation law in this problem: as $\mathcal{L}$ does not explicitly depend on t, the Hamiltonian is conserved. As there are no time-dependent constraints ($\mathcal{L}$ is quadratic in generalised velocities), $\mathcal{H} = T + V$, and the full energy is conserved. Thus, one finds

$$\mathcal{H} = \frac{1}{2}\mu\left(\dot{r}^2 + r^2\dot{\phi}^2\right) + V(r) = E_0 \quad \Rightarrow \quad \frac{1}{2}\mu\dot{r}^2 + \frac{\ell^2}{2\mu r^2} + V(r) = E_0\,. \tag{4.9}$$

Both Eqs. (4.8) and (4.9) describe the trajectory $r(t)$ of a particle in the field of the central force - so they should be equivalent. By an obvious analogy with the problem with external time-dependent constraints, we introduce the effective potential by

Note Eq. (4.9) is the *first integral* of Eq. (4.8).

$$V^{\text{eff}}(r) = V(r) + \frac{\ell^2}{2\mu r^2}\,, \quad \Rightarrow \quad \mathcal{H} = E_0 = \frac{\mu}{2}\dot{r}^2 + V^{\text{eff}}(r) \tag{4.10}$$

The problem is, therefore, reduced to that with one degree of freedom only!

Aside: The term $\dfrac{\ell^2}{2\mu r^2}$ is often referred to as the "centrifugal barrier" because the conservation of l prevents particles from reaching $r = 0$.

We can also get V^{eff} directly from Lagrange's equation for r, using the fact that $\frac{\partial V}{\partial r} = \frac{\mathrm{d}V}{\mathrm{d}r}$:

We can replace the partial with the full derivative as V depends only on r

$$\mu\ddot{r} = \frac{\ell^2}{\mu r^3} - \frac{\mathrm{d}V}{\mathrm{d}r} = -\frac{\mathrm{d}}{\mathrm{d}r}\left(V + \frac{\ell^2}{2\mu r^2}\right) \equiv -\frac{\mathrm{d}V^{\text{eff}}}{\mathrm{d}r}\,. \qquad (4.11)$$

Indeed, this is just the equation of motion of a particle moving in an *effective* potential, $V^{\text{eff}}(r)$, in full correspondence to Eq. (4.10) above, obtained from the energy conservation. Note that

$$E_0 - V^{\text{eff}} = \frac{1}{2}\mu\dot{r}^2 \geq 0 \quad \Rightarrow \quad E_0 \geq V^{\text{eff}}, \qquad (4.12)$$

i.e. a particle's motion is only possible with energies exceeding V^{eff} – exactly as we saw for the full potential $V(r)$ previously.

As the problem has now been formally reduced to that previously studied, the general solution is found as before by expressing the velocity in terms of the conserved energy in Eq. (4.9) which leads (with $m \to \mu$) to

Useful: In general, the effective potential can be composed by re-allocating to it all terms from the kinetic energy that are not proportional to the 2nd power of the generalised velocities

$$\frac{\mathrm{d}r}{\mathrm{d}t} = \sqrt{\frac{2}{\mu}\left[E - V^{\text{eff}}(r)\right]} \quad \Rightarrow \quad t = \sqrt{\frac{\mu}{2}}\int_{r_0}^{r(t)} \frac{\mathrm{d}r}{\sqrt{\left[E - V^{\text{eff}}(r)\right]}} \qquad (4.13)$$

However, one important difference to last time is that this motion is planar (e.g. in the equatorial plane) rather than truly 1D. One can use the conservation of generalised momenta conservation to find an expression for an orbit in quadratures. First note that

Here the conservation of generalised momenta corresponds to conserving the modulus of the angular momentum

$$\frac{\dot{r}}{\dot{\phi}} = \underbrace{\frac{\mathrm{d}r}{\mathrm{d}t}\frac{\mathrm{d}t}{\mathrm{d}\phi}}_{\text{chain rule}} = \frac{\mathrm{d}r}{\mathrm{d}\phi}\,.$$

As both derivatives on the l.h.s. are known (see Eqs. (4.7) and (4.13)), we find

Remember: this is an example of "dot cancellation".

$$\frac{\mathrm{d}r}{\mathrm{d}\phi} = \frac{\mu r^2}{\ell}\sqrt{\frac{2}{\mu}\left[E - V^{\text{eff}}(r)\right]} \qquad \Rightarrow \qquad \phi = \frac{\ell}{\mu}\int_{r_0}^{r(t)} \frac{\mathrm{d}r}{r^2\sqrt{\frac{2}{\mu}\left[E - V^{\text{eff}}(r)\right]}} \qquad (4.14)$$

This integral can be attempted for concrete problems, once we know $V(r)$. However, even without going on to further calculations, quite a few things can be found in general just from the statement that $E > V^{\text{eff}}$, as before. However, we postpone such a qualitative analysis for a later stage of this course. Now we shall switch (first time in this course) to not only deriving equations, but fully solving them for the most famous problem of classical mechanics, ***the Kepler problem***.

In summary for motion in central forces: $|\mathbf{L}|$ and the direction of $\mathbf{L}$ are conserved $\Rightarrow$ the problem reduces to that with one degree of freedom, r (although it is still convenient to describe it in terms of both r and θ). By introducing V^{eff}, Eq. (4.10), we explicitly reduce the problem to that (considered in the previous lecture) of motion in the effective $1d$ potential $V^{\text{eff}}(r)$, $r > 0$.

5 Kepler orbits

5.1 Kepler Orbits

We will now consider motion under the influence of an inverse square law force, such as gravity. This example is very important - both for orbits of satellites, planets stars etc and also for the motion of charged particles on an atomic scale, where classical results often illuminate the quantum mechanical calculations. Here we will derive the *shape* of the orbit, revealing that it is elliptical. We mean by 'shape' $r(\phi)$, *i.e.*eliminating time, t, from $r(t)$ and $\phi(t)$.

5.1.1 Dimensionless Variables

For the case of an inverse square law, such as gravity,

$$V(r) = -\frac{Gm_1m_2}{r} \equiv -\frac{\mu M}{r} \equiv -\frac{k}{r} \quad \Rightarrow \quad V^{\text{eff}}(r) = \frac{\ell^2}{2\mu r^2} - \frac{k}{r}\,. \tag{5.1}$$

We can now use our previous equation with the effective potential for Kepler's problem:

$$\phi = \frac{\ell}{\mu}\int_{r_0}^{r(t)} \frac{\mathrm{d}r}{r^2\sqrt{\frac{2}{\mu}\left[E - V^{\text{eff}}(r)\right]}} = \ell\int_{r_0}^{r(t)} \frac{\mathrm{d}r}{r^2}\left\{2\mu\left[E - \frac{\ell^2}{2\mu r^2} + \frac{k}{r}\right]\right\}^{-\frac{1}{2}}\,.$$

First of all, let us choose "natural" dimensionless variables. By comparing the two terms in the effective potential $V^{\text{eff}} = 0$ we find that $r_0 = \ell^2/\mu k$ fully defines the scale. For example, $V^{\text{eff}} = 0$ at $r = r_0/2$.) If we introduce a dimensionless variable $\rho = r/r_0$, V^{eff} becomes

Note: This does not work for $\ell = 0$ which is a trivial case of falling into the centre along a straight line.

$$V^{\text{eff}} = \frac{\ell^2}{2\mu r_0^2}\left(\frac{1}{\rho^2} - \frac{2}{\rho}\right) \equiv \mathcal{E}_0\left(\frac{1}{\rho^2} - \frac{2}{\rho}\right)$$

It is natural now to measure the full energy E - one of the constants of motion - in units of $\mathcal{E}_0$, so that we introduce dimensionless $\epsilon \equiv E/\mathcal{E}_0$. Now the equation for ϕ is simplified considerably:

$$\phi = \int \frac{\mathrm{d}\rho}{\rho^2\sqrt{\epsilon - \frac{1}{\rho^2} + \frac{2}{\rho}}}\,.$$

The easiest way to calculate this integral is to change variables to $u = 1/\rho$ so that $\mathrm{d}\rho/\rho^2 = -\,\mathrm{d}u$:

$$\phi = -\int \frac{\mathrm{d}u}{\sqrt{\epsilon - u^2 + 2u}} = -\int \frac{\mathrm{d}u}{\sqrt{\epsilon + 1 - (u-1)^2}} = \phi_0 + \arccos\frac{u-1}{\sqrt{\epsilon+1}}\,. \tag{5.2}$$

Here ϕ_0 is an irrelevant const of integration . Therefore, choosing $\phi_0 = 0$, we obtain the orbit equation as

ϕ_0 gives the starting point of the orbit which could always be rotated to $\phi_0 = 0$)

$$u = 1 + \sqrt{\epsilon + 1}\cos\phi\,. \tag{5.3}$$

Before analysing this equation, it's worth a diversion to obtain it in a different way. Let us consider directly Lagrange's equation for an arbitrary central force potential, $V(r)$, which is

$$\mu\ddot{r} = \frac{\ell^2}{\mu r^3} - \frac{\mathrm{d}V}{\mathrm{d}r}\,.$$

First, let us convert this into an equation for the orbit, by changing differentiation with respect to t to that with respect to ϕ and then using the same "magic" substitution $u = r_0/r$:

$$\dot{r} = \frac{\mathrm{d}r}{\mathrm{d}\phi}\frac{\mathrm{d}\phi}{\mathrm{d}t} = \frac{\mathrm{d}r}{\mathrm{d}\phi}\frac{\ell}{\mu r^2} = -\frac{\ell}{\mu}\frac{\mathrm{d}}{\mathrm{d}\phi}\left(\frac{1}{r}\right) = -\frac{\ell}{\mu r_0}\frac{\mathrm{d}u}{\mathrm{d}\phi}$$

$$\Rightarrow \quad \ddot{r} = \frac{\mathrm{d}\dot{r}}{\mathrm{d}t} = \frac{\mathrm{d}\dot{r}}{\mathrm{d}\phi}\frac{\mathrm{d}\phi}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}\phi}\left(-\frac{\ell}{\mu r_0}\frac{\mathrm{d}u}{\mathrm{d}\phi}\right)\frac{\ell}{\mu r^2} = -\left(\frac{u^2\ell^2}{\mu^2 r_0^3}\right)\frac{\mathrm{d}^2u}{\mathrm{d}\phi^2}\,.$$

Lagrange's equations becomes

$$-\left(\frac{u^2\ell^2}{\mu r_0^3}\right)\frac{\mathrm{d}^2u}{\mathrm{d}\phi^2} = \frac{\ell^2}{\mu r_0^3}u^3 + f(u) \quad \Rightarrow \quad u''_{\phi\phi} + u = -\frac{1}{u^2}f(u)\frac{\mu r_0^3}{\ell^2}\,.$$

For the Kepler problem, $V = -k/r \Rightarrow f(r) = -k/r^2 \Rightarrow f(u) = -ku^2/r_0^2$, and so the r.h.s. is simply the constant $k\mu r_0/\ell^2$. Since r_0 was arbitrary, we can now choose it in the simplest possible way, $r_0 = \ell^2/k\mu$, reducing the orbital equation to $u''_{\phi\phi} + u = 1$. This is like the SHO equation with a constant force, and the solution is to be found immediately:

$$u = 1 + e\cos(\phi - \phi_0)\,. \tag{5.4}$$

As it should, it has two integration constants, e and ϕ_0, and it coincides with Eq. (5.3), where, however, the physical meaning of the constant attached to the cosine was clear from the very beginning. To make it clear now, independently of the previous page, we write down the Hamiltonian in terms of u:

$$\mathcal{H} = \frac{1}{2}\mu\dot{r}^2 + \frac{\ell^2}{2\mu r^2} + V(r) = \frac{\ell^2}{2\mu r_0^2}\left[\left(u'_\phi\right)^2 + u^2\right] + V(u) = E_0$$

For the Kepler problem, $V(u) = -ku/r_0 = -(\ell^2/\mu r_0^2)$ so that

$$E_0 = \mathcal{E}_0\left[\left(u'_\phi\right)^2 + u^2 - 2u\right]\,, \quad \mathcal{E}_0 \equiv \frac{\ell^2}{2\mu r_0^2} \quad \text{is a natural unit of energy}$$

We know that E_0 must be a constant. Indeed, substituting the solution, we find $E_0/\mathcal{E}_0 = e^2 - 1$ so that the eccentricity $e^2 = E_0/\mathcal{E}_0 + 1 \equiv \epsilon + 1$ where ϵ is the dimensionless energy, *i.e.*energy measured in the units of $\mathcal{E}_0$.

5.1.2 Summary: Analysing Kepler's orbits

We found the orbital equation

$$u = 1 + e\cos(\phi - \phi_0)\,,$$

where the eccentricity $e \equiv \sqrt{1+\epsilon}$ with $\epsilon \equiv E/\mathcal{E}_0$. We use dimensionless variables ϵ for energy and $\rho = r/r_0$ for distance, where units of measurements are

$$r_0 \equiv \frac{\ell^2}{\mu\kappa} \qquad\qquad \mathcal{E}_0 \equiv \frac{\ell^2}{2\mu r_0^2} = \frac{\mu\kappa^2}{2\ell^2} = \frac{\kappa}{r_0}\,.$$

Both the energy ϵ and the amplitude of the angular momentum ℓ are conserved quantities, or constants (integrals) of motion.

5.1.3 Conical sections

Remember: a conical section is a curve obtained by intersecting a plane with a cone, yielding a circle, ellipse, parabola, or hyperbola.

Let's now check that Eqs. (5.3) and (5.4) describe conical sections, as we qualitatively stated earlier. Note that $e = \sqrt{\epsilon+1}$ in these equations is always non-negative. Let us now discuss what happens for different values of e

- The smallest value $e = 0$ corresponds to $\epsilon = -1$, or in usual dimensional units $E = -\mu k^2/2\ell^2$. As discussed above, the energy obeys $E \geq V^{\text{eff}}$ for classically-allowed motion. The value $E = -\mu k^2/2\ell^2$ is exactly the minimal value of V^{eff} (where the motion is circular with $r = r_0$). Naturally, one see this from the solution: at $e = 0$ one has $u = 1$, *i.e.*$r = r_0$ as expected.

- From our qualitative analysis, we expect that for $E < 0 \;\Leftrightarrow\; \epsilon < 0 \;\Leftrightarrow\; e < 1$ the orbit is elliptic. Indeed, for $e < 1$ one has $1 + e\cos\phi > 0$ for any ϕ. It tells one that the motion is periodic, as all values of the angle ϕ are possible. This is actually the most natural equation for the ellipse of ***eccentricity*** e or for any conical section.

- For $e = 1$, we see that $u = 0 \Rightarrow r = \infty$ at $\phi - \phi_0 = \pi(2n+1)$ – therefore, the orbit becomes unbound, albeit only for a discrete set of values of ϕ. This is – as we shall see shortly – a parabolic orbit.

- Finally, for $e > 1$, u reaches 0 before ϕ goes from 0 to π, and there is an interval of unphysical angles corresponding to negative u. This corresponds to open hyperbolic orbits.

To see this all more explicitly, we reduce the orbital equation to that in Cartesian coordinates ($x = r\cos\phi$ and $r = \sqrt{x^2+y^2}$) which look more familiar. Restoring the usual units, $u \equiv r_0/r$, one has

$$\frac{r_0}{r} = 1 + e\cos\phi \quad \Rightarrow \quad r_0 = r + er\cos\phi = \sqrt{x^2+y^2} + ex$$

which gives

$$\Rightarrow \quad x^2 + y^2 = r_0^2 - 2exr_0 + e^2x^2$$

This must be the equation for a conical section. To see this more clearly, one completes the square to obtain

$$(1-e^2)x^2 + 2exr_0 + y^2 = r_0^2 \quad \Rightarrow \quad r_0^2 = (1-e^2)\left[x + \frac{er_0}{1-e^2}\right]^2 + y^2 - \frac{(er_0)^2}{1-e^2} \tag{5.5}$$

Therefore, for $e^2 - 1 \neq 0$ one finds

$$\frac{\left[x + \frac{er_0}{1-e^2}\right]^2}{\frac{r_0^2}{\left(1-e^2\right)^2}} + \frac{y^2}{\frac{r_0^2}{1-e^2}} = 1$$

The next step depends on the sign of $e^2 - 1$. For $e^2 - 1 < 0$, we've got a re-scaled ***sum*** of squares of x and y while for $e^2 - 1 > 0$ – a re-scaled ***difference***. Introducing new notations,

$$a \equiv \frac{r_0}{|1-e^2|}, \qquad \text{and} \qquad b \equiv \frac{r_0}{\sqrt{|1-e^2|}}, \tag{5.6}$$

we cast finally the orbital equation into the canonical form:

$$\begin{aligned}
\frac{(x+ea)^2}{a^2} + \frac{y^2}{b^2} &= 1 \qquad \text{for } e^2 < 1 \qquad \text{(ellipse)} \\
\frac{(x-ea)^2}{a^2} - \frac{y^2}{b^2} &= 1 \qquad \text{for } e^2 > 1 \qquad \text{(hyperbola)} \\
\pm x = \frac{r_0}{2} - \frac{y^2}{r_0^2} &= 1 \qquad \text{for } e \;\; = \pm 1 \qquad \text{(parabola)}
\end{aligned} \tag{5.7}$$

5.1.4 Time of orbits; Kepler's laws

Let now consider motion in time for the Kepler problem. We have obtained previously the general solution for $t(r)$ in central forces:

$$
\begin{aligned}
t &= \sqrt{\frac{\mu}{2}} \int_{r_0}^{r(t)} \frac{\mathrm{d}r}{\sqrt{[E - V^{\mathrm{eff}}(r)]}} = \sqrt{\frac{\mu}{2}} \int_{r_0}^{r(t)} \frac{\mathrm{d}r}{\sqrt{\left[E + \frac{\kappa}{r} - \frac{\ell^2}{2\mu r^2}\right]}} \\
&= T_0 \int \frac{\mathrm{d}\rho}{\sqrt{\frac{2}{\rho} - \frac{1}{\rho^2} + \epsilon}}
\end{aligned}
$$

where we have introduce a natural unit of time:

$$T_0 \equiv \sqrt{\frac{\mu r_0^2}{2\mathcal{E}}} \equiv \frac{\ell^3}{\mu\kappa^2}\,. \tag{5.8}$$

We can use the orbital equation, $u = 1 + e\cos(\phi - \phi_0)$ to rewrite this as an integral over ϕ:

$$t = \int_{\phi_1}^{\phi_2} \frac{\mathrm{d}\phi}{\left[1 + e\cos\left(\phi - \phi_0\right)\right]^2}\,. \tag{5.9}$$

Either of these equations can be integrated; however, this is not particularly useful as the resulting expressions are rather involved. We limit ourselves to finding the period of motion in case $e < 1$, *i.e.*for an ellipse.

Let us recall the conservation law $\frac{1}{2}\mu r^2\dot{\phi} = \ell$ which we recast as

$$\frac{\mathrm{d}\mathcal{A}}{\mathrm{d}t} \equiv \frac{1}{2} r^2 \dot{\phi} = \frac{\ell}{2\mu}\,. \tag{5.10}$$

Here $\mathrm{d}\mathcal{A}$ is area of a part of the orbit swept by an angle $\mathrm{d}\phi$. The fact that in an orbital motion such an area swept per unit time is a constant for any given orbit is known as ***2*nd *Kepler's law: the law of equal areas***. It is actually valid for any motion in central forces.

The ***1*st *Kepler's law*** we have already – implicitly – derived:

The planets move in elliptical orbits about the sun at one of its foci.

In contrast to the 2nd law, and in line with the 3rd which we are about to derive, it is valid only for the inverse square law for central force.

To find the period T, we integrate Eq. (5.10) over the entire period:

$$\int_0^T \frac{\mathrm{d}\mathcal{A}}{\mathrm{d}t}\,\mathrm{d}t = \mathcal{A} = \frac{\ell T}{2\mu}\,.$$

Here $\mathcal{A}$ is the total area of the elliptic orbit.

Now we use a simple geometric fact: the area of an ellipse is

$$\mathcal{A} = \pi ab$$

where, as we have found, the semiminor axis b is related to a by

$$b = a\sqrt{1-e^2} = \sqrt{\frac{a\ell^2}{\mu\kappa}}\,.$$

Hence we have $T^2 \propto a^3$ as desired.

Kepler's 3rd law states that for bodies orbiting the same central mass, the square of the orbital period T T is proportional to the cube of the semi-major axis a a of the orbit.

5.1.5 Kepler's orbits: Summary

Shape of the orbits

$$
\begin{aligned}
\frac{(x+ea)^2}{a^2} + \frac{y^2}{b^2} &= 1 \qquad \text{for } e^2 < 1 \qquad \text{(ellipse)} \\
\frac{(x-ea)^2}{a^2} - \frac{y^2}{b^2} &= 1 \qquad \text{for } e^2 > 1 \qquad \text{(hyperbola)} \\
\pm x = \frac{r_0}{2} - \frac{y^2}{r_0^2} &= 1 \qquad \text{for } e \ = \pm 1 \qquad \text{(parabola)}
\end{aligned}
\tag{5.11}
$$

Kepler's 1st Law: planets move along elliptic orbit with the Sun being in one of the foci of the ellipse.

Parameters of the orbit

$$
a = \frac{r_0}{|e^2-1|} \equiv \frac{r_0}{|\epsilon|}\,, \quad r_0 \equiv \frac{\ell^2}{\mu k}\,, \quad \epsilon \equiv \frac{E}{\mathcal{E}}\,, \quad \mathcal{E} \equiv \frac{\ell^2}{2\mu r_0^2} = \frac{\mu k^2}{2\ell^2}\,.
$$

We have also derived 2nd and 3rd Kepler's laws.

6 Small oscillations around equilibrium

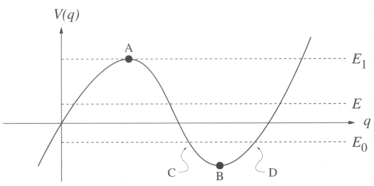
In this section, we shall explore the motion that takes place close to a stable equilibrium configuration. When the displacements from equilibrium are small, the equations of motion can be linearised, leading to small oscillations that are accurately described by simple harmonic motion. As we shall explore, this approximation allows complicated systems to be analysed in terms of normal modes and characteristic frequencies, providing deep insight into their stability and dynamical behaviour.

6.1 Systems with 1 degree of freedom

Although any system with only one degree of freedom is integrable, with the solution given (as we previously showed) by

$$t = \sqrt{\frac{m}{2}} \int_0^{q_1} \frac{\mathrm{d}q}{\sqrt{\mathcal{E} - V(q)}}, \tag{6.1}$$

it is worth considering how to approximate the behaviour in some specific regimes, namely near the minima and maxima of effective potential V^{eff}. The reason is that the motion in these regions has some universal features, knowledge of which will be very useful in describing systems with more than one degrees of freedom in the following section.



A generic 1D potential

Let us again consider a "generic" (effective) 1d potential as shown in this diagram. Instead of directly addressing Eq. (6.1) we first write again the Lagrangian,

$$\mathcal{L} = \frac{1}{2} m \dot{q}^2 - V^{\text{eff}}(q)$$

and expand $V^{\text{eff}}(q)$ near extremal points q_0 (using the extremal condition):

$$\frac{\mathrm{d}V^{\text{eff}}(q)}{\mathrm{d}q} = 0 \quad \Rightarrow \quad V^{\text{eff}}(q) = V^{\text{eff}}(q_0) + \frac{1}{2} \left. \frac{\partial^2 V^{\text{eff}}(q)}{\partial q^2} \right|_{q_0} q^2 \equiv V_0 + \frac{1}{2} \kappa q^2 ,$$

Note in passing that in this case $E_0 = V_0 + \frac{1}{2} m \dot{q}^2 + \frac{1}{2} \kappa q^2$.

where obvious new notations are introduced. However, it is more convenient to use Lagrange's equation directly:

$$m\ddot{q} + \kappa q = 0 \quad \Leftrightarrow \quad \ddot{q} + {\omega_0}^2 q = 0 , \qquad \omega_0 = \sqrt{\frac{\kappa}{m}} . \tag{6.2}$$

The sign of κ could be either positive (at B in the diagram), or negative (at A), producing real or imaginary ω_0. The former case correspond to

stable equilibrium: when the system is driven out of the equilibrium point, there are returning forces so that it starts oscillating around the equilibrium. When ω is imaginary, the equilibrium is ***unstable***: the system is out of equilibrium, it is pushed farther away.

Formally, this can be seen by solving the SHO equation Eq. (6.2). We look for a solution in a trigonometrical form:

$$q = q_0 \sin(\omega t + \phi_0) \ .$$

Substituting this in to Eq. (6.2), one finds

$$\left(-\omega^2 + \omega_0^2\right) q_0 \sin(\omega t + \phi_0) = 0 \quad \Rightarrow \quad \omega = \omega_0 .$$

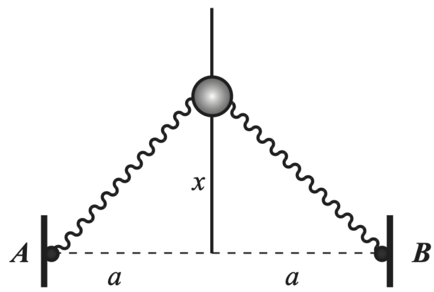
When ω_0 is real, the solution is, indeed, trigonometrical. When it is imaginary, sin $\to$ sinh, and the exponentially growing part of the solution, which drives the system away from the equilibrium, prevails.

Note that q_0 and ϕ_0 can only be found from the initial conditions: we must have two such conditions as this is a 2[nd] order differential equation.

Of course, all the results obtained by such an expansion are equally easy to extract from the exact solution given by Eq. (6.1). However, this expansion method works equally efficiently for systems with more than one degrees of freedom when we do not have an exact solution. Before going to such a situation, we consider one example of a system with 1 degree of freedom.

6.1.1 Example: Small Oscillation of Mass on a Rod with Springs

Consider the system of a mass constrained to move along a rod, while it is connected by two springs to two boundaries. We assume that the system lies in the horizontal plane (i.e. there is no force of gravity acting along the x direction). We also assume that two springs are identical, with natural lengths ℓ and spring constants κ. In this problem, there is only one degree of freedom: the distance along the rod, x. The Lagrangian is



$$\mathcal{L} = \frac{1}{2} m\dot{x}^2 - V^{\text{eff}}(x), \quad V^{\text{eff}}(x) \qquad = \kappa\left(\sqrt{x^2+a^2} - \ell\right)^2$$

i.e. $V^{\text{eff}} \equiv V$ is the sum of the two spring energies. The equilibrium positions of the mass obey

$$\frac{\mathrm{d}V^{\text{eff}}(x)}{\mathrm{d}x} = 0 \quad \Rightarrow \quad 2\kappa\left(\sqrt{x^2+a^2} - \ell\right)\frac{x}{\sqrt{x^2+a^2}} = 0 \, .$$

There are three solutions: $x_1 = 0$ and $x_{2,3} = \pm\sqrt{\ell^2 - a^2}$ but the last two exist only for $\ell > a$. To get ω_0, one needs to expand the effective potential near the equilibria positions:

$$\frac{\partial^2 V^{\text{eff}}}{\partial x^2} = 2\kappa \left[1 - \frac{\ell}{\sqrt{x^2+a^2}} + \frac{\ell x^2}{\sqrt{\left(x^2+a^2\right)^3}}\right]$$
$$= \begin{cases} 2\kappa\left(1 - \dfrac{\ell}{a}\right), & x = 0 \\ 2\kappa\left(1 - \dfrac{a^2}{\ell^2}\right) & x = \pm\sqrt{\ell^2 - a^2} \end{cases}$$

For $\ell < a$, only the first solution exists and the corresponding frequency is real so that there are oscillations around $x = 0$ with

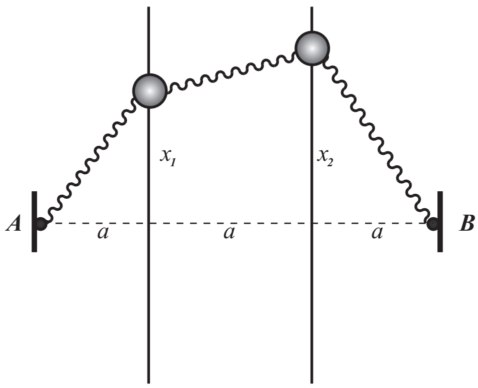
$$\omega_0 = \sqrt{\frac{2\kappa}{m}\left(1 - \frac{\ell}{a}\right)}.$$

For $\ell > a$, this frequency becomes imaginary, and therefore the $x = 0$ equilibrium position becomes unstable. A new stable equilibrium arises at two symmetric points $x = \pm\sqrt{\ell^2 - a^2}$. The oscillations around this equilibrium take place with the frequency

$$\omega_0 = \sqrt{\frac{2\kappa}{m}\left(1 - \frac{a^2}{\ell^2}\right)}.$$

6.2 Systems with 2 degrees of freedom

6.2.1 Normal modes: Beads on parallel horizontal spokes connected by a spring



Now let us consider two beads of mass m constrained to move along two rods, while being connected by springs as shown in the diagram. Each mass can move only in one (x) direction and is connected by identical springs of natural length $\ell(< a)$ and spring constant k. We again assume that the entire setup is in horizontal plane so that gravity is irrelevant. As the potential energy of each spring is $(k/2)(\Delta\ell)^2$, and $(\Delta\ell_1)^2 = x_1^2 + a^2$ *etc*, we have

$$\mathcal{L} = \frac{m}{2}\left(\dot{x}_1^2 + \dot{x}_2^2\right) - \frac{k}{2}\left[\left(\sqrt{x_1^2+a^2} - \ell\right)^2 + \left(\sqrt{x_2^2+a^2} - \ell\right)^2\right]$$
$$- \frac{k}{2}\left[\left(\sqrt{(x_2-x_1)^2+a^2} - \ell\right)^2\right]$$

Although finding a complete solution is hopeless, it is still reasonably straightforward to consider small oscillations near the equilibrium points. Our first task is therefore to find all equilibria positions. Here,

Usually one should formally look for extremal points of V by finding its partial derivatives, as above.

the straightforward symmetry guarantees that equilibrium is reached at $x_1 = x_2 = 0$, when all the three strings' length equal ℓ.

Note that $\ell \neq a$, so that the elastic energy does not vanish even at equilibrium.

Expanding each of the spring energies up to the quadratic order in x as

$$\left(\sqrt{x_1^2 + a^2} - \ell\right)^2 = \left[a\left(1 + \frac{x^2}{a^2}\right)^{1/2} - \ell\right]^2 = \left[a - \ell + \frac{x^2}{2a^2}\right]^2 = (a-\ell)^2 + \frac{a-\ell}{a}x^2\,, \quad \text{etc}\ ,$$

one finds

$$\frac{1}{m}\mathcal{L} \approx \frac{1}{2}\left(\dot{x}_1^2 + \dot{x}_2^2\right) - \frac{\omega_0^2}{2}\left[x_1^2 + x_2^2 + (x_1 - x_2)^2\right]\,, \quad \omega_0^2 \equiv \frac{k}{m}\,\frac{a-\ell}{a}\,,$$

where an irrelevant constant (the springs energy at the equilibrium) was omitted.

Now the two Lagrange's equations are

$$\left.\begin{aligned} \frac{\partial \mathcal{L}}{\partial x_1} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{x_1}}\right) \quad &\Rightarrow \quad \ddot{x}_1 = -\omega_0^2\,(2x_1 - x_2)\ , \\ \frac{\partial \mathcal{L}}{\partial x_2} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{x_2}}\right) \quad &\Rightarrow \quad \ddot{x}_2 = -\omega_0^2\,(-x_1 + 2x_2)\ , \end{aligned}\right\} \quad \Leftrightarrow \quad \begin{pmatrix}\ddot{x}_1(t)\\ \ddot{x}_2(t)\end{pmatrix} = \begin{pmatrix}2 & -1\\ -1 & 2\end{pmatrix}\begin{pmatrix}x_1(t)\\ x_2(t)\end{pmatrix}$$

One looks for a solution in oscillatory form, *e.g.*

$$\begin{pmatrix}x_1(t)\\ x_2(t)\end{pmatrix} = \begin{pmatrix}x_1\\ x_2\end{pmatrix}\cos(\omega t + \phi) \quad \Rightarrow \quad \begin{pmatrix}\ddot{x}_1(t)\\ \ddot{x}_2(t)\end{pmatrix} = -\omega^2\begin{pmatrix}x_1(t)\\ x_2(t)\end{pmatrix}$$

Therefore, Lagrange's equations in matrix form reduce to

$$\begin{aligned} -\omega^2\begin{pmatrix}x_1\\ x_2\end{pmatrix} &= -\omega_0^2\begin{pmatrix}2 & -1\\ -1 & 2\end{pmatrix}\begin{pmatrix}x_1\\ x_2\end{pmatrix} \\ \Rightarrow \quad &\begin{pmatrix}2\omega_0^2 - \omega^2 & -\omega_0^2\\ -\omega_0^2 & 2\omega_0^2 - \omega^2\end{pmatrix}\begin{pmatrix}x_1\\ x_2\end{pmatrix} = 0 \end{aligned}$$

The last equation has a nonzero solution only when the determinant of the ω-matrix is nonzero, so that one obtains the *characteristic equation*:

$$\left(2\omega_0^2 - \omega^2\right)^2 = \omega_0^4 \quad \Rightarrow \quad \begin{cases}\omega_1^2 = \omega_0^2\\ \omega_1^2 = 3\omega_0^2\end{cases}.$$

The solutions found above are called ***normal frequencies*** or ***eigenfrequencies***. We can also find the ***normal modes*** which basically characterise a relative value of oscillations in each degree of freedom (x_1 and x_2 in the present case). For this, we substitute the normal frequencies back into the matrix equations. We expect both matrix

Remember you have met "normal modes" before, e.g. in Quantum Approach to Solids.

rows to be proportional to each other (as det of the matrix is zero), so that multiplying any of them by the matrix column gives the same result. In the present case: The first normal mode with $\omega_1 = \omega_0$:

$$\begin{pmatrix} \omega_0^2 & -\omega_0^2 \\ -\omega_0^2 & \omega_0^2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0 \qquad \Rightarrow \qquad \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \text{const} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

These are called "breathing oscillations". The second normal mode with $\omega_2 = \omega_0\sqrt{3}$:

$$\begin{pmatrix} -\omega_0^2 & -\omega_0^2 \\ -\omega_0^2 & -\omega_0^2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 0 \qquad \Rightarrow \qquad \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \text{const} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

These are called "anti-breathing oscillations".

Any small oscillation near equilibrium can be represented as a sum of normal modes. The complete solution, $\{\mathbf{q}(t), \dot{\mathbf{q}}(t)\}$ can then be found given the initial (or boundary) conditions. Such a solution in the case considered looks like

$$\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = A_1 \sin(\omega_1 t + \phi_1) \begin{pmatrix} 1 \\ -1 \end{pmatrix} + A_2 \sin(\omega_2 t + \phi_2) \begin{pmatrix} 1 \\ 1 \end{pmatrix} ,$$

$$\begin{pmatrix} \dot{x}_1(t) \\ \dot{x}_2(t) \end{pmatrix} = A_1\omega_1 \cos(\omega_1 t + \phi_1) \begin{pmatrix} 1 \\ -1 \end{pmatrix} + A_2\omega_2 \cos(\omega_2 t + \phi_2) \begin{pmatrix} 1 \\ 1 \end{pmatrix} .$$

This depends on four constants of integration, the amplitudes $A_{1,2}$ and phases $\phi_{1,2}$ of the oscillations, which should be found from four initial conditions (for $x_{1,2}(0)$ and $\dot{x}_{1,2}(0)$ respectively).

Let us, for example, consider $\dot{x}_1(0) = v$, $\dot{x}_2(0) = x_{1,2}(0) = 0$. Then we have 2 matrix, or 4 usual, equations to determine the constants:

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} = A_1 \sin\phi_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} + A_2 \sin\phi_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} ,$$

$$\begin{pmatrix} v \\ 0 \end{pmatrix} = A_1\omega_1 \cos\phi_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} + A_2\omega_2 \cos\phi_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} .$$

$$\Rightarrow \quad \begin{aligned} 0 &= A_1 \sin\phi_1 + A_2 \sin\phi_2 \\ 0 &= -A_1 \sin\phi_1 + A_2 \sin\phi_2 \\ v &= A_1\omega_1 \cos\phi_1 + A_2\omega_2 \cos\phi_2 \\ 0 &= -A_1\omega_1 \cos\phi_1 + A_2\omega_2 \cos\phi_2 \end{aligned}$$

Adding and subtracting the 1st two equations, we find $A_1 \sin\phi_1 = A_2 \sin\phi_2 = 0$. As neither of the amplitudes vanish, this gives $\phi_1 =$

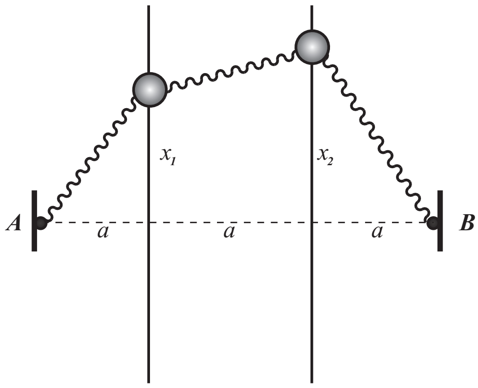
$\phi_2 = 0$. Then from the 2nd two equations, we see that $A_2 = (v/2\omega_2)$ and $A_1 = -(v/2\omega_1)$.

Note that symmetric initial conditions (*e.g.*, $x_1(0) = x_2(0) = b$) will excite only the breathing mode (*i.e.*in this case $A_1 = 0$), while anti-symmetric ones (like $x_1(0) - = x_2(0) = b$) will excite only the anti-breathing mode.

7 More Normal Modes

For the sake of breaking the lecture notes into roughly equal size chunks, this is a new Section but one which by focusing on"More Normal Modes", continues the problem that we were studying previously.

7.1 Normal modes: Beads on parallel horizontal spokes connected by a spring (again)



The main difference with the previous section is that now we are considering the case of $\ell > a$, i.e. the natural length is greater than the distance between the spokes. Otherwise, the underlying recipe/mechanistic steps are the same as before.

Above we evaluated the behaviour of the system in the vicinity of the equilibrium point of $x_1 = x_2 = 0$. All the previous considerations remain valid for $\ell > a$ except that we can now see that this equilibrium point must become unstable. (We can see that both $\omega_1 = \omega_0$ and $\omega_2 = \sqrt{3}\omega_0$ become purely imaginary as $\omega_0^2 = (k/m)(1 - \ell/a) < 0$.) Therefore, we should first look for new equilibrium configurations.

We consider solutions of the equilibrium equations

$$\frac{\partial V(x_1^*, x_2^*)}{\partial x_1} = \frac{\partial V(x_1^*, x_2^*)}{\partial x_2} = 0\,.$$

and take advantage of the symmetry of the potential

$$\begin{aligned} V(x_1, x_2) =& \frac{k}{2}\left(\sqrt{x_1^2 + a^2} - \ell\right)^2 + \frac{k}{2}\left(\sqrt{(x_1 - x_2)^2 + a^2} - \ell\right)^2 \\ &+ \frac{k}{2}\left(\sqrt{x_2^2 + a^2} - \ell\right)^2 . \end{aligned}$$

w.r.t the interchange of the variables $x_1 \leftrightarrow x_2$. There will be two types of solutions: symmetric $x_1^* = x_2^*$ and anti-symmetric $x_1^* = -x_2^*$ ones.

Let us concentrate on the symmetric solution $x_1^* = x_2^* = x^*$, where $x^* = \pm\sqrt{\ell^2 - a^2}$ is found to be the same as in the example of one degree of freedom discussed above. To proceed, we need to expand the potential $V(x_1, x_2) = V(x^* + q_1, x^* + q_2)$ with respect to small deviations q_1, q_2 from equilibrium. We have

$$V(x^* + q_1, x^* + q_2) \simeq V(x^*, x^*) + \frac{1}{2}\begin{pmatrix} q_1, & q_2 \end{pmatrix}\begin{pmatrix} V''_{11} & V''_{12} \\ V''_{21} & V''_{22} \end{pmatrix}\begin{pmatrix} q_1 \\ q_2 \end{pmatrix} ,$$

where the second derivatives $V''_{kl} = \partial^2 V/\partial x_k \partial x_l$ are evaluated at $x_1 = x_2 = x^* = \sqrt{\ell^2 - a^2}$, so that

Note: the calculations needed to obtain this are tedious but straightforward and are left as an exercise.

$$\begin{aligned} V''_{11} &= V''_{22} = k\left(2 - \frac{a^2}{\ell^2} - \frac{\ell}{a}\right) , \\ V''_{12} &= V''_{21} = k\left(\frac{\ell}{a} - 1\right) . \end{aligned}$$

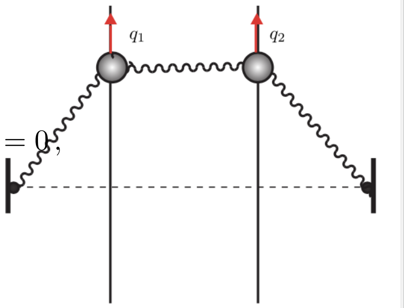
The equations of motion near the extremum of the potential are linear:

$$m\begin{pmatrix}\ddot{q}_1\\ \ddot{q}_2\end{pmatrix} = -\begin{pmatrix}V''_{11} & V''_{12}\\ V''_{21} & V''_{22}\end{pmatrix}\begin{pmatrix}q_1\\ q_2\end{pmatrix}.$$

and can be solved by the substitution

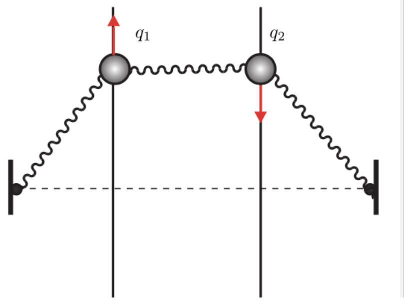
$$\begin{pmatrix}q_1\\ q_2\end{pmatrix} = \Re\begin{pmatrix}a\\ b\end{pmatrix}e^{\mathrm{i}\omega t}, \qquad \begin{pmatrix}\ddot{q}_1\\ \ddot{q}_2\end{pmatrix} = -\omega^2\Re\begin{pmatrix}a\\ b\end{pmatrix}e^{\mathrm{i}\omega t}$$

leading to the condition on the frequencies



$$\begin{vmatrix}\frac{1}{m}V''_{11}-\omega^2 & \frac{1}{m}V''_{12}\\ \frac{1}{m}V''_{21} & \frac{1}{m}V''_{22}-\omega^2\end{vmatrix} = \begin{vmatrix}\frac{k}{m}\left(2-\frac{a^2}{\ell^2}-\frac{\ell}{a}\right)-\omega^2 & \frac{k}{m}\left(\frac{\ell}{a}-1\right)\\ \frac{k}{m}\left(\frac{\ell}{a}-1\right) & \frac{k}{m}\left(2-\frac{a^2}{\ell^2}-\frac{\ell}{a}\right)-\omega^2\end{vmatrix} = 0,$$

which gives



$$\omega^2 - \frac{k}{m}\left(2-\frac{a^2}{\ell^2}-\frac{\ell}{a}\right) = \pm\frac{k}{m}\left(\frac{\ell}{a}-1\right) \quad\Rightarrow\quad \omega^2_{1,2} = \begin{cases}\frac{k}{m}\left(1-\frac{a^2}{\ell^2}\right)\\ \frac{k}{m}\left(3-\frac{2\ell}{a}-\frac{a^2}{\ell^2}\right)\end{cases}$$

By substituting the values of $\omega_{1,2}$ into the matrix we obtain the normal modes:

$\omega_1^2 = \frac{k}{m}\left(1-\frac{a^2}{\ell^2}\right) > 0$

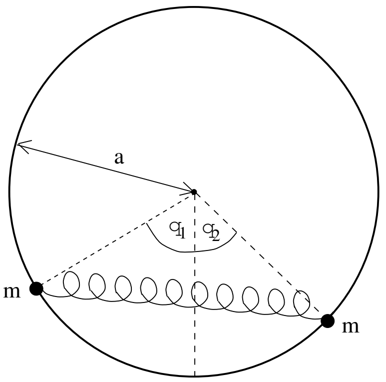
$$\begin{pmatrix}\frac{k}{m}\left(1-\frac{\ell}{a}\right) & \frac{k}{m}\left(\frac{\ell}{a}-1\right)\\ \frac{k}{m}\left(\frac{\ell}{a}-1\right) & \frac{k}{m}\left(1-\frac{\ell}{a}\right)\end{pmatrix}\begin{pmatrix}q_1^0\\ q_2^0\end{pmatrix} = 0 \quad\Rightarrow\quad \begin{pmatrix}x_1^0\\ x_2^0\end{pmatrix} = \text{const}\begin{pmatrix}1\\ 1\end{pmatrix}$$

$\omega_2^2 = \frac{k}{m}\left(3-\frac{2\ell}{a}-\frac{a^2}{\ell^2}\right)$

$$\begin{pmatrix}\frac{k}{m}\left(\frac{\ell}{a}-1\right) & \frac{k}{m}\left(\frac{\ell}{a}-1\right)\\ \frac{k}{m}\left(\frac{\ell}{a}-1\right) & \frac{k}{m}\left(\frac{\ell}{a}-1\right)\end{pmatrix}\begin{pmatrix}q_1^0\\ q_2^0\end{pmatrix} = 0 \quad\Rightarrow\quad \begin{pmatrix}x_1^0\\ x_2^0\end{pmatrix} = \text{const}\begin{pmatrix}1\\ -1\end{pmatrix}$$

It can be seen that $\omega_2^2 \leq 0$ since it is clearly negative for large ℓ/a and reaches zero for $\ell = a$. Therefore the second normal mode is unstable. Thus the equilibrium for $x_1 = x_2 = \sqrt{\ell^2 - a^2}$ is neither a minimum nor a maximum but rather a saddle point with one stable and one unstable direction.

7.2 Normal modes: Two beads on a ring connected by a spring



In the previous example, the equilibrium position was obvious, which made calculations relatively straightforward. Consider now a situation where this position should be first determined: we have two beads of mass m constrained to move on a vertical loop of radius a under the influence of gravity. They are also connected by a spring of natural length a and spring constant $k = mg/a$. The goal is to find small oscillations near an - as yet unknown - equilibrium. One starts, as always, with determining the Lagrangian.

Firstly, the kinetic energy for a particle moving in the plane in terms of polar coordinates (r, θ) is

$$T = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\theta}^2\right).$$

If $r = a$ is constant, this becomes $T = \frac{1}{2}ma^2\dot{\theta}^2$, so that the kinetic energy for our system is

$$T(\dot{\theta}_1, \dot{\theta}_2) = \frac{1}{2}ma^2\left(\dot{\theta}_1^2 + \dot{\theta}_2^2\right).$$

Secondly, the potential energy has two components. The first is due to gravity and has the form

$$V_{\text{grav}} = -mga\cos\theta_1 - mga\cos\theta_2$$

since the vertical distance from the centre of the hoop is $a\cos\theta$ in the downward direction. The second component is due to the spring, and has the form $\frac{1}{2}k(x - x_0)^2$ where k is the spring constant, x is the spring length, and x_0 its natural length. For our problem, $k = mg/a$, $x_0 = a$, and $x = 2a\sin\left(\frac{\theta_1+\theta_2}{2}\right)$. It follows that

To determine x, we note that the spring forms the third side of an isosceles triangle with equal sides a and internal angle $\theta_1 + \theta_2$.

$$V_{\text{spring}} = \frac{1}{2}mga\left[2\sin\left(\frac{\theta_1+\theta_2}{2}\right) - 1\right]^2$$

Putting this all together, the Lagrangian, $\mathcal{L} = T - V$, is given by

$$\mathcal{L} = \frac{1}{2}ma^2(\dot{\theta}_1^2 + \dot{\theta}_2^2) + mga\left(\cos\theta_1 + \cos\theta_2\right) - \frac{1}{2}\left[2\sin\left(\frac{\theta_1+\theta_2}{2}\right) - 1\right]^2.$$

The condition for static equilibrium is that $V(\theta_1, \theta_2)$ is stationary. For our system this leads to the equations (omitting for now mga)

$$\begin{aligned} 0 = \frac{\partial V}{\partial \theta_1} &= \sin\theta_1 + \cos\left(\frac{\theta_1+\theta_2}{2}\right)\left[2\sin\left(\frac{\theta_1+\theta_2}{2}\right) - 1\right] \\ &= \sin\theta_1 + \sin\left(\theta_1+\theta_2\right) - \cos\left(\frac{\theta_1+\theta_2}{2}\right) \end{aligned}$$

and

$$\begin{aligned} 0 = \frac{\partial V}{\partial \theta_2} &= \sin\theta_2 + \cos\left(\frac{\theta_1+\theta_2}{2}\right)\left[2\sin\left(\frac{\theta_1+\theta_2}{2}\right) - 1\right] \\ &= \sin\theta_2 + \sin(\theta_1+\theta_2) - \cos\left(\frac{\theta_1+\theta_2}{2}\right) \end{aligned}$$

As the 2nd terms in these two equations coincide, it follows that the solution must obey $\sin\theta_1 = \sin\theta_2$. This is possible in two cases:

1. $\theta_1 = \theta_2 \equiv \theta_0$ where θ_0 is the equilibrium angle
2. $\theta_1 + \theta_2 = \pi$.

Let us consider these in turn:

- ***The 1st case,*** $\theta_1 = \theta_2 \equiv \theta_0$: Then both equations reduce to

$$\sin\theta_0 + 2\sin\theta_0\cos\theta_0 - \cos\theta_0 = 0\,.$$

Note that $(\sin\theta_0 - \cos\theta_0)^2 = 1 - 2\sin\theta_0\cos\theta_0$.

Introducing $\lambda \equiv \sin\theta_0 - \cos\theta_0$, we have to solve a quadratic equation:

$$\lambda^2 + \lambda - 1 = 0 \qquad \to \qquad \lambda = -1 \pm \frac{\sqrt{5}}{2}.$$

Throwing away a wrong solution, one finds $\sin 2\theta_0 = 1 - \lambda^2 = (\sqrt{5}-1)/2 \approx 0.618$. This gives $2\theta_0 \approx 38.2^o$. For further calculations, we will need to know

$$\sin 2\theta_0 \approx 0.618, \quad \cos 2\theta_0 \approx 0.786, \quad \sin\theta_0 \approx 0.327, \quad \cos\theta_0 \approx 0.945\,.$$

Now we can formally Taylor-expand the potential near the equilibrium point $\theta_1 = \theta_2 = \theta_0$ with respect to small deviations from equilibrium which we denote $\alpha_1 \equiv \theta_1 - \theta_0$ and $\alpha_2 \equiv \theta_2 - \theta_0$:

$$V(\theta_1,\theta_2) - V(\theta_0,\theta_0) = \frac{1}{2}\left[\left.\frac{\partial^2 V}{\partial\theta_1^2}\right|_{\substack{\theta_1=\theta_0\\ \theta_2=\theta_0}} \alpha_1^2 + \left.\frac{\partial^2 V}{\partial\theta_2^2}\right|_{\substack{\theta_1=\theta_0\\ \theta_2=\theta_0}} \alpha_2^2 + 2\left.\frac{\partial^2 V}{\partial\theta_1\,\partial\theta_2}\right|_{\substack{\theta_1=\theta_0\\ \theta_2=\theta_0}} \alpha_1\alpha_2\right]$$

We therefore need to calculate the 2nd derivatives around the point of equilibrium, which are:

$$\frac{\partial^2 V}{\partial\theta_1^2} = \cos\theta_1 + \cos(\theta_1+\theta_2) + \frac{1}{2}\sin\left(\frac{\theta_1+\theta_2}{2}\right) \qquad \Rightarrow \qquad \underbrace{\cos\theta_0}_{\equiv A} + \underbrace{\cos 2\theta_0 + \tfrac{1}{2}\sin\theta_0}_{\equiv B} \equiv A + B$$

$$\frac{\partial^2 V}{\partial\theta_1\,\partial\theta_2} = \cos(\theta_1+\theta_2) + \frac{1}{2}\sin\left(\frac{\theta_1+\theta_2}{2}\right) \qquad \Rightarrow \qquad \cos 2\theta_0 + \frac{1}{2}\sin\theta_0 \equiv B$$

Numerically, $B \approx 0.945$ and $A \approx 0.950$. Now the Lagrangian around the equilibrium becomes

$$\mathcal{L} = \frac{1}{2}m\left\{(\dot{\alpha}_1^2 + \dot{\alpha}_2^2) - \omega_0^2\left[(A+B)(\alpha_1^2+\alpha_2^2) + 2B\alpha_1\alpha_2\right]\right\}, \qquad \omega_0 \equiv \sqrt{\frac{g}{a}}$$

and Lagrange's equations are

$$\left.\begin{aligned}\ddot{\alpha}_1 &= \omega_0^2\left[(A+B)\alpha_1 + B\alpha_2\right]\\ \ddot{\alpha}_2 &= \omega_0^2\left[B\alpha_1 + (A+B)\alpha_2\right]\end{aligned}\right\} \Leftrightarrow \begin{pmatrix}\ddot{\alpha}_1\\ \ddot{\alpha}_2\end{pmatrix} = \omega_0^2\begin{pmatrix}A+B & B\\ B & A+B\end{pmatrix}\begin{pmatrix}\alpha_1\\ \alpha_2\end{pmatrix}$$

Now we do exactly as in the previous problem, looking for the solution in oscillatory form:

$$\begin{pmatrix}\alpha_1(t)\\ \alpha_2(t)\end{pmatrix} = \begin{pmatrix}\alpha_1\\ \alpha_2\end{pmatrix}\cos(\omega t + \phi) \quad \Rightarrow \quad \begin{pmatrix}\ddot{\alpha}_1(t)\\ \ddot{\alpha}_2(t)\end{pmatrix} = -\omega^2\begin{pmatrix}\alpha_1(t)\\ \alpha_2(t)\end{pmatrix}$$

The matrix equation to solve is straightforward:

$$\begin{pmatrix}A+B-\omega^2/\omega_0^2 & B\\ B & A+B-\omega^2/\omega_0^2\end{pmatrix}\begin{pmatrix}\alpha_1\\ \alpha_2\end{pmatrix} = 0\,.$$

The characteristic equations ($\det M = 0$) gives

$$\omega_1^2 = \omega_0^2(A+2B) \quad \Rightarrow \quad \begin{pmatrix}\alpha_1\\ \alpha_2\end{pmatrix}_1 = C_1\begin{pmatrix}1\\ -1\end{pmatrix}\cos(\omega_1 t + \phi_1)$$

$$\omega_2^2 = \omega_0^2 A \quad \Rightarrow \quad \begin{pmatrix}\alpha_1\\ \alpha_2\end{pmatrix}_2 = C_2\begin{pmatrix}1\\ -1\end{pmatrix}\cos(\omega_1 t + \phi_1)\,.$$

This solution is stable: the beads are ***oscillating*** near the equilibrium position.

- ***The 2nd case,***

. In this case, $\cos[(\theta_1+\theta_2)/2] = \cos(\pi/2) = 0$, so that the equilibrium condition is simply $\sin\theta_1 = \sin\theta_2 = 0$, which gives for the present case $\theta_1 = 0$ and $\theta_2 = \pi$ – obviously, this is an equilibrium, with the beads being along the vertical axis. It is also obvious that such an equilibrium is unstable. Any small perturbation will drive this system out of equilibrium. In general, whether the equilibrium is stable or not is not that obvious, so we should have general rules, as we shall now discuss.

7.3 Conditions for Stability

If the normal frequencies are real, then the system ***oscillates*** around the equilibrium which is, therefore, stable. If there is instead a complex (or purely imaginary) frequency, then this indicates an exponentially increasing deviation from equilibrium, which is, therefore, unstable. So, to establish stability, one first must find the normal frequencies; if all of these are real, then the equilibrium position is stable.

For example, for $\omega = \beta + i\gamma$, after some time the oscillation

$$\cos\omega t = \frac{1}{2}\left[\mathrm{e}^{i\beta}\,\mathrm{e}^{-\gamma t} + \mathrm{e}^{-i\beta}\,\mathrm{e}^{\gamma t}\right] \rightarrow \frac{1}{2}\mathrm{e}^{-i\beta}\,\mathrm{e}^{\gamma t}$$

which is exponentially growing and corresponds to unstable motion.

Let's find "frequencies" near the manifestly unstable equilibrium found in the previous example (corresponding to the vertical position of the spring), i.e. with $\theta_1 = 0$ and $\theta_2 = \pi$, to see how the instability reveals itself. We again need to expand the potential near the equilibrium. The corresponding expansion coefficients are as found above, except now we have to take into account that the 2nd derivatives with respect to θ_1 and θ_2 are now different. We then find:

To show this, just plug the values of θ_1, θ_2 into the previously calculated second derivatives.

$$\left.\frac{\partial^2 V}{\partial\theta_1^2}\right|_{\substack{\theta_1=0\\ \theta_2=\pi}} = \frac{1}{2}\,, \quad \left.\frac{\partial^2 V}{\partial\theta_2^2}\right|_{\substack{\theta_1=0\\ \theta_2=\pi}} = -\frac{3}{2}\,, \quad \left.\frac{\partial^2 V}{\partial\theta_1\,\partial\theta_2}\right|_{\substack{\theta_1=0\\ \theta_2=\pi}} = -\frac{1}{2}\,.$$

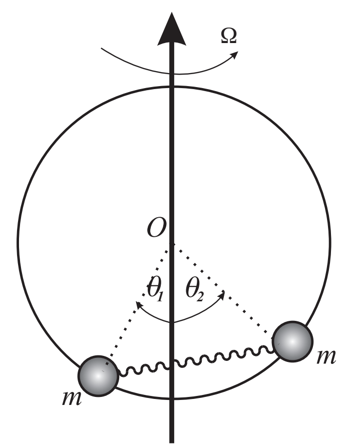
In contrast to the case with 1 d.o.f., the fact that some of the 2nd derivatives are negative does not mean anything – equilibrium still might have been stable. Instead to determine stability, one needs to find the frequencies. The characteristic equations in this case becomes (with $\lambda \equiv \omega^2/\omega_0^2$)

$$\begin{vmatrix} \frac{1}{2}-\lambda & -\frac{1}{2} \\ -\frac{1}{2} & -\frac{3}{2}-\lambda \end{vmatrix} = 0 \quad\Rightarrow\quad \lambda^2+\lambda-1=0 \quad\Rightarrow\quad \lambda = \frac{-1\pm\sqrt{2}}{2}\,.$$

The negative root corresponds to the imaginary frequency which, indeed, means the instability.

7.4 Normal modes: effective potential

Now, let us consider an example which is not as straightforward, in which we have an effective potential. This enters when constraints are time-dependent (usually, some rotation is involved). For our example, we shall consider the previous problem but now in the absence of gravity (e.g. on board of a space station). Instead, we assume that the loop is rotating around the vertical axis at a constant angular velocity Ω. The elastic potential energy remains the same (while the gravitational energy disappears). However, we do not assume any special value for the elastic strength k. In the kinetic energy, there appears an additional (centrifugal) term, as the beads are rotating together with the loop. The linear velocity of rotation is $r\Omega$, where



the radius of rotation $r = a\sin\theta$. Substituting this for both beads, we obtain the Lagrangian:

$$\begin{aligned}\mathcal{L} =& \frac{1}{2}ma^2\left[(\dot\theta_1^2+\dot\theta_2^2)+\Omega^2\left(\sin^2\theta_1+\sin^2\theta_2\right)\right]\\ &-\frac{1}{2}ka^2\left[2\sin\left(\theta_1+\frac{\theta_2}{2}\right)-1\right]^2.\end{aligned}$$

Introducing $\omega_0 \equiv \sqrt{\frac{k}{m}}$, we can identify the effective potential as usual as

$$\begin{aligned}V^{\text{eff}} &= \frac{1}{2}ma^2\left\{\omega_0^2\left[2\sin\left(\theta_1+\frac{\theta_2}{2}\right)-1\right]^2-\Omega^2\left(\sin^2\theta_1+\sin^2\theta_2\right)\right\}\\ &\equiv \frac{ma^2\omega_0^2}{2}\left\{\left[2\sin\left(\theta_1+\frac{\theta_2}{2}\right)-1\right]^2-\lambda\left(\sin^2\theta_1+\sin^2\theta_2\right)\right\},\end{aligned}$$

Note that non-symmetric equilibria also exist as we will see later.

where $\lambda \equiv \Omega^2/\omega_0^2$. Let's first investigate symmetric equilibria: $\theta_1 = \theta_2 \equiv \theta_0$. In this case we write only one equilibrium condition, say $\frac{\partial V}{\partial\theta_1} = 0$, and substitute in the above equality (the 2nd condition after the same substitution will be identical). This condition (now omitting the overall factor $ma^2\omega_0^2/2$) is

$$\begin{aligned}\frac{\partial V^{\text{eff}}}{\partial\theta_1} =& 2\left[\sin(\theta_1+\theta_2)-\cos\frac{\theta_1+\theta_2}{2}\right]-\lambda\sin 2\theta_1\\ &\mapsto 2\left(\sin 2\theta_0-\cos\theta_0\right)-\lambda\sin 2\theta_0 = 0.\end{aligned} \tag{7.1}$$

As $\sin 2\theta_0 = 2\sin\theta_0\cos\theta_0$, the above equation has a trivial solution

$$\cos\theta_0 = 0 \qquad \Rightarrow \qquad \theta_1=\theta_2=\theta_0=\frac{\pi}{2}.$$ ***1st equilibrium***

Its meaning is obvious: both beads have the largest possible rotation radius, and thus the rotation contribution to V^{eff} (coming with the negative sign!) is minimal. The elastic energy is also minimal, as we have chosen $\ell = a$, so that the strings are at their natural length.

A non-trivial solution is found from the $V_\theta^{'} = 0$ equation above after cancellation of $\cos\theta_0$:

$$2(2\sin\theta_0-1)-2\lambda\sin\theta_0 = 0 \quad \Rightarrow \quad \sin\theta_0 = \frac{1}{2-\lambda}.$$ ***2nd equilibrium***

From $0 < \sin\theta \le 1$ follows that $\lambda \le 1$, *i.e.*$\Omega \le \omega_0$. In the opposite case of a very fast rotation the 2nd equilibrium does not exist.

Now we can evaluate normal modes around both equilibria and determine whether these equilibria are stable or not. First, we find the 2nd derivatives of V^{eff} (again, omitting the overall factor $ma^2/2$):

$$
\begin{aligned}
\frac{\partial^2 V}{\partial\theta_2^2} &= 2\left[\cos\left(\theta_1+\theta_2\right)+\frac{1}{2}\sin\left(\theta_1+\frac{\theta_2}{2}\right)\right]-2\lambda\cos 2\theta_1 \\
&\mapsto 2\left[\cos 2\theta_0+\frac{1}{2}\sin\theta_0\right]-2\lambda\cos 2\theta_0 \equiv A+B \\
\frac{\partial^2 V}{\partial\theta_1\,\partial\theta_2} &= 2\left[\cos\left(\theta_1+\theta_2\right)+\frac{1}{2}\sin\left(\theta_1+\frac{\theta_2}{2}\right)\right]\mapsto 2\left[\cos 2\theta_0+\frac{1}{2}\sin\theta_0\right]\equiv B
\end{aligned}
$$

These derivatives – in terms of θ_0 – are the same for both equilibria. Therefore, Taylor's expansion and the resulting Lagrangian and Lagrange's equations are the same – in terms of A and B – for both equilibria. Moreover, Lagrange's equations are practically the same (apart from slightly different notations) as in the previous example of beads on the loop in the field of gravity. Therefore, the characteristic equation is the same as for the previous problem (and, therefore, normal modes are also the same):

$$
\begin{pmatrix} A+B-\omega^2/\omega_0^2 & B \\ B & A+B-\omega^2/\omega_0^2 \end{pmatrix}\begin{pmatrix}\alpha_1\\ \alpha_2\end{pmatrix}=0
$$

$$
\Rightarrow \frac{\omega_{1,2}^2}{\omega_0^2}=\begin{cases} A & =-2\lambda\cos 2\theta_0 \\ A+2B & =2\left(2-\lambda\right)\cos 2\theta_0+2\sin\theta_0\,. \end{cases}
$$

The stability conditions depend on the value of θ_0 and are different for the two equilibria.

1*st *equilibrium $\qquad \theta_0=\frac{\pi}{2}: \qquad \frac{\omega_{1,2}^2}{\omega_0^2}=\begin{cases} 2\lambda \\ 2\left(\lambda-1\right)\end{cases}$

Here ω_2 is real **only** when $\lambda>1 \quad\Leftrightarrow\quad \Omega>\omega_0$ – precisely the condition of the absence of the 2nd equilibrium. When $\Omega<\omega_0$, then ω_2 is imaginary, and this equilibrium becomes unstable.

2*nd *equilibrium $\quad \sin\theta_0=\frac{1}{2-\lambda}: \quad \frac{\omega_{1,2}^2}{\omega_0^2}=\begin{cases} -2\lambda\cos 2\theta_0 \\ 2\left(\frac{\cos 2\theta_0}{\sin\theta_0}+\sin\theta_0\right)=2\frac{\cos^2\theta_0}{\sin\theta_0}>0 \end{cases}$

Note immediately that if θ_0 is a solution, so is $\pi-\theta_0$. This reflects the symmetry with respect to the "horizontal" axis (*i.e.*the axis perpendicular to the rotation one): such two solutions are fully equivalent.

Stability: to investigate stability, one must find whether normal frequencies are real or not. In the present case, ω_2 is always real. However, $\omega_1^2>0$ only when $\cos 2\theta_0<0$, *i.e.*for $\theta_0>\pi/4$. It is easy to see that $\sin\theta_0\geq\frac{1}{2}\Rightarrow\theta_0\geq\pi/6$. Therefore, for $\Omega<\omega_0$ and $\pi/6<\theta_0<\pi/4$, the 2nd equilibrium does exists, but turns out to be unstable.

Since $\cos 2\theta_0 = 1 - 2\sin^2\theta_0$, on substituting here $\sin\theta_0 = 1/(2-\lambda)$, one finds that the equilibrium becomes unstable at $\lambda < 2 - \sqrt{2}$. But the trivial one is unstable for all $\lambda < 1$. So, when both equilibria are unstable, where does the system go?

7.5 Spontaneous breaking of symmetry

By asymmetric extremal points, we mean ones that do not obey $\theta_1 = \theta_2$.

Before answering this, we note that we have actually missed asymmetric extremal points; however, this turns out not to be important. Indeed, looking at the previously calculated second derivatives of V, one sees that the condition for having an extremum is $\sin 2\theta_1 = \sin 2\theta_2$, from which it follows that either $\theta_1 = \theta_2$ (i.e. the case already considered), or $\theta_1 + \theta_2 = \pi/2$, or, finally, for $\sin 2\theta_1 = \sin 2\theta_2 = 0$, $\theta_1 = 0$ and $\theta_2 = \pi$. The last one is clearly unstable. However, less obviously, we find the solution with $\theta_1 = \pi/2 - \theta_2$ is also unstable. This solution is found from Eq. (7.1) where we see that $\sin 2\theta_1 = (2-\sqrt{2})/\lambda$, which exists in the region $\lambda < 2 - \sqrt{2}$.

To show that this solution is unstable, one can solve an appropriate characteristic equation.

We have now investigated all stationary points, but the question remains: what does the system do for a relatively slow rotation, $\lambda < 2 - \sqrt{2}$ where the equilibria are unstable? Using a one-dimensional analogy, we can see that a potential can have a minimum not only where its 1st derivative vanish, but also at the edges, *i.e.*at values of parameters beyond which one cannot get (for this or that reason). In fact, that is the case here. In the absence of rotation, $\lambda = 0$, the system can be in ***neutral*** equilibrium: the spring is unstretched, *i.e.*$\theta_1 + \theta_2 = 60^o$, but the value of θ_1 is arbitrary. When the rotation is very slow, the centrifugal energy is still unable to compete with the elastic one; however, the $\theta_1 + \theta_2 = 60^o$ equilibrium is no longer neutral.

Neutral equilibrium means that after being slightly displaced, the system stays in its new position instead of returning to where it started or moving farther away.

Physical insight: maybe a boundary state with the constraint that $\theta_1 + \theta_2 = 60^o$ (where V_{elastic} is minimal) could be the lowest energetic state? One still needs to find the minimal possible centrifugal energy, subject to this constraint. It is easy to see that this minimum is reached for $\theta_1 = 2\pi/3$ and $\theta_2 = -\pi/3$ (with angles measured in the ***opposite*** directions as in the picture). This gives symmetric positions of the two masses with respect to the "horizontal" axis (*i.e.*perpendicular to the axis of rotation): each is at an arc of $\pi/6$ off this axis. The value of potential at this point,

$$V_0 = -\frac{3}{2}\lambda$$

should be compared to that at the minima. Firstly, for the 1st equilibrium, we have $V_1 = 1 - 2\lambda$, which is lower than V_0 for $\lambda > 2$. Therefore while this does exist as a stable equilibrium for $\lambda > 1$, it is only actually ***metastable*** unless $\lambda > 2$. Secondly, for the 2nd equilibrium, we have $V_2 = -\lambda/(2-\lambda)$, which becomes lower than V_0 only for $\lambda > 4/3$, where it simply does not exist.Therefore, the 2^{nd} equilibrium is ***metastable*** at $2-\sqrt{2} < \lambda < 1$, and totally unstable at $\lambda < 2-\sqrt{2}$ when the system – with any initial conditions – would fall into the "boundary minimum" V_0.

7.5.1 General rules for small oscillations

- Write down $\mathcal{L}$ and identify V^{eff} where relevant, *i.e.*where T includes parts not proportional to $\dot{q}^2$ or cyclic variables exist. In the latter case, calculate the appropriate conserved generalised momenta and express the appropriate $\dot{q}$ in terms of the conserved quantities.

- Find extremal points of V^{eff} by solving the system of n equations ($n = 2$ in the case below):

$$\frac{\mathrm{d}V^{\text{eff}}}{\mathrm{d}q_i} = 0\,,\ i = 1\dots n \Rightarrow \text{gives solutions for the extrema}$$

- Calculate the 2^{nd} derivatives of the potential ***at each equilibrium point*** and thus Taylor-expand your potential ***around the equilibrium***, q_1^0, q_2^0 (repeat this for each equilibrium):

$$V^{\text{eff}}(q_1, q_2) - V_0^{\text{eff}} = \frac{1}{2}\left\{a_{11}\,(\delta q_1)^2 + 2a_{12}\delta q_1\,\delta q_2 + a_{22}\,(\delta q_2)^2\right\}$$

where $\delta q_i \equiv q_i - q_i^0$,

$$a_{11} = \left.\frac{\partial^2 V(q_1,q_2)}{\partial q_1^2}\right|_{\substack{q_1 = q_1^0\\ q_2 = q_2^0}} \qquad a_{22} = \left.\frac{\partial^2 V(q_1,q_2)}{\partial q_2^2}\right|_{\substack{q_1 = q_1^0\\ q_2 = q_2^0}}$$

$$a_{12} = a_{21} = \left.\frac{\partial^2 V(q_1,q_2)}{\partial q_1\,\partial q_2}\right|_{\substack{q_1 = q_1^0\\ q_2 = q_2^0}}$$

The generalisation to higher dimensions is straightforward.

- Thus reduce Lagrange's equations to the following matrix differential equation to be solved:

$$\mathsf{M}\begin{pmatrix}\ddot{q}_1\\ \ddot{q}_2\end{pmatrix} + \mathsf{A}\begin{pmatrix}q_1\\ q_2\end{pmatrix} = 0 \qquad \text{where}$$

$$\mathsf{A} = \begin{pmatrix}a_{11} & a_{12}\\ a_{21} & a_{22}\end{pmatrix} \qquad \text{and usually} \qquad \mathsf{M} = \begin{pmatrix}m_1 & 0\\ 0 & m_2\end{pmatrix}$$

M is usually diagonal as above, but not always – see the last problem in Problem Sheet 2. The solution is always reduced by the harmonic substitution

$$\begin{pmatrix} q_1(t) \\ q_2(t) \end{pmatrix} = \begin{pmatrix} q_1 \\ q_2 \end{pmatrix} \cos(\omega t + \phi) \quad \Rightarrow$$

$$\begin{pmatrix} \ddot{q}_1(t) \\ \ddot{q}_2(t) \end{pmatrix} = -\omega^2 \begin{pmatrix} q_1 \\ q_2 \end{pmatrix} \cos(\omega t + \phi)$$

to solving the ***characteristic equation***:

$$\det\left[-\mathsf{M}\omega^2 + \mathsf{A}\right] = 0$$

- The solution gives directly the eigenfrequencies (normal frequencies). If they are all real, the appropriate equilibrium is stable. If any contains an imaginary part, the equilibrium is unstable. A zero frequency is a sign of a neutral equilibrium.

8 Calculus of Variations leading to Hamilton's Principle

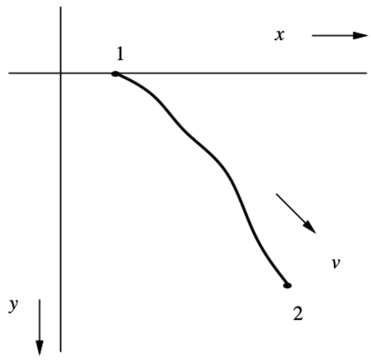
8.1 Calculus of variations

The derivation of Lagrange's equations earlier in the course was somewhat inelegant and long-winded, and was based on the original Newton laws and d'Alembert virtual work principle. Historically, it is right to put Newton's laws at the foundation of classical mechanics. But logically, one could look for a "better" principle to start with. An excellent historical example is the Fermat principle in optics: the ray of light propagates in deflecting media in such a way that the propagation time is the shortest. It took a while before a similar principle was formulated for mechanics by Hamilton. The Hamilton (or minimal action) principle will be considered next week.

Aside: One of numerous advantages of Hamilton's approach to mechanics is in its natural connection with quantum mechanics, or general theory of relativity, quantum field theory and other areas of modern theoretical physics.

However we must first construct the necessary mathematical framework: namely, the *calculus of variations.* We will define this theory in a general way as the subject is of great utility outside the realm of mechanics.

8.1.1 The brachistochrone problem



The brachistochrone problem lead to the founding of the calculus of variations and is the following: imagine a bead sliding down a wire between two given points, under the influence of gravity; what is the form of the wire that minimises the time taken on the descent? John Bernoulli proposed this problem as a challenge to the mathematicians of Europe in 1696, allowing six months for its solution. The two Bernoullis, Newton, Leibnitz, and de l'Hôpital all found the solution; Newton apparently solved the problem overnight after hearing about it. We will follow the later approach of Euler and Lagrange.

If v is the speed along the curve then the time taken over an interval of arc-length $\mathrm{d}s$ of the wire is $\mathrm{d}s/v$ and hence the time taken overall is the integral of this:

$$t_{12} = \int_1^2 \frac{\mathrm{d}s}{v}$$

We may deduce the velocity by using energy conservation, measuring the vertical displacement, y, from the initial point of release (choosing the axis so that $y > 0$):

$$\frac{1}{2}mv^2 = mgy \quad \Rightarrow \quad v = \sqrt{2gy}$$

Then using the fact that the differential of the arc-length, ds is:

$$\mathrm{d}s = \sqrt{dx^2 + dy^2} = \mathrm{d}x\sqrt{1 + y^{'2}}$$

where prime indicates differentiation with respect to x, then the expression for the time taken becomes:

$$t_{12} = \int_1^2 \frac{\mathrm{d}s}{v} = \frac{1}{\sqrt{2g}} \int_1^2 \frac{\sqrt{1 + y^{'2}}}{\sqrt{y}} \, \mathrm{d}x \tag{8.1}$$

and we must minimise this expression, with respect to the 'shape of the wire', $y(x)$.

The first peculiar feature of this problem is that the left hand side of Eq. (8.1) does not contain the variable x. So although t_{12} depends on $y(x)$, it is *not* a function of a certain variable, but an entity termed a *functional* which maps a *function* ($y(x)$ in this case) onto a *number* (t_{12} in this case). To denote this relation we will use the notation $t_{12}[y(x)]$ (or $t_{12}[y]$ in brief), and in general $F[\phi]$ where F is a functional of the function ϕ.

Note: A common mathematical notation is $y(x) \overset{F}{\longmapsto} t_{12}$.

Secondly the minimisation that we must perform is on a grander scale than the minimisation of a function of a single variable, $f(x)$, as we must minimise $t_{12}[y(x)]$ with respect to each of the values $y(x)$ for all x. This leads us to an answer that is a curve (or function) rather than just one value of x in the minimisation of $f(x)$.

It was not until the middle of the eighteenth century that Euler and Lagrange formulated the general theory behind the calculus of variations. We will now see how the minimising function, $y(x)$ in the above example, satisfies a differential equation, the Euler-Lagrange equation. Presently we will see that Lagrange's equations are an example of such equations and determine the *functional* which, upon minimisation, yields those equations.

8.2 Euler-Lagrange equations

Let us consider the general problem of minimising a functional $F[y]$ with respect to $y(x)$, and assume that the expression for F has the form:

$$F[y] = \int_{x_1}^{x_2} f(y, y^{'}, x) \, \mathrm{d}x \tag{8.2}$$

where we take the endpoints, x_1 and x_2, to be given, so that $y(x_1) = y_1$ and $y(x_2) = y_2$.

We investigate the minimisation of Eq. (8.2) by analogy with ordinary calculus, by writing $y(x, \varepsilon) = y(x, 0) + \varepsilon h(x)$ where $y(x, 0)$ is the minimising function. We pick

$$h(x_1) = 0 \quad \text{and} \quad h(x_2) = 0 \tag{8.3}$$

Then substituting into Eq. (8.2)

$$F[y](\varepsilon) = \int_{x_1}^{x_2} f(y(x,\varepsilon), y^{'}(x,\varepsilon), x)\ \mathrm{d}x \tag{8.4}$$

and taking $\partial/\partial\varepsilon$ of the resulting expression. The condition for the minimum is the familiar one:

$$\left(\frac{\mathrm{d}F}{\mathrm{d}\varepsilon}\right)_{\varepsilon=0} = 0.$$

Using the chain rule, we may express $\mathrm{d}F/\,\mathrm{d}\varepsilon$ as an integral:

$$\frac{\mathrm{d}F}{\mathrm{d}\varepsilon} = \int_{x_1}^{x_2} \left(\frac{\partial f}{\partial y}\frac{\partial y}{\partial \varepsilon} + \frac{\partial f}{\partial y^{'}}\frac{\partial y^{'}}{\partial \varepsilon}\right)\mathrm{d}x$$

The second term in this integral may be fruitfully re-expressed using integration by parts:

$$\begin{aligned}\int_{x_1}^{x_2} \frac{\partial f}{\partial y^{'}}\frac{\partial y^{'}}{\partial \varepsilon}\ \mathrm{d}x &= \int_{x_1}^{x_2} \frac{\partial f}{\partial y^{'}}\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial y}{\partial \varepsilon}\right)\mathrm{d}x \\ &= \left.\frac{\partial f}{\partial y^{'}}\frac{\partial y}{\partial \varepsilon}\right|_{x_1}^{x_2} - \int_{x_1}^{x_2} \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial f}{\partial y^{'}}\right)\frac{\partial y}{\partial \varepsilon}\, dx\end{aligned}$$

Note that Eq. (8.3) means that $y(x,\varepsilon)$ does not depend on ε at $x = x_1$ and $x = x_2$. Thus the first term on the right hand side vanishes and we have:

$$\frac{\mathrm{d}F}{\mathrm{d}\varepsilon} = \int_{x_1}^{x_2} \left(\frac{\partial f}{\partial y} - \frac{\mathrm{d}}{\mathrm{d}x}\frac{\partial f}{\partial y^{'}}\right)\frac{\partial y}{\partial \varepsilon}\ \mathrm{d}x = \int_{x_1}^{x_2} \left(\frac{\partial f}{\partial y} - \frac{\mathrm{d}}{\mathrm{d}x}\frac{\partial f}{\partial y^{'}}\right) h(x)\ \mathrm{d}x$$

Now since we want this expression to vanish for *arbitrary* $h(x)$, *i.e.*an arbitrary deviation from the minimising path, we require that the coefficient of $h(x)$ vanishes:

$$\frac{\partial f}{\partial y} - \frac{\mathrm{d}}{\mathrm{d}x}\frac{\partial f}{\partial y^{'}} = 0 \tag{8.5}$$

This Eq. (8.5) is the Euler-Lagrange equation associated with the minimisation of the functional $F[y]$.

The analogy to Lagrangian mechanics for systems with one degree of freedom is obvious. We've got the following set of identifications:

$$\begin{array}{lcl} f(y, y^{'}, x) & \longleftrightarrow & \mathcal{L}(q, \dot{q}, t) \\ y & \longleftrightarrow & q \\ x & \longleftrightarrow & t \end{array} \tag{8.6}$$

We'll come to this set later again. For now, let us note that anything already proved for $\mathcal{L}(q,\dot{q};t)$ must be also valid – in the appropriate terms – for $f(y,y^{'};x)$. Thus, we come to the existence of the first integrals of the Euler-Lagrange equation.

8.2.1 First integrals of the Euler-Lagrange equations.

We have learned that for $\mathcal{L}$ not explicitly depending on time, *i.e.*$\partial\mathcal{L}/\partial t = 0$, the Hamiltonian $\mathcal{H}$ is conserved. By analogy, we formulate the appropriate conservation law for the function f obeying the Euler-Lagrange equation when it does not explicitly depend on x, *i.e.*$\partial f/\partial x = 0$. Therefore, we come to the identification less trivial than in Eq. (8.6) above:

$$\mathcal{H} = \dot{q}\frac{\partial\mathcal{L}}{\partial\dot{q}} - \mathcal{L}(q,\dot{q}) = \text{const} \quad \longleftrightarrow \quad \mathcal{J} = y^{'}\frac{\partial f}{\partial y^{'}} - f(y,y^{'}) = \text{const}\,. \quad (8.7)$$

Both the equations above should be understood in the sense that $\mathcal{H}$ (or $\mathcal{J}$) is constant when calculated with $q\,,\dot{q}$ (or y, $y^{'}$) which satisfy the Euler-Lagrange equations. In many cases, using Eq. (8.7) will considerably ease finding a solution to Eq. (8.5).

8.3 Solution to the brachistochrone problem

Let's use Eq. (8.7) to solve the brachistochrone problem. In this case,

$$f(y,y^{'};x) = \frac{\sqrt{1+y^{'2}}}{\sqrt{y}}\,, \qquad\qquad \frac{\partial f}{\partial x} = 0\,,$$

so that one has

$$\begin{aligned}\mathcal{J} &= y^{'}\frac{\partial}{\partial y^{'}}\left(\frac{\sqrt{1+y^{'2}}}{\sqrt{y}}\right) - \frac{\sqrt{1+y^{'2}}}{\sqrt{y}} \\ &= \frac{y^{'2}}{\sqrt{y}\sqrt{1+y^{'2}}} - \frac{\sqrt{1+y^{'2}}}{\sqrt{y}} = -\frac{1}{\sqrt{y}\sqrt{1+y^{'2}}}\,.\end{aligned}$$

Renaming a negative constant $\mathcal{J} \equiv -1/c$, one obtains after raising to the 2^{nd} power and some re-arranging:

$$1+y^{'2} = \frac{c^2}{y} \quad \Rightarrow \quad y^{'} = \pm\frac{\sqrt{c^2-y}}{\sqrt{y}} \quad \Rightarrow \quad x = \int\frac{\sqrt{y}\,\mathrm{d}y}{\sqrt{c^2-y}}\,.$$

The integration is performed with the help of the substitution $y = c^2\cos^2\theta \quad \Rightarrow \quad \mathrm{d}y = -c^2 2\cos\theta\sin\theta$ which yields

$$x = \pm c^2\int\cos^2\theta\,\mathrm{d}\theta = \pm c^2\int(1+\cos 2\theta)\,\mathrm{d}\theta = \pm c^2\left(\theta + \tfrac{1}{2}\sin 2\theta\right) + x_0\,.$$

The constants of integration, c and x_0 are defined by the boundary conditions, $y(x_1) = 0$ and $y(x_2) = y_2$. The solution is found in *parametric form*, which is especially useful when x and y cannot be explicitly found either in form $x = \phi(y)$ or in form $y = \varphi(x)$, but they are absolutely uniquely defined in terms of the parameter θ:

$$\begin{aligned} y &= c^2 \cos^2\theta, \\ x &= \pm c^2\left(\theta + \tfrac{1}{2}\sin 2\theta\right) + x_0\,. \end{aligned}$$

For this problem, x can be actually expressed via y explicitly (but not vice versa). However, such an expression would be neither more transparent nor more useful than that above.

8.4 Geodesic lines

One of the important applications of calculus is finding the so called geodesics (or geodesic lines), *i.e.*the shortest (or the optimal) path on a curved surface. It is not simply analogous to the brachistochrone problem. In a much deeper way, the motion in the field of gravity can be reduced to finding the shortest path in the "curved" space-time. For illustrative purpose, we consider a straightforward problem of finding geodesics on a sphere.

Of course we know the answer from the very start: the geodesic lines on a sphere are great circles.

8.4.1 Geodesics on a spherical surface

One always starts with defining the element of length in proper coordinates, knowing that in Cartesian coordinates

$$\mathrm{d}s^2 = \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2\,.$$

As usual,

$$\mathbf{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = r\begin{pmatrix} \sin\theta\cos\phi \\ \sin\theta\sin\phi \\ \cos\theta \end{pmatrix}$$

$$\Rightarrow \quad \begin{aligned} \mathrm{d}x &= \mathrm{d}r\sin\theta\cos\phi + r(\cos\theta\cos\phi\,\mathrm{d}\theta - \sin\theta\sin\phi\,\mathrm{d}\phi) \\ \mathrm{d}y &= \mathrm{d}r\sin\theta\sin\phi + r(\cos\theta\sin\phi\,\mathrm{d}\theta + \sin\theta\cos\phi\,\mathrm{d}\phi) \\ \mathrm{d}z &= \mathrm{d}r\cos\theta - r\sin\theta\,\mathrm{d}\theta \end{aligned}$$

Since $r = \mathrm{const}$, *i.e.*$\mathrm{d}r = 0$, one finds that

$$\mathrm{d}s^2 = \mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\phi^2 \quad \Rightarrow \quad \mathrm{d}s = \mathrm{d}\phi\sqrt{\theta'^2 + \sin^2\theta}\,, \qquad \theta' \equiv \frac{\mathrm{d}\theta}{\mathrm{d}\phi}\,.$$

In other words, we are minimizing the total length of the path on the sphere.

Therefore, the functional to minimise is

$$S[\theta] = \int_{\phi_1}^{\phi_2} \sqrt{\theta'^2 + \sin^2\theta}\,\mathrm{d}\phi \equiv \int_{\phi_1}^{\phi_2} f(\theta, \theta'; \phi)\,\mathrm{d}\phi\,.$$

Since f does not explicitly depends on ϕ, there exists the following 'conserved quantity' (first integral):

$$\mathcal{I} = \alpha = \theta' \frac{\partial f}{\partial \theta'} - f(\theta, \theta') = \frac{\theta'^2}{\sqrt{\theta'^2 + \sin^2\theta}} - \sqrt{\theta'^2 + \sin^2\theta}$$

$$= -\frac{\sin^2\theta}{\sqrt{\theta'^2 + \sin^2\theta}} \equiv -\frac{1}{a}\,.$$

Solving for the derivative, one finds

$$\frac{\mathrm{d}\theta}{\mathrm{d}\phi} = \pm\sin\theta\sqrt{a^2\sin^2\theta - 1} \quad \Rightarrow \quad \phi = \pm\int \frac{\mathrm{d}\theta}{\sin\theta\sqrt{a^2\sin^2\theta - 1}}\,.$$

The substitution $t = \cot\theta \;\Rightarrow\; \mathrm{d}\theta = -\mathrm{d}t/(t^2+1)$ and $\sin\theta = (t^2+1)^{-1/2}$ reduces the integral above to

$$\phi = \mp\int \frac{\mathrm{d}t}{\sqrt{(a^2-1) - t^2}} = \mp\arcsin\frac{t}{\sqrt{a^2-1}} + \phi_0\,.$$

Renaming $\sqrt{a^2-1} \to b$ and taking sin of both parts, one obtains

$$b\sin(\phi - \phi_0) = \cot\theta = t\,,$$

with b and ϕ_0 being the integration constants. We can convert this expression to Cartesians in order to interpret it, by multiplying through by $r\sin\theta$ and expanding $\sin(\phi-\phi_0)$ to get

$$(b\cos\phi_0)r\sin\theta\sin\phi - (b\sin\phi_0)r\sin\theta\cos\phi = r\cos\theta$$

Denoting $b\cos\phi_0 = A$ and $b\sin\phi_0 = B$ we see:

$$A(r\sin\theta\sin\phi) - B(r\sin\theta\cos\phi) = r\cos\theta \;\Leftrightarrow\; Ay - Bx = z$$

i.e. the equation of a plane going through the origin (which is the centre of our sphere). The plane intersects the sphere in 2 circular arcs - these are the great circles.

8.5 Hamilton's Principle

We have shown that the minimisation of a functional

$$F[y] = \int_{x_1}^{x_2} f(y, y^{'}, x)\,\mathrm{d}x$$

resulted in the Euler-Lagrange equation for $f(y, y^{'}; x)$. Without any further derivation, we can immediately conclude that the appropriate result in a mechanical context, *i.e.*the standard Lagrange's equation, would follow from the minimisation of the following functional:

$$S[q] = \int_{t_1}^{t_2} \mathcal{L}(q, \dot{q}; t)\,\mathrm{d}t\,. \tag{8.8}$$

The functional $S[q]$ is called ***action***. As Lagrange's equation follow from its minimisation, the entire foundation of mechanics could be ***the principle of least action***: this is also known as ***Hamilton's Principle***.

The variational approach paves the way for the most transparent formulation of quantum mechanics, due to Feynman, called the *path integral* formulation. There the 'particle' behaves like a wave in the sense that it propagates along *all* of the paths, not just the one that minimises the action. However the different paths are weighted by a complex exponential which is the action associated with the path:

$$\psi(q, t) \propto \sum_{\text{paths} \quad q(t)} e^{iS[q]/\hbar}\psi(q^{'}, t = 0)$$

where $\psi(q, t)$ is the *wave function* of the particle at time t. The sum over the paths runs over all paths connecting $q^{'}$ at $t = 0$ to q at time t. The attractive feature about this is the description of the 'classical' limit, in which Planck's constant, $\hbar$, vanishes. Then the phase in the exponential varies very rapidly from path to path, but varies most slowly in the vicinity of the path which minimises the action—the classical path. Thus the main contribution comes from that path, where the cancellation is least severe. So the only contribution is from the path that the particle would move along obeying classical mechanics.

8.5.1 General functional with many degrees of freedom

This is naturally generalised to a system with many degrees of freedom, for which the action would be

$$S[\{q_i\}] = \int_{t_1}^{t_2} \mathcal{L}(q_1, \dot{q}_1 \ldots q_n, \dot{q}_n; t)\,\mathrm{d}t\,. \tag{8.9}$$

Before going any further, we need to generalise the Euler-Lagrange equations to a functional that depends on many degrees of freedom. This is as straightforward as for Lagrange's equations in mechanics, since different degrees of freedom are mutually independent. A general task is to find a curve in the $n+1$ dimensional space spanned by $y_1, \ldots, y_n, x$ that minimises a functional

$$F[\mathbf{y}(x)] = \int_{x_1}^{x_2} f(y_1, y_1' \ldots, y_n, y_n'; x)\,\mathrm{d}x\,.$$

A step-by-step repetition of the procedure in Section 8.1 leads to a set of n Euler-Lagrange equations:

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial f}{\partial y_i'}\right) - \frac{\partial f}{\partial y_i} = 0 \tag{8.10}$$

Another clear analogy here with Lagrange's equations in mechanics: if f does not explicitly depends on y_i (*i.e.*the latter is an ignorable, or cyclic, variable), than there is a "conservation law":

$$p_i \equiv \frac{\partial f}{\partial y_i'} = \text{const}\,,\ \ ie \quad \text{does not depend on } x\ \ .$$

8.6 Relativistic Dynamics

We may use a variational principle to derive the equations of motion for a relativistic particle. The relativity principle can be formulated in a geometric way as the statement of the invariance of the ***interval*** in 4-dimensional space-time. The infinitesimal interval is defined as

$$\mathrm{d}s^2 = c^2\,\mathrm{d}t^2 - \mathrm{d}x^2 - \mathrm{d}y^2 - \mathrm{d}z^2 \equiv c^2\,\mathrm{d}t^2 - \mathrm{d}\mathbf{r}^2\,,$$

where c is the speed of light. The invariance means that it remains the same in all inertial frames. But it contains also all necessary knowledge for motion of a free particle in a given frame in space-time. A particle moves along ***geodesic lines*** that optimise the interval between given points.

The finite interval can be written down as

$$S[\mathbf{r}] = \int \mathrm{d}s = c\int_{t_1}^{t_2} \sqrt{1 - (\dot{\mathbf{r}}/c)^2}\ \mathrm{d}t\,.$$

An overall factor attached to a functional to be optimised is never relevant. We choose then to change $S \mapsto S^{\text{Rel}} \equiv -m_0 c S$ where m_0 is the particle's rest mass. Thus we arrive at the functional

$$S^{\text{Rel}}[\mathbf{r}] = -m_0 c^2 \int_{t_1}^{t_2} \sqrt{1 - (\dot{\mathbf{r}}/c)^2}\ \mathrm{d}t \equiv \int_{t_1}^{t_2} \mathcal{L}^{\text{Rel}}(\dot{\mathbf{r}})\ \mathrm{d}t\,.$$

Note that the relativistic Lagrangian

$$\mathcal{L}^{\mathrm{Rel}}(\dot{\mathbf{r}}) \equiv -m_0 c^2 \sqrt{1-(\dot{\mathbf{r}}/c)^2}$$

does not explicitly depends only on time, but also on the coordinates $\mathbf{r}$. This means firstly that $\mathbf{r} = (x,\, y,\, z)$ is an ignorable coordinate(s), so that the corresponding generalised momentum is conserved:

$$\mathbf{p} = \frac{\partial \mathcal{L}}{\partial \dot{\mathbf{r}}} = -m_0 c^2 \frac{\mathrm{d}\sqrt{1-(\dot{\mathbf{r}}/c)^2}}{\mathrm{d}\dot{\mathbf{r}}} = \frac{m_0 \dot{\mathbf{r}}}{\sqrt{1-(\dot{\mathbf{r}}/c)^2}} \equiv \frac{m_0 \mathbf{v}}{\sqrt{1-(v/c)^2}} \,.$$

But we also have a conserved “Hamiltonian”:

$$\mathcal{H}^{\mathrm{Rel}} = \mathbf{p}\cdot\dot{\mathbf{r}} - \mathcal{L}^{\mathrm{Rel}}(\dot{\mathbf{r}}) = \frac{m_0 \mathbf{v}^2}{\sqrt{1-(v/c)^2}} + m_0 c^2 \sqrt{1-(v/c)^2} = \frac{m_0 c^2}{\sqrt{1-(v/c)^2}} \,.$$

Thus, as the conserved Hamiltonian should be equal to the full energy $\mathcal{E}$ (if dimensions are right - that is why we multiplied the action by $m_0 c$), one arrives at the most famous formula in the world:

$$\mathcal{E} = mc^2 \quad \text{where} \quad m \equiv \frac{m}{\sqrt{1-\dfrac{v^2}{c^2}}} \quad \text{is the total mass .}$$

For non-relativistic (small) v’s, Taylor’s expansion gives – with the accuracy up to an undetectable, although huge, constant $m_0 c^2$ – the standard Newtonian expression for the kinetic energy.

9 Hamilton's Equations

9.1 Conservation Laws

We have seen many times how the existence of the conservation laws (i.e. first integrals) help to solve problems. We have encountered two kinds of these:

1. $\mathcal{H}$ is conserved when $\partial\mathcal{L}/\partial t = 0$. This is absolutely general and does not depend on the choice of coordinates.

For example, for motion in the field of a central force, one immediately sees cyclic variables in polar coordinates but not in the Cartesian ones.

2. If any q_i is ignorable, then $p_i \equiv \partial\mathcal{L}/\partial q_i = \text{const}$. This much more depends on our lucky – or unlucky – choice of coordinates.

Hamiltonian dynamics – in contrast to Lagrangian – incorporates conserving quantities in a natural way from the start. The main reason is that the Hamiltonian $\mathcal{H}$ is "naturally" expressed in terms of generalised coordinates and momenta (rather than velocities), and thus the existence of cyclic coordinates is immediately taken into account. All one needs to do is to formulate equations of motion in terms of $\mathcal{H}$ rather than of $\mathcal{L}$, as we shall now do.

Note that the Hamiltonian formulation of classical mechanics is closest to the usual formulation of quantum mechanics, in the form of Schrödinger's equation with operators corresponding to physical observables. Conserved quantities are very important in quantum mechanics where the quantum numbers associated with conserved quantities label the states of the quantum system. The Hamilton-Jacobi equation, which is the culmination of the course, is remarkably close to the Schrödinger equation - indeed it is identical to the form of the Schrödinger equation in the limit that Planck's constant is imagined to approach zero.

9.2 Hamilton's equations

As usual we will consider initially the simple case with only one generalised coordinate. Firstly let us recall the definition of the conjugate momentum to q and of the Hamiltonian:

$$p = \frac{\partial\mathcal{L}}{\partial\dot{q}}\,, \qquad \mathcal{H} = \dot{q}p - \mathcal{L}\,. \tag{9.1}$$

Here $\mathcal{H}$ remains a function of all $q, \dot{q}, p$, and we need to change variables formally. We have done this before for particular cases, but we shall now do it in general.

To see how to do this, note the following: A function of, say, x and y is always Taylor-expanded in terms of $\mathrm{d}x$, $\mathrm{d}y$, and higher differentials of x and y:

$$\mathrm{d}f(x,y) = \frac{\partial f}{\partial x}\,\mathrm{d}x + \frac{\partial f}{\partial y}\,\mathrm{d}y\,.$$

The Lagrangian $\mathcal{L}(q,\dot{q},t)$ can similarly be expanded in terms of differentials of $q, \dot{q}$ and t:

$$\mathrm{d}\mathcal{L} = \frac{\partial \mathcal{L}}{\partial q}\,\mathrm{d}q + \frac{\partial \mathcal{L}}{\partial \dot{q}}\,\mathrm{d}\dot{q} + \frac{\partial \mathcal{L}}{\partial t}\,\mathrm{d}t = \frac{\partial \mathcal{L}}{\partial q}\,\mathrm{d}q + p\,\mathrm{d}\dot{q} + \frac{\partial \mathcal{L}}{\partial t}\,\mathrm{d}t$$

The Hamiltonian, $\mathcal{H}(q_i, p_i;\, t)$, can also be expanded in terms of differentials of q, p and t as:

$$\mathrm{d}\mathcal{H}(q_i, p_i;\, t) \equiv \frac{\partial \mathcal{H}}{\partial p}\,\mathrm{d}p + \frac{\partial \mathcal{H}}{\partial q}\,\mathrm{d}q + \frac{\partial \mathcal{H}}{\partial t}\,\mathrm{d}t \tag{9.2}$$

On the other hand, using Eq. (9.1), one obtains

$$\begin{aligned}\mathrm{d}\mathcal{H} &= \mathrm{d}(p\dot{q}) - \mathrm{d}\mathcal{L} = \dot{q}\,\mathrm{d}p + p\,\mathrm{d}\dot{q} - \left(\frac{\partial \mathcal{L}}{\partial q}\,\mathrm{d}q + p\,\mathrm{d}\dot{q} + \frac{\partial \mathcal{L}}{\partial t}\,\mathrm{d}t\right)\\ &= \dot{q}\,\mathrm{d}p - \frac{\partial \mathcal{L}}{\partial q}\,\mathrm{d}q - \frac{\partial \mathcal{L}}{\partial t}\,\mathrm{d}t = \dot{q}\,\mathrm{d}p - \dot{p}\,\mathrm{d}q - \frac{\partial \mathcal{L}}{\partial t}\,\mathrm{d}t,\end{aligned} \tag{9.3}$$

where the last equality follows from the definition of p, Eq. (9.1), and Lagrange's equation in the form

$$\frac{\partial \mathcal{L}}{\partial q} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right) = \dot{p}\,.$$

Comparing Eqs. (9.2) and (9.3), one obtains ***Hamilton's equations***:

$$\dot{p} = -\frac{\partial \mathcal{H}}{\partial q}, \qquad \dot{q} = \frac{\partial \mathcal{H}}{\partial p} \tag{9.4}$$

Reminders:

- $\mathcal{H} = \mathcal{E} = T + V$ if and only if there are no time-dependent constraints on motion
- When such constraints exist, $\mathcal{H} = T^{\text{eff}} + V^{\text{eff}}$, where T^{eff} is the quadratic in $\dot{q}_i$ (or p_i) part of the kinetic energy, while its p independent part is included into V^{eff} with the opposite sign.

When $\mathcal{L}$ depends on time, one additional equation is

$$\frac{\partial \mathcal{H}}{\partial t} = -\frac{\partial \mathcal{L}}{\partial t}\,.$$

Naturally, a generalisation for a system with many degrees of freedom is straightforward. As we have already used many times,

$$\mathcal{H} = \sum_i p_i\,\dot{q}_i - \mathcal{L}\,,$$

so that Hamilton's equations become

$$\dot{p}_i = -\frac{\partial \mathcal{H}}{\partial q_i}, \qquad \dot{q}_i = \frac{\partial \mathcal{H}}{\partial p_i} \qquad i = 1, \dots n \tag{9.5}$$

In other words, the $2n$ Hamilton's (1st order) differential equations are fully equivalent to the n Lagrange's (2nd order) differential equations.

Important!

9.3 Variational Principle for the Hamiltonian

We have derived Lagrange's equations from the least action principle:

$$\delta S = \delta \int_{t_1}^{t_2} \mathcal{L}(q_i, \dot{q}_i;\, t)\, \mathrm{d}t$$

Let's now vary the action expressed in term of $\mathcal{H}$:

$$\delta S = \delta \int_{t_1}^{t_2} \left[\sum_i p_i \dot{q}_i - \mathcal{H}(q_i, p_i;\, t) \right] \mathrm{d}t \equiv \delta \int_{t_1}^{t_2} f(q_i, \dot{q}_i, p_i, \dot{p}_i; t)\, \mathrm{d}t$$

The Euler-Lagrange equation for the function of $4n$ variables (omitting the indices i):

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial f}{\partial \dot{q}}\right) = \frac{\partial f}{\partial q} \quad \Rightarrow \quad \dot{p} = -\frac{\partial \mathcal{H}}{\partial q}$$
$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial f}{\partial \dot{p}}\right) = \frac{\partial f}{\partial p} \quad \Rightarrow \quad 0 = \dot{q} - \frac{\partial \mathcal{H}}{\partial p}$$

As expected, Hamilton's equations also follow from the variational principle. Clearly, $\mathcal{H}$ is also not uniquely defined: any full derivative can be added to the action. That explains how different Hamilton's equations (obtained from Hamiltonians related by canonical transformations) can describe the same system.

9.4 Examples

Let us now revisit some of the classic examples we have used repeatedly during the course so far to see how Hamilton's equations can lead us to the equations of motion.

- **Particle in 1D Potential**

  The Lagrangian

  $$\mathcal{L} = \frac{1}{2} m \dot{q}^2 - V(q)\,.$$

  Lagrange's equation:

  $$\frac{\partial \mathcal{L}}{\partial q} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right) \quad \Rightarrow \quad p = \frac{\partial \mathcal{L}}{\partial \dot{q}} = m\dot{q}.$$

  The Hamiltonian:

  $$\mathcal{H} = p\dot{q} - \mathcal{L} = \frac{1}{2} m \dot{q}^2 + V(q) = \frac{p^2}{2m} + V(q)\,.$$

  Hamilton's equations:

  $$\dot{q} = \frac{\partial \mathcal{H}}{\partial p} = \frac{p}{m}, \quad \dot{p} = -\frac{\partial \mathcal{H}}{\partial q} = -\frac{\partial V}{\partial q} \quad \Rightarrow \quad \ddot{q} = \frac{\dot{p}}{m} = -\frac{1}{m}\frac{\partial V}{\partial q}\,.$$

- **Particle in Central Potential**

  The Lagrangian (for motion in the equatorial plane, $\theta = \text{const}$):

$$\mathcal{L} = \frac{1}{2}m\dot{\mathbf{r}}^2 - V(r) = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\phi}^2\right) - V(r)\,.$$

  Generalised momenta:

$$p_r = \frac{\partial \mathcal{L}}{\partial \dot{r}} = m\dot{r} \qquad\qquad p_\phi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = mr^2\dot{\phi}$$
$$\Rightarrow \quad \dot{r} = \frac{p_r}{m} \qquad\qquad \dot{\phi} = \frac{p_\phi}{mr^2}$$

  The Hamiltonian

$$\mathcal{H} = p_r\dot{r} + p_\phi\dot{\phi} - \mathcal{L} = \frac{1}{2}m\left(\dot{r}^2 + r^2\dot{\phi}^2\right) + V(r) = T + V = \mathcal{E}$$
$$\Rightarrow \quad \mathcal{H} = \frac{p_r^2}{2m} + \frac{p_\phi^2}{2mr^2} + V(r)$$

  Hamilton's equations:

$$\dot{r} = \frac{\partial \mathcal{H}}{\partial p_r} = \frac{p_r}{m} \qquad\qquad \dot{p}_r = -\frac{\partial \mathcal{H}}{\partial r} = \frac{p_\phi^2}{mr^3} - \frac{\partial V}{\partial r}$$
$$\dot{\phi} = \frac{\partial \mathcal{H}}{\partial p_\phi} = \frac{p_\phi}{mr^2} \qquad\qquad \dot{p}_\phi = -\frac{\partial \mathcal{H}}{\partial \phi} = 0\,.$$

  To convert Hamilton's equations to the Lagrange equations of motion, one takes the second derivative of q with respect to t and substitutes the equation for $\dot{p}$, unless the appropriate coordinate is ignorable – then one just uses the appropriate conservation law:

$$\ddot{r} = \frac{\dot{p}_r}{m} = \frac{1}{r^3}\left(\frac{p_\phi}{m}\right)^2 - \frac{1}{m}\frac{\partial V}{\partial r} = r\dot{\phi}^2 - \frac{1}{m}\frac{\partial V}{\partial r}$$
$$p_\phi = mr^2\dot{\phi} = \text{const}$$

- **Bead on Rotating Spoke**

$$\mathcal{L} = \frac{1}{2}m(\dot{r}^2 + r^2\omega^2\sin^2\theta - gr\cos\theta) \quad\Rightarrow\quad p_r = \frac{\partial \mathcal{L}}{\partial \dot{r}} = m\dot{r}$$
$$\Rightarrow \quad \mathcal{H} = \underbrace{\dot{r}\frac{\partial \mathcal{L}}{\partial \dot{r}}}_{= m\dot{r}^2} - \mathcal{L} = \frac{1}{2}m(\dot{r}^2 - r^2\omega^2\sin^2\theta + gr\cos\theta) \neq \mathcal{E}$$

$$\Rightarrow \quad \mathcal{H} = \frac{p^2}{2m} + \frac{1}{2}m\left(gr\cos\theta - r^2\omega^2\sin^2\theta\right) = T^{\text{eff}} + V^{\text{eff}}$$

  Here $\mathcal{H}$ is still conserved, but $\mathcal{E}$ is not: rotating the spoke requires some work done on the system.

- **Relativistic Motion**

  We have already considered the relativistic Lagrangian

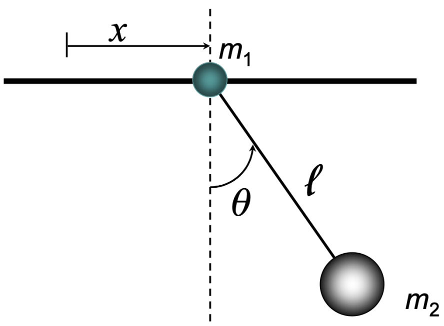
$$\mathcal{L}^{\text{Rel}}(\dot{\mathbf{r}}) \equiv -m_0 c^2 \sqrt{1-(\dot{\mathbf{r}}/c)^2}\,, \qquad \mathbf{p} = \frac{\partial \mathcal{L}}{\partial \dot{\mathbf{r}}} = \frac{m_0 \mathbf{v}}{\sqrt{1-(v/c)^2}}\,.$$

$$\mathcal{H} = \mathbf{p}\cdot\dot{\mathbf{r}} - \mathcal{L} = \frac{m_0 c^2}{\sqrt{1-(v/c)^2}}\,.$$

  However, $\mathcal{H}$ here is not in natural variables, which are $\mathbf{p}$ and $\mathbf{r}$. One needs to reverse the equation for $\mathbf{p}$ to express $\mathbf{v}$ (or more precisely, γ) in terms of $\mathbf{p}$:

$$p^2 + m_0^2 c^2 = \frac{m_0^2 v^2}{1-(v/c)^2} + m_0^2 c^2 = \frac{m_0^2 c^2}{1-(v/c)^2}$$
$$\Rightarrow \quad \mathcal{H} = c\sqrt{p^2 + m_0^2 c^2}\,.$$

- **Sliding pendulum**



  Consider a bob of mass m_1 moving without friction along a rod. Another bob of mass m_2 is attached to the first one by a weightless wire. We have calculated the Lagrangian for this problem previously as:

$$\mathcal{L} = \frac{1}{2}m_1\dot{x}^2 + \frac{1}{2}m_2\left(\dot{x}^2 + \ell^2\dot{\theta}^2 + 2\ell\dot{x}\dot{\theta}\cos\theta\right) + m_2 g\ell\cos\theta\,.$$

  Now, the generalised momenta are

$$p_x = \frac{\partial \mathcal{L}}{\partial \dot{x}} = (m_1+m_2)\dot{x} + m_2\ell\dot{\theta}\cos\theta$$
$$p_\theta = \frac{\partial \mathcal{L}}{\partial \dot{\theta}} = m_2\ell\dot{x}\cos\theta + m_2\ell^2\dot{\theta},$$

  and the Hamiltonian is,

$$\begin{aligned}\mathcal{H} &= p_x\dot{x} + p_\theta\dot{\theta} - \mathcal{L} = T + V \\ &= \frac{1}{2}m_1\dot{x}^2 + \frac{1}{2}m_2\left(\dot{x}^2 + \ell^2\dot{\theta}^2 + 2\ell\dot{x}\dot{\theta}\cos\theta\right) - m_2 g\ell\cos\theta\,.\end{aligned}$$

  However, we still need to express the generalised velocities appearing here in terms of the generalised momenta. To finish the problem, we therefore should substitute

$$\dot{x} = \frac{p_x}{m_1} - \frac{p_\theta}{m_1\ell\cos\theta}, \qquad \ell\dot{\theta} = \frac{p_\theta(m_1+m_2)}{m_1 m_2} - \frac{p_x\cos\theta}{m_1\ell}$$

into the expression for $\mathcal{H}$ above. This example shows that finding $\mathcal{H}$ can be much more taxing than finding $\mathcal{L}$, since it is not always easy to write down the kinetic energy directly in terms of the generalised momenta. However, now we will go through a much more general approach to finding convenient generalised coordinates and momenta.

9.5 Legendre Transformation

The main idea in introducing the Hamiltonian instead of the Lagrangian is to change variables from $\{q, \dot{q}\}$ to $\{q, p\}$. Therefore, the full set of actions in deriving $\mathcal{H}$ from $\mathcal{L}$ is the following

Aside: you will also come across Legendre transformations e.g. in statistical physics where they are used to replace one natural variable with its conjugate variable (for example, energy with temperature) in a thermodynamic potential, yielding a new potential that is more appropriate for a given experimental situation.

1. Find out all the generalised momenta from $\mathcal{L}$:
$$p_i \equiv \frac{\partial \mathcal{L}}{\partial q_i} \qquad i = 1 \dots n\,.$$
2. Use the above to express all the generalised velocities $\dot{q}_i$ in terms of the generalised coordinates and momenta
3. Write down the Hamiltonian, $\mathcal{H} = p_i \dot{q}_i - \mathcal{L}$
4. Substitute all the generalised velocities found in step (b) into the expression (c) for $\mathcal{H}$ thus expressing it in its ***natural***, or ***canonical*** variables, q_i and p_i

The change of variable of this sort is an example of more general, Legendre transformations.

9.6 Canonical Transformations

The main – yet to be exploited – advantage of Hamiltonian mechanics is our ability to choose generalised momenta and coordinates with a great degree of flexibility. However, to appreciate this, we must first show that for any particular problem either $\mathcal{L}$ or $\mathcal{H}$ can be chosen in a great variety of ways.

9.6.1 Non-uniqueness of the Lagrangian

Consider

$$\widetilde{\mathcal{L}}(q, \dot{q}; t) = \mathcal{L}(q, \dot{q}; t) + \frac{\mathrm{d}F(q;t)}{\mathrm{d}t} \tag{9.6}$$

where F is an arbitrary differentiable function which does not depend on $\dot{q}$. Then, $\widetilde{\mathcal{L}}$ and $\mathcal{L}$ describe identical mechanical systems, *i.e.*they result in the same Lagrange's equation.

Note that we cannot add a full derivative of a function depending on $\dot{q}$: this would with necessity contain $\ddot{q}$ violating our condition that $\mathcal{L}$ depends only on generalised coordinates and velocities (but not accelerations).

This can be verified directly, using the "dot cancellation" rule we introduce previously, which is valid for any $\dot{q}$-independent function :

Note it was proved for $\mathbf{r}(q,t)$ but, of course, remains valid for any function of these variables

$$\frac{\partial F}{\partial q} = \frac{\partial \dot{F}}{\partial \dot{q}}\,.$$

Then

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \widetilde{\mathcal{L}}}{\partial \dot{q}} - \frac{\partial \widetilde{\mathcal{L}}}{\partial q} = \underbrace{\frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \mathcal{L}}{\partial \dot{q}} - \frac{\partial \mathcal{L}}{\partial q}}_{=0} + \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial \dot{F}}{\partial \dot{q}} - \frac{\partial \dot{F}}{\partial q} = \frac{\mathrm{d}}{\mathrm{d}t}\frac{\partial F}{\partial q} - \frac{\partial}{\partial q}\frac{\mathrm{d}F}{\mathrm{d}t} = 0\,,$$

the latter is true as the order of differentiation for any "good" function can be changed . Note that this is even more straightforward to see in the derivation from the least action principle: adding a full derivative to the action does not change the result of variation, thus leading to the same Euler-Lagrange equations.

Note this has already been used in proving the dots cancellation previously.

Simple Example: For a $1d$ system with a t-independent $\mathcal{L}$, one can simply add to $\mathcal{L}$ any product $\Phi(q)\dot{q}$, without changing Lagrange's equations. A direct calculation shows that $\frac{\mathrm{d}\Phi}{\mathrm{d}t}$ is cancelled with $\frac{\partial \Phi}{\partial q}\dot{q}$.

9.7 Generating Function

The non-uniqueness of $\mathcal{L}$ naturally results in the non-uniqueness of $\mathcal{H}$. However, our aim is not simply to exploit possible changes in $\mathcal{H}$ or $\mathcal{L}$, but to find out how to change coordinates, as our eventual aim is obtain Hamilton's equations in new coordinates.

We start with $\mathcal{L}$ and demonstrate how the function $F(q,t)$ could lead us to generate a set of new coordinates. Let us make F in Eq. (9.6) depend not only on q but on some ***parameter*** Q which will eventually serve as a new generalised coordinate:

$$F(q,t) \quad \longrightarrow \quad F_1(Q,q;t)\,.$$

From the viewpoint of the "old" Lagrange's equation, Q is some parameter, so that

$$\widetilde{\mathcal{L}}(Q,\dot{Q}) \equiv \widetilde{\mathcal{L}}\Big(q(Q,\dot{Q}),\dot{q}(Q,\dot{Q})\Big) = \mathcal{L}(q,\dot{q}) - \frac{\mathrm{d}F_1}{\mathrm{d}t}. \tag{9.7}$$

Note that the Lagrangian $\widetilde{\mathcal{L}}$ is fully equivalent to $\mathcal{L}$ where the old coordinates are replaced by the new. This replacement is easier in Hamilton's language. Substituting $\widetilde{\mathcal{L}} = P\dot{Q} - \widetilde{\mathcal{H}}$ and $\mathcal{L} = p\dot{q} - \mathcal{H}$ into Eq. (9.7), one has

$$P\dot{Q} - \widetilde{H}(P,Q) = p\dot{q} - \mathcal{H}(p,q) - \dot{F}_1\,. \tag{9.8}$$

Now we differentiate the above equation with respect to $\dot{q}$ and $\dot{Q}$, using the ***dot cancellation rule*** and the fact that neither $\mathcal{H}$ nor $\widetilde{\mathcal{H}}$ depend on the velocities:

$$\begin{aligned}
&\text{Apply} \quad \frac{\partial}{\partial \dot{q}} \quad \Rightarrow \quad 0 = p - \underbrace{\frac{\partial \dot{F}_1}{\partial \dot{q}}}_{=\partial F_1/\partial q} \quad \Rightarrow \quad p = \frac{\partial F_1}{\partial q}\,, \\
&\text{Apply} \quad \frac{\partial}{\partial \dot{Q}} \quad \Rightarrow \quad P = -\underbrace{\frac{\partial \dot{F}_1}{\partial \dot{Q}}}_{=\partial F_1/\partial Q} \quad \Rightarrow \quad P = -\frac{\partial F_1}{\partial Q}\,.
\end{aligned} \tag{9.9}$$

These relations define (implicitly) Q and P as functions of q and p and vice versa.

This generates a ***canonical transformations***:

> A transformation from $\mathcal{H}(p,q)$ to $\widetilde{\mathcal{H}}(P,Q)$ is called canonical if Hamilton's equations for $\widetilde{\mathcal{H}}(P,Q)$ have the standard (i.e. ***canonical***) form

In the case considered, this is guaranteed as we have started from a transformation of $\mathcal{L}$ that does not change Lagrange's equations.

To explicitly find $\widetilde{\mathcal{H}}$, one substitutes p and P from Eq. (9.9) into Eq. (9.8) and uses a chain rule for $\frac{\mathrm{d}F_1}{\mathrm{d}t}$ which gives $\dot{F}_1 = p\dot{q} - P\dot{Q} - \frac{\partial F_1}{\partial t}$:

$$\widetilde{\mathcal{H}}(Q,P) = \mathcal{H}\Big(q(Q,P),\, p(Q,P)\Big) + \frac{\partial F_1}{\partial t}\,.$$

Note that in many practical cases F_1 does not depend on t. The most important part here is the change of variables in the arguments of $\mathcal{H}$.

9.8 Examples

1. A trivial but important example: $F_1 = qQ$. This just leads to interchanging the role of p and q in any $\mathcal{H}$. Indeed,

$$\begin{aligned} P &= -\frac{\partial F_1}{\partial Q} = -q \\ p &= \frac{\partial F_1}{\partial q} = \quad Q \end{aligned} \qquad \Rightarrow \qquad \widetilde{\mathcal{H}}(P,Q) = \mathcal{H}(-q,p)\,.$$

   Any differentiable function of q and Q that obeys the condition

$$\frac{\partial^2 F_1}{\partial q\,\partial Q} \neq 0$$

   generates a canonical transformation. However, an arbitrary function would tend to make equations more complicated. The "best" way to find a function which simplifies Hamilton's equations is an educated guess. It's good to have foresight (or insight) but it is usually done by hindsight.

2. The steps are standard for any examples using generating functions. As an involved example, we choose the generating function which is useful for the Hamiltonian of the simple harmonic oscillator (SHO):

$$F_1(q,Q) = \frac{1}{2} q^2 \cot 2\pi Q.$$

   (a) Find expressions for p and P in terms of q and Q.

$$p = \frac{\partial F_1}{\partial q} = q \cot 2\pi Q$$
$$P = -\frac{\partial F_1}{\partial Q} = \frac{\pi q^2}{\sin^2 2\pi Q}$$

   (b) Invert this to find $p(P,Q)$ and $q(P,Q)$ to substitute them into the "old" $\mathcal{H}$. The solution for q is immediate from the 2nd equation, and that for p follows upon the substitution of q into the 1st equation:

$$q = \sqrt{\frac{P}{\pi}} \sin 2\pi Q \qquad \text{(the sign is irrelevant)}$$
$$p = \sqrt{\frac{P}{\pi}} \cos 2\pi Q$$

   Note the inverse relations which define the canonical transformation in this case:

$$\begin{aligned} Q &= \frac{1}{2\pi} \arctan \frac{q}{p} \\ P &= \pi \left(p^2 + q^2\right) \end{aligned} \tag{9.10}$$

   (c) Substitute $p(P,Q)$ and $q(P,Q)$ into $\mathcal{H}$ and thus find $\widetilde{\mathcal{H}}$. Up to now, the transformation was not related to any $\mathcal{H}$ and it could be applied to any $\mathcal{H}(p,q)$. It is particularly useful, though, for the Hamiltonian of SHO with unit mass and frequency, $\mathcal{H} = \frac{1}{2}\left(p^2 + q^2\right)$

$$\begin{aligned} \widetilde{\mathcal{H}} &= \mathcal{H}\left([p(P,Q)]^2 + [q(P,Q)]^2\right) \\ &= \frac{P}{2\pi}\left(\sin^2 2\pi Q + \cos^2 2\pi Q\right) = \frac{P}{2\pi} \end{aligned}$$

   (d) Write down and solve Hamilton's equations for $\widetilde{\mathcal{H}}$.

   We have found that Q is ignorable, so that Hamilton's equations are particularly simple in the case:

$$\dot{Q} = \frac{\partial \widetilde{\mathcal{H}}}{\partial P} = \frac{1}{2\pi} \qquad \Longrightarrow \qquad Q = \frac{t}{2\pi} + Q(0)\,,$$
$$\dot{P} = \frac{\partial \widetilde{H}}{\partial Q} = 0 \qquad \Longrightarrow \qquad P = P(0)$$

(e) Sometimes, one also needs to find the solution in terms of the original variables. Then one should substitute into the (inverse) canonical transformations found in (b). Renaming the constants of integration,

$$Q(0) \equiv \varphi_0 \qquad\qquad \sqrt{\frac{P}{\pi}} \equiv a\,,$$

one finds in the case considered:

$$\begin{aligned} q &= a\sin(t+\varphi_0) \\ p &= a\cos(t+\varphi_0) \end{aligned}$$ as expected for the simple harmonic oscillator.

9.9 Other types of the generating function

The generating function $F_1\,(q,Q,t)$ had both new and old generalised coordinates independently. In general, we can choose a generating function that has one of the old, q, p, and one of the new, Q, P, coordinates which serve as independent variables. Therefore, there are three more types of generating functions. $F_1(q,Q,t)$ is called the generating function of the first type. The other three are

$$F_2(q,P,t) \qquad\qquad F_3(p,Q,t) \qquad\qquad F_4(p,P,t)\,. \tag{9.11}$$

To get each of the three types of generating functions from F_1, we shall use a so-called Legendre transformation to the new coordinates. First, let us write Taylor's expansion for a t-independent generating function of the 1st type, $F_1(q,Q)$:

This is precisely the type of transformation we saw above that is used for changing from $\mathcal{L}$ to $\mathcal{H}$.

$$\mathrm{d}F_1(q,Q) = \frac{\partial F_1}{\partial q}\,\mathrm{d}q + \frac{\partial F_1}{\partial Q}\,\mathrm{d}Q = p\,\mathrm{d}q - P\mathrm{d}Q\,,$$

where we have used the generating equations,

$$P = -\frac{\partial F_1}{\partial Q}\,, \qquad\qquad p = \frac{\partial F_1}{\partial q}\,.$$

Let us now introduce the 2nd type generating function:

$$F_2 = F_1(q,Q) + QP.$$

The role of the QP term here is to interchange the variables Q and P, since

$$\mathrm{d}F_2 = \mathrm{d}F_1 + \mathrm{d}(QP) = p\,\mathrm{d}q - P\mathrm{d}Q + P\mathrm{d}Q + Q\,\mathrm{d}P = p\,\mathrm{d}q + Q\,\mathrm{d}P.$$

The fact that the r.h.s. contains differentials of q and P means that F_2 is a natural function of q and P. We employ Taylor's expansion

$$\mathrm{d}F_2(q,P) = \frac{\partial F_2}{\partial q}\,\mathrm{d}q + \frac{\partial F_2}{\partial P}\,\mathrm{d}P \equiv p\,\mathrm{d}q + Q\,\mathrm{d}P,$$

which gives the generating equations

$$p = \frac{\partial F_2}{\partial q}\,, \qquad\qquad Q = \frac{\partial F_2}{\partial P}\,.$$

These two equations again implicitly define relations between new and old variables.

In general, a generating function should depend on one new and one old variable, which gives two more possibilities:

$$\begin{aligned}
&F_3(p,Q,t) = F_1(q,Q,t) - pq,\\
&F_4(p,P,t) = F_3(p,Q,t) + QP = F_2(q,P,t) - pq,
\end{aligned}$$

which leads to the relation between the new and old variables for both these functions:

$$\begin{aligned}
q &= -\frac{\partial F_3}{\partial p} & \qquad q &= -\frac{\partial F_4}{\partial p}\\
P &= -\frac{\partial F_3}{\partial Q} & \qquad Q &= \frac{\partial F_4}{\partial P}\,.
\end{aligned} \tag{9.12}$$

Any differentiable function F with nonzero mixed derivative with respect to their natural variables can serve as a generating function, but finding a useful one is an art.

Note that a generalisation for many degrees of freedom is straightforward as usual. All one needs is to supply indices i to each of the variables. We still stick to the family of 4 generating functions, not trying different combinations of new and old variables for different values of i, although it is possible in principle.

10 Poission brackets and Hamilton-Jacobi theory

10.1 Poisson Brackets

The 4 types of generating functions discussed above is quite a general way of arriving at canonical transformations. However, one can put a different question: given a set of transformations from the "old" to "new" variables, how can one find whether they are canonical or not?

To solve this problem, a particular mathematical construction, called Poisson brackets, turns out to be the key. For two arbitrary functions that depend on the same set of canonically conjugate variables, $\{p_i, q_i\}$, the Poisson bracket (PB) is defined as

$$\{f, g\} \equiv \sum_i \left(\frac{\partial f}{\partial q_i} \frac{\partial g}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial g}{\partial q_i} \right)$$

10.2 Conserved quantities

The Poisson bracket gives the simplest indication if any particular quantity is conserved. Consider the full time derivative of f depending on a set of canonically conjugate variables:

$$\begin{aligned} \frac{\mathrm{d}f}{\mathrm{d}t} &= \sum_{i=1}^{n} \left(\frac{\partial f}{\partial q_i} \dot{q}_i + \frac{\partial f}{\partial p_i} \dot{p}_i \right) + \frac{\partial f}{\partial t} = \sum_i \left(\frac{\partial f}{\partial q_i} \frac{\partial \mathcal{H}}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial \mathcal{H}}{\partial q_i} \right) + \frac{\partial f}{\partial t} \\ &\equiv \{f, \mathcal{H}\} + \frac{\partial f}{\partial t} , \end{aligned} \tag{10.1}$$

Important: this approach has the advantage that one does not need to solve the equations of motion to determine whether quantities are conserved!

where we have used Hamilton's equations to get rid of derivatives of $\mathcal{H}$. If F is not explicitly time-dependent, so that $\partial f / \partial t = 0$, then whether f is conserved or not is determined by whether the Poisson bracket with the Hamiltonian vanishes.

This is, of course, more general then just noticing that some variable is ignorable and there exists a conservation law. But, of course, it is easy to see that the Poisson bracket with $\mathcal{H}$ of the generalised momentum (p_k) conjugate to cyclic variable (q_k) vanishes. Indeed,

$$\{p_k, \mathcal{H}\} = \sum_i \left(\frac{\partial p_k}{\partial q_i} \frac{\partial \mathcal{H}}{\partial p_i} - \frac{\partial p_k}{\partial p_i} \frac{\partial \mathcal{H}}{\partial q_i} \right) = -\frac{\partial \mathcal{H}}{\partial q_k} = 0 ,$$

Obviously, $\{\mathcal{H}, \mathcal{H}\} = 0$ so that using PB confirms what we have already learn: $\mathcal{H}$ is conserved when it is not explicitly t-dependent.

as expected.

The Poisson bracket also gives us a direct route to quantum mechanics. Let us call $\hat{f}$, $\hat{g}$, *etc*, the QM operators corresponding to the classical quantities f and q. Then there exists the following relation:

$$\hat{f}\hat{g} - \hat{h}\hat{f} \equiv \left[\hat{f}, \hat{g} \right] = i\hbar \left\{ \widehat{f, g} \right\} .$$

Therefore, if the Poisson bracket of two classical quantities equals zero, then the appropriate quantum operators commute. If a classical quantity is the integral of motion, so is its quantum generalisation. As the PB of the conserved quantity (the integral of motion) with $\mathcal{H}$ equals zero, then the operator of the appropriate quantum quantity commutes with the quantum Hamiltonian.

10.3 Properties of Poisson Brackets

Simple properties: the Poisson brackets involving canonical variables:

$$\begin{aligned} \{f, q_k\} &= \sum_i \left(\frac{\partial f}{\partial q_i} \frac{\partial q_k}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial q_k}{\partial q_i} \right) = -\frac{\partial f}{\partial p_k} \\ \{f, p_k\} &= \sum_i \left(\frac{\partial f}{\partial q_i} \frac{\partial p_k}{\partial p_i} - \frac{\partial f}{\partial p_i} \frac{\partial p_k}{\partial q_i} \right) = \frac{\partial f}{\partial q_k} \end{aligned} \tag{10.2}$$

It follows, in particular, that

$$\{q_k, q_\ell\} = 0 \qquad \{p_k, p_\ell\} = 0 \qquad \{q_\ell, p_k\} = -\{p_k, q_\ell\} = \delta_{k\ell}\,. \tag{10.3}$$

The last property is so important that we repeat it in words:

The Poisson bracket of canonically conjugate variables equals 1

Simple algebraic properties (c is a const):

$$\begin{aligned} \{f, c\} &= 0 \\ \{f, g\} &= -\{g, f\} \\ \{f + h, g\} &= \{f, g\} + \{h, g\} \\ \{fh, g\} &= f\{h, g\} + h\{f, g\} \end{aligned}$$

Jacobi's identity:

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0\,.$$

All these properties could be proven by direct calculations but we leave them without a proof.

A very important consequence of the Jacobi identity (that follows if one puts $h \equiv \mathcal{H}$): the Poisson bracket of two conserved quantities generates another conserved quantity:

If f and g are integrals of motion, so is $\{f, g\}$

This quantity is not necessarily new, as it can be a trivial function of the same f and g (or just a constant). However, sometimes this is a way to generate new integrals of motion. Example: angular momentum:

$$\{L_x, L_y\} = \{L_z\}$$

10.3.1 Hamilton's equations via Poisson Brackets

Equations 10.2 allow us to rewrite Poisson equations in a form that is explicitly symmetric with respect to momenta and coordinates. If one substitutes $f = \mathcal{H}$ into Eq. (10.2), one finds using Hamilton's equations that $\dot{q}_k = -\{\mathcal{H}, q_k\}$ and $\dot{p}_k = -\{\mathcal{H}, p_k\}$ so that all the pairs of Hamilton's equations could be written as

$$\dot{\xi}_i = \{\xi_i, \mathcal{H}\} \;, \qquad i = 1 \ldots s \tag{10.4}$$

where ξ_i is either a generalised momentum, or a generalised coordinate. We have seen that one can be canonically transformed into another so that it is useful to have the above form which is identical for both. Eq. (10.4) directly follows also from the general expression for the time derivative, Eq. (10.1).

10.4 Poisson brackets and canonical transformations

A very important property of the Poisson brackets:

> *Poisson brackets remain* ***invariant*** *under the canonical transformations*

It means that the result of calculation of the bracket does not depend on whether one uses old or new variables:

$$\{f, g\}_{p,\,q} = \{f, g\}_{P,\,Q}$$

One can prove this without a calculation but using the following speculation: Suppose, both f and g do not explicitly depend on t. Let us consider g formally as a Hamiltonian of some fictitious system. Then

If they do depend on t, the result would not be changed as the time-derivative does not explicitly enter the Poisson brackets, but plays a role of the parameter of the transformation.

$$\frac{\mathrm{d}f}{\mathrm{d}t} = \{f, g\}$$

The fact that f is a conserved quantity cannot depend on the choice of variables – therefore, it must remain the same when calculated with the help of new variables.

Of course, one can check this by a direct calculation, using the fact that the transformation is canonical, *i.e.*Hamilton's equations describe the system both in new and old variables.

This immediately gives the key to testing whether a given transformation is canonical. Eq. (10.3) is straightforward when the brackets of canonical variables are calculated with respect to themselves; the invariance guarantees that the PB of the new canonical variables calculated with respect to the old ones, or of the old calculated with respect to the new ones must be the same, which gives the mathematical conditions of the transformations being canonical:

$$\{Q_i, P_k\}_{q,p} = \delta_{ik} \qquad \{Q_i, Q_k\}_{q,p} = 0 \qquad \{P_i, P_k\}_{q,p} = 0$$

10.4.1 Examples

1. Let's consider the canonical transformation discussed previously generated by the function

$$F(q,Q)=\frac{1}{2}q^2\cot 2\pi Q$$

We have obtained that transformation as

$$Q=\frac{1}{2\pi}\arctan\frac{q}{p},\qquad P=\pi\left(p^2+q^2\right).$$

As this is a system with one degree of freedom, the conditions $\{Q,Q\}=\{P,P\}=0$ are trivial, and we only need to calculate the single fundamental Poisson bracket:

$$\{Q,P\}=\frac{\partial Q}{\partial q}\frac{\partial P}{\partial p}-\frac{\partial Q}{\partial p}\frac{\partial P}{\partial q}=\frac{1}{2\pi}\left[\frac{1}{\dfrac{q^2}{p^2}+1}\frac{1}{p}\cdot 2\pi p-\frac{1}{\dfrac{q^2}{p^2}+1}\left(-\frac{q}{p^2}\right)\cdot 2\pi q\right]$$

$$=\frac{p^2}{p^2+q^2}+\frac{q^2}{p^2+q^2}=1$$

so that this transformation is, indeed, canonical, as expected.

Naturally, the Poisson bracket of the old variables with respect to new also must be equal 1 which, indeed, it does:

$$\begin{aligned}
&q=\sqrt{\frac{P}{\pi}}\sin 2\pi Q,\qquad p=\sqrt{\frac{P}{\pi}}\cos 2\pi Q\\
\Rightarrow\quad &\{q,p\}_{QP}=\frac{\partial q}{\partial Q}\frac{\partial p}{\partial P}-\frac{\partial q}{\partial P}\frac{\partial p}{\partial Q}\\
&=2\pi\left(\sqrt{\frac{P}{\pi}}\cos 2\pi Q\cdot\frac{1}{2}\sqrt{\frac{1}{\pi P}}\cos 2\pi Q-\frac{1}{2}\sqrt{\frac{1}{\pi P}}\sin 2\pi Q\cdot\sqrt{\frac{P}{\pi}}\left[-\sin 2\pi Q\right]\right)=1\,.
\end{aligned}$$

2. Another example of a canonical transformation:

$$Q=\ln\left(\frac{1}{q}\sin p\right),\qquad P=q\cot p.$$

This time we don't know the generating function, and must use the PB to prove that it is canonical:

$$\begin{aligned}
\{Q,P\}&=\frac{\partial Q}{\partial q}\frac{\partial P}{\partial p}-\frac{\partial Q}{\partial p}\frac{\partial P}{\partial q}=\left(-\frac{1}{q}\right)\left(-\frac{q}{\sin^2 p}\right)-\cot^2 p\\
&=\left(\cot^2 p+1\right)-\cot^2 p=1\,.
\end{aligned}$$

10.5 Hamilton-Jacobi theory

The main reason for introducing canonical transformations is to make solving particular mechanical problems easier. One way of doing so is transforming to a set of coordinates that includes as many cyclic ones as possible. Then the integration will become trivial with respect to these coordinates.

An alternative possibility seems even more attractive. One can formulate an ultimate goal: to find, at least in principle, the canonical transformation (CT) that solves any problem. To this end, note that the transformation from a set of the coordinates at time t to their values at the initial moment of time is a CT by itself. To see that this is the case, let us rewrite our defining expression for a CT, Eq. (9.8), *i.e.*the statement that $\mathcal{L}$ (or $\mathcal{H}$) is not defined uniquely:

$$\begin{aligned} P\dot{Q} - \widetilde{H}(P,Q) &= p\dot{q} - \mathcal{H}(p,q) - \dot{F}_1 \\ \Rightarrow \quad \mathrm{d}F_1 &= (p\,\mathrm{d}q - P\mathrm{d}Q) - \left(\widetilde{H} - H\right)\mathrm{d}t \end{aligned} \tag{10.5}$$

Let's compare this with the defining expression for the action, S:

$$\begin{aligned} S &= \int_1^2 \mathrm{d}t\mathcal{L} \to \int_1^2 (p\,\mathrm{d}q - \mathcal{H}\,\mathrm{d}t) \\ \Rightarrow \quad \mathrm{d}S &= p_{t+\mathrm{d}t}\,\mathrm{d}q_{t+\mathrm{d}t} - p_t\,\mathrm{d}q_t - \left(\mathcal{H}_{t+\mathrm{d}t} - \mathcal{H}_t\right)\mathrm{d}t \end{aligned}$$

This is equivalent to Eq. (10.5), with S playing the role of a generating function and $(p,q)_{t+\mathrm{d}t}$ being the "old" variables while q,p are 'new'. Choosing $t \equiv t_0$, we can relate our new variables to the initial conditions, this solving the problem.

Note: the choice of what is old and what is new is a matter of convenience.

Then, the equations expressing the "old" coordinates, q,p in terms of "new" ones, q_0, p_0, give the sought solution of the problem at hand:

$$q = q(q_0, p_0; t), \qquad p = p(q_0, p_0; t).$$

Our aim now is to find the CT that gives such a solution.

10.6 Transformation to a zero Hamiltonian

Let us require that the new Hamiltonian, $\mathcal{H}$, is identically zero. Then its solutions are t-independent P and Q, so that we can choose them to coincide with the initial conditions and then relate the old variables to them. Hamilton's equations for P and Q:

$$\begin{aligned} Q_i &= \frac{\partial\widetilde{\mathcal{H}}}{\partial P_i} = 0 \qquad &\Rightarrow \quad Q_i = \text{const} \\ P_i &= -\frac{\partial\widetilde{\mathcal{H}}}{\partial Q_i} = 0 \qquad &\Rightarrow \quad P_i = \text{const} \end{aligned}$$

As we have seen, any new $\widetilde{\mathcal{H}}$ can be generated from $\mathcal{H}$ with the help of a generating function. This is true also for $\widetilde{\mathcal{H}} = 0$:

$$0 = \widetilde{\mathcal{H}}(P_i, Q_i) = \mathcal{H}\Big(p_i(P,Q), q_i(P,Q), t\Big) + \frac{\partial S}{\partial t}. \tag{10.6}$$

Thus requiring $\widetilde{H} = 0$ results in Eq. (10.6) for the old $\mathcal{H}$ and some unknown generating function. Any S contains one of the old, and one of the new variables. It is convenient to choose S to depend on q and P which makes it the generating function of the 2$^{\text{nd}}$ type, $F_2(q, P)$. We can then express the old momenta p in terms of the generating function $S \equiv F_2$ by

Note: we call this function S anticipating its relation to the action.

Here we have used a customary relabelling $P_i \equiv \alpha_i$

$$p_i = \frac{\partial S(q_i, \alpha_i; t)}{\partial q_i}$$

Thus, we arrive at the equation

$$\mathcal{H}\left(q_i, \frac{\partial S}{\partial q_i}; t\right) + \frac{\partial S}{\partial t} = 0 \tag{10.7}$$

$$\boxed{\textsf{\textit{This equation is called the Hamilton-Jacobi equation}}}$$

It is a partial differential equation in $n+1$ variables (q_i, t).

Thus one should have

$$\boxed{\begin{aligned} &\frac{\partial S}{\partial t} + \mathcal{H}\left(q_i, \frac{\partial S}{\partial q_i}; t\right) = 0 \\ &S = S\left(q_1 \ldots q_n, \alpha_1 \ldots \alpha_n; t\right) \end{aligned}} \tag{10.8}$$

The second line of the equation above gives the desired solution which is called ***Hamilton's principle function***. The HJ approach can be described as follows:

$$\boxed{\textsf{\textit{Substitute }} p = \partial S/\partial q \textsf{\textit{ into }} \mathcal{H}(q,p) \textsf{\textit{ and thus get }} \mathcal{H} + \frac{\partial S}{\partial t} = 0}$$

Let's see now that the letter "S" was chosen for a good reason. The full time derivative of S:

$$\frac{\mathrm{d}S}{\mathrm{d}t} = \frac{\partial S}{\partial t} + \sum_i \frac{\partial S}{\partial q_i}\dot{q}_i = \frac{\partial S}{\partial t} + \sum_i p_i \dot{q}_i = -\mathcal{H} + \sum_i p_i \dot{q}_i \equiv \mathcal{L} \tag{10.9}$$

since (i) all the α_i are constant in time; (ii) $\frac{\partial S}{\partial q_i} = p_i$, (iii) $\frac{\partial S}{\partial t} = -\mathcal{H}$ for $\widetilde{\mathcal{H}} = 0$. It follows immediately from Eq. (10.9) that $S = \int \mathcal{L}\,\mathrm{d}t$, *i.e.*this is, indeed, the action.

The partial differential equation 10.8 is fully equivalent to the $2n$ first-order Hamilton's differential equations. Although S depends on $2n + 1$ variables, those which are called α (formerly P) are simply constants of motion, by the definition of new variables. Thus, effectively S is a function of $n+1$ variables. Knowing all the α's effectively gives us solutions to the equations of motion. In general, solving such an equation is at least equally (if not more) difficult than the original equations. However, such an equivalence not only gives us new insight but provides one with a valuable tool for a few practical problems. Notably, this approach is very important for a so called "quasi-classical solution" of a quantum-mechanical problem, *i.e.*the solution in the situation where the action is quite smooth (large on the scale of $\hbar$) – this, however, lies well beyond the scope of the present course.

Another area where the HJ approach is useful is geometrical optics – but this is an example in many ways equivalent to the previous: the limit which is called geometrical optics, where the wave nature of light is less (or totally un-) important is methodologically the same as the quasi-classical limit in quantum mechanics.

10.7 Hamilton-Jacobi equation for SHO

As usual, SHO is the standard workhorse for any new techniques. For later comparisons, we keep m and ω_0 arbitrary:

$$\begin{aligned} \mathcal{H} &= \frac{1}{2m}p^2 + \frac{1}{2}m\omega_0^2 q^2 \quad \text{substitute} \quad p = \frac{\partial S}{\partial q} \\ \Rightarrow \frac{1}{2m}\left(\frac{\partial S}{\partial q}\right)^2 &+ \frac{1}{2}m\omega_0^2 q^2 = -\frac{\partial S}{\partial t} \end{aligned} \tag{10.10}$$

Note a certain resemblance with the Schrödinger equation:

$$-\frac{\hbar^2}{2m}\left(\frac{\partial \Psi}{\partial q}\right)^2 + V(q) = i\frac{\partial \Psi}{\partial t}$$

Apart from the constants, the only essential difference is the presence of i in front of the derivative with respect to time.

In modern language, the HJ equation is the Schrödinger equation in the imaginary time (not the other way around!).

Solving it is similar to going from the time to the energy representation in quantum mechanics. As only the last term in Eq. (10.10) depends on t, one solves it by equating both parts to the same constant, α. In other words, one makes a substitution $S(q, \alpha; t) = -\alpha t + W(q, \alpha)$. This gives the following equation for W (putting $m = \omega_0 = 1$ – correct units can be restored at any time using the dimensional analysis):

$$\frac{1}{2}\left(\frac{\partial W}{\partial q}\right)^2 + \frac{1}{2}q^2 = \alpha$$

This can be immediately integrated:

$$W = \int\sqrt{2\alpha - q^2}\,\mathrm{d}q \quad \Rightarrow \quad S = -\alpha t + \int\sqrt{2\alpha - q^2}\,\mathrm{d}q$$

It's not too difficult to carry out the integration, but there is no particular need in this, as we will need to know all the constants of motion that are just the derivatives of S (which is the generating function). We had for the 2nd type of the generating function

$$Q = \frac{\partial F}{\partial P} \qquad \Rightarrow \qquad \beta = \frac{\partial S}{\partial \alpha}$$

where it is customary to rename the conserved generalised coordinate Q into β after we have renamed P into α. Therefore,

$$\beta = \frac{\partial S}{\partial \alpha} = -t + \int \frac{\mathrm{d}q}{\sqrt{2\alpha - q^2}} = -t + \arcsin \frac{q}{\sqrt{2\alpha}}\,.$$

This contains already all the information about the harmonic motion. Inverting this, one finds

$$q(t) = \sqrt{2\alpha}\sin(t + \beta)$$

where new generalised coordinates just play the role of the constants of the integration. The old momentum can also be determined from S (or W) by

$$p = \frac{\partial S}{\partial q} = \frac{\partial}{\partial q}\int^q \sqrt{2\alpha - \tilde{q}^2}\,\mathrm{d}\tilde{q} = \sqrt{2\alpha - q^2} \mapsto \sqrt{2\alpha}\cos(t + \beta)\,.$$

To complete the story, one should connect α and β to the initial conditions of the original problem. Although this is straightforward, it's worth noting that α is nothing more than the conserved full energy of the system, *i.e.*the value of $\mathcal{H}$ for the solution to the equations of motion, which simply follows from the fact that the HJ equation is $\partial S/\partial t + \mathcal{H} = 0$, so that equating $\partial S/\partial t = -\alpha$ means $\mathcal{H} = \alpha$.

11 Applications of Hamilton-Jacobi equation

The following is not assessed in the module, but is left for interest at a later date as it connects with future modules.

11.1 Particle in a constant force

The Hamiltonian is:

$$\mathcal{H} = \frac{1}{2m}p^2 - Fq, \tag{11.11}$$

while Hamilton's equations are:

$$\dot{q} = \frac{\partial \mathcal{H}}{\partial p} = \frac{p}{m}, \qquad \dot{p} = -\frac{\partial \mathcal{H}}{\partial q} = F \qquad \Rightarrow \qquad m\ddot{q} = F$$

i.e. this is indeed a particle moving under a constant force.

The HJ equation is obtained by the standard substitution that follows from the philosophy of HJ approach: $\widetilde{H} = 0$ and the standard CT formula $\widetilde{H} = H + \frac{\partial S}{\partial t}$ where the special notation S is used for the generating function, which is in this case called Hamilton's principal function (and is equal to the action when calculated with the Lagrangian obeying the least action principle).

$$\mathcal{H} = -\frac{\partial S}{\partial t}, \qquad p = \frac{\partial S}{\partial q}.$$

On substituting this into Eq. (11.11), one writes down the HJ equation:

$$\frac{\partial S}{\partial t} + \frac{1}{2m}\left(\frac{\partial S}{\partial q}\right)^2 - Fq = 0 \quad \Leftrightarrow \quad -\frac{\partial S}{\partial t} = \frac{1}{2m}\left(\frac{\partial S}{\partial q}\right)^2 - Fq = \alpha$$

As above, both functions have been equated to a constant, as this is the only way that a function of t can be equal to a function of q.

where, as usual, separation of variables was used. Therefore, $S = -\alpha t + W(q)$ and W obeys

$$\frac{1}{2m}\left(\frac{\partial W}{\partial q}\right)^2 - Fq = \alpha \qquad \Rightarrow W = \int^q \mathrm{d}\tilde{q}\sqrt{2m\,(\alpha + F\tilde{q})}$$

$$\Rightarrow S = -\alpha + \int^q \mathrm{d}\tilde{q}\sqrt{2m\,(\alpha + F\tilde{q})}$$

The constant "coordinate" is found again from the general equation for the generating function – in this case Hamilton's principal function:

$$\beta = \frac{\partial S}{\partial \alpha} = -t\sqrt{\frac{m}{2}}\int^{q}\frac{\mathrm{d}\tilde{q}}{\sqrt{\alpha + F\tilde{q}}} = -\,t + \frac{1}{F}\sqrt{2m(\alpha + Fq)}$$

$$\Rightarrow 2m\left(\alpha + Fq\right) = F^2\left(t+\beta\right)^2$$

$$\Rightarrow q = \frac{1}{2}\frac{F}{m}\left(t+\beta\right)^2 - \frac{\alpha}{F},$$

which is quite a recognisable result, apart from unusual notations for the constants. We can also get P from the standard $p = \partial S/\partial q$.

11.2 Hamilton-Jacobi equation for central forces

The Hamiltonian is:

$$\mathcal{H} = \frac{1}{2m}\left(p_r^2 + \frac{p_\theta^2}{r^2}\right) + V(r)\,.$$

Hamilton's principal function S becomes in this case a function of r, θ and two constants of motion, $\alpha_{1,2}$. The HJ equation is obtained by the following set of standard substitutions:

$$\mathcal{H} = -\frac{\partial S}{\partial t}\,, \qquad p_r = \frac{\partial S}{\partial r}\,, \qquad p_\theta = \frac{\partial S}{\partial \theta}\,,$$

which yields

$$-\frac{\partial S}{\partial t} = \frac{1}{2m}\left(\frac{\partial S}{\partial r}\right)^2 + \frac{1}{2mr^2}\left(\frac{\partial S}{\partial \theta}\right)^2 + V(r) = \alpha_1 \equiv \mathcal{E} \quad \text{full energy}$$

Here θ is cyclic, and the dependence on θ is as simple as that on t. However, we will not use this simplicity from the very beginning but try to find a solution in a fully separable form:

$$S = -\alpha_1 t + W_r(r) + W_\theta(\theta).$$

On substituting this form, we have

$$\alpha_1 = \frac{1}{2m}\left(\frac{\mathrm{d}W_r}{\mathrm{d}r}\right)^2 + \frac{1}{2mr^2}\left(\frac{\mathrm{d}W_\theta}{\mathrm{d}\theta}\right)^2 + V(r)$$

$$\Rightarrow 2mr^2\left(\alpha_1 - V(r)\right) - r^2\left(\frac{\mathrm{d}W_r}{\mathrm{d}r}\right)^2 = \left(\frac{\mathrm{d}W_\theta}{\mathrm{d}\theta}\right)^2 = \alpha_2^2$$

As the l.h.s. is a function of r only, and the r.h.s. is a function of θ, they can be equal to each other only if they are constant, this constant being denoted as α_2^2. As p_θ is actually the angular momentum, $\alpha_2 \equiv \ell$. Then one obtains $W_\theta = \alpha_2 \theta$ and

$$\frac{1}{2}\left(\frac{\mathrm{d}W_r}{\mathrm{d}r}\right)^2 + \frac{\alpha_2^2}{2r^2} + V(r) = \alpha_1$$

so that

Here the integration parameter is renamed to ρ.

$$\begin{aligned}
\frac{\mathrm{d}W_r}{\mathrm{d}r} &= \pm\sqrt{2\Big[\alpha_1 - V(r)\Big] - \frac{\alpha_2^2}{r^2}} \\
\Rightarrow W_r &= \int^r \mathrm{d}\rho \sqrt{2\Big[\alpha_1 - V(\rho)\Big] - \frac{\alpha_2^2}{\rho^2}} \\
\Rightarrow S &= -\alpha_1 t + \alpha_2 \theta + \int^r \mathrm{d}\rho \sqrt{2\Big[\alpha_1 - V(\rho)\Big] - \frac{\alpha_2^2}{\rho^2}}
\end{aligned}$$

Now constant coordinates, $\beta_i = \partial S/\partial \alpha_i$, are found as

$$\beta_1 = \frac{\partial S}{\partial \alpha_1} = -t + \int^r \frac{\mathrm{d}\rho}{\left[2\left(\alpha_1 - V(\rho)\right) - \frac{\alpha_2^2}{\rho^2}\right]^{1/2}}$$

$$\beta_2 = \frac{\partial S}{\partial \alpha_2} = \theta - \int^r \frac{\alpha_2 \, \mathrm{d}\rho}{\rho^2 \left[2\left(\alpha_1 - V(\rho)\right) - \frac{\alpha_2^2}{\rho^2}\right]^{1/2}}$$

These are precisely "quadratures" for the time and polar angle found previously. The difference in notations is obvious: the first integral, α_1, is found because $\mathscr{H}$ is t independent; not surprisingly, it is the same as the energy constant. The second integral, α_2, was found here because θ is ignorable; it coincides with p_ϕ found previously.

11.3 Hamilton-Jacobi and quantum mechanics

We start from the Schrödinger equation:

$$i\hbar\frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} + V(x)\Psi \qquad (= \hat{\mathscr{H}}\Psi)\,. \tag{11.12}$$

The complex wavefunction can be represented as

$$\Psi(x,t) = \sqrt{\rho(x,t)}\mathrm{e}^{i\frac{S(x,t)}{\hbar}} \tag{11.13}$$

where $\sqrt{\rho}$ and S are, respectively, its amplitude and phase. Now we substitute the representation 11.13 into Eq. (11.12):

$$-\frac{\hbar^2}{2m}\left\{\frac{\partial^2\sqrt{\rho}}{\partial x^2}+\frac{2i}{\hbar}\frac{\partial\sqrt{\rho}}{\partial x}\frac{\partial S}{\partial x}-\frac{1}{\hbar^2}\sqrt{\rho}\left(\frac{\partial S}{\partial x}\right)^2+\frac{i}{\hbar}\sqrt{\rho}\frac{\partial^2 S}{\partial x^2}\right\}+V(x)\sqrt{\rho}$$
$$=i\hbar\left\{\frac{\partial\sqrt{\rho}}{\partial t}+\frac{i}{\hbar}\sqrt{\rho}\frac{\partial S}{\partial t}\right\}$$

where we have cancelled the phase factor on both sides of this equation.

The semi-classical condition is

$$\hbar\left|\frac{\partial^2 S}{\partial x^2}\right|\ll\left|\frac{\partial S}{\partial x}\right|^2$$

In some way, one can say that $\hbar$ is small (physically, this means that any spatial scale of motion is much bigger than λ_{F}).

Eliminating all the terms that depend on $\hbar$, one reduces this equation to the Hamilton-Jacobi equation for the phase:

$$\frac{1}{2m}\left(\frac{\partial S}{\partial x}\right)^2+V(x)+\frac{\partial S}{\partial t}=0\,.$$

This still contains very important QM information. This approximation in QM is called the WKB approximation.